

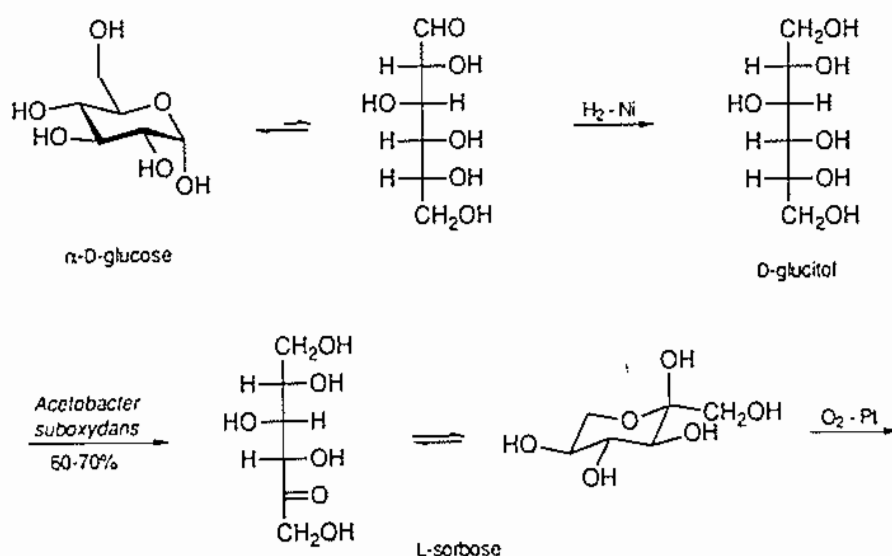
Oxidation 19

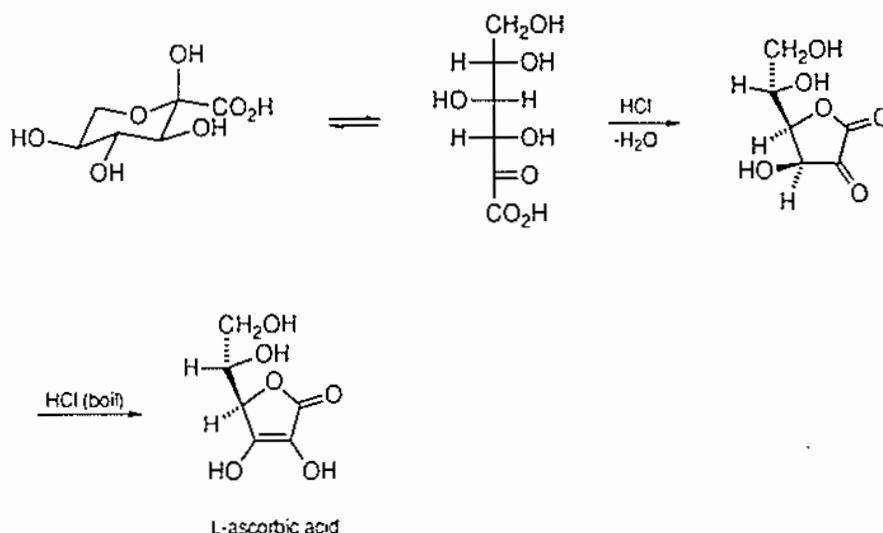
19.1 Introduction

The scope of the term oxidation has been discussed earlier (section 4.9). The processes described in this chapter are those which lead to the incorporation of oxygen or the removal of hydrogen from a molecule; those which result in the partial removal of electrons (e.g. the bromination of methane) are included only where they are employed *en route* to the introduction of oxygen or the removal of hydrogen, as in the conversion of toluene into benzaldehyde *via* benzal chloride (p. 599).

The different techniques used in the laboratory and in large-scale industrial processes are stressed. In the former, complex polyfunctional molecules are often involved; regioselective and/or stereoselective reagents are required, many of which are expensive, and high temperatures have usually to be avoided. In industrial high-tonnage processes, relatively simple compounds are often involved which can withstand vigorous conditions; the first essential is cheapness. Air is the natural industrial oxidizing agent, as in the conversions of isopropanol into acetone, propylene into propenal, and naphthalene into phthalic anhydride, each over metal oxide catalysts at high temperatures.

Bacterial oxidation is also employed industrially. For example, *Acetobacter suboxydans* is the best reagent for the specific oxidation of the C₂-carbon of D-glucitol in the production of vitamin C (L-ascorbic acid) from glucose:





Both bacterial and enzymatic methods may eventually prove to be powerful tools for selective oxidation in the laboratory.

This chapter is arranged according to the type of system to be oxidized, so that whereas the appropriate method for a particular oxidation can readily be found, the properties of any one oxidizing agent are not collected together.

19.2 Hydrocarbons

(a) Alkene double bonds

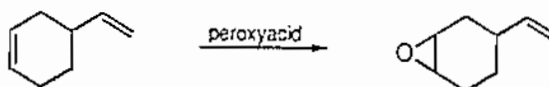
(i) *Epoxidation.* Alkenes react with peroxyacids to give epoxides. The reaction is usually expressed as



but this does not reflect the fact that electron-releasing groups in the alkene increase the reactivity (e.g. $\text{RCH}=\text{CHR} \sim \text{R}_2\text{C}=\text{CH}_2 > \text{RCH}=\text{CH}_2 > \text{CH}_2=\text{CH}_2$), as do electron-attracting groups in the peroxyacid: that is, there is a partial transfer of electrons from alkene to peroxyacid at the transition state.

The important characteristics of the reaction are:

- It is regioselective for the more electron-rich double bond when more than one is present, e.g.



- It is stereospecific, as expected from the above mechanism; for example, *cis*-2-butene gives only the *cis* product:

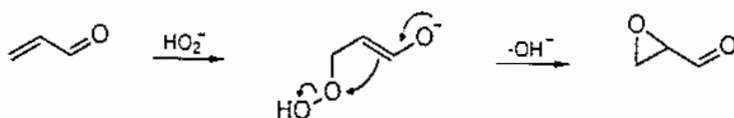


- Cyclic alkenes are attacked predominantly from the less hindered side, e.g.

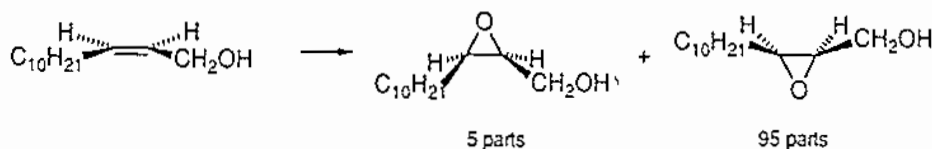


The peroxyacid can be prepared *in situ* from the parent acid and hydrogen peroxide. However, it is generally more convenient to use a commercially available compound. One which has been widely used and gives good yields is *m*-chloroperoxybenzoic acid but safety considerations have led to its replacement by the magnesium salt of monoperoxyphthalic acid. A peroxyacid from a relatively strong acid such as $\text{CF}_3\text{CO}_2\text{H}$ is unsuitable since the acid released during epoxidation is sufficiently strong to bring about ring opening of the epoxide (see p. 591).

$\alpha\beta$ -Unsaturated carbonyl compounds. These can be epoxidized by an alkaline solution of hydrogen peroxide. Nucleophilic addition of HO_2^- to $\text{C}=\text{C}$ is facilitated by the $\text{C}=\text{O}$ group (cf. Michael reaction, p. 240):



Enantioselective epoxidation: Sharpless reaction. An alkene which is achiral inevitably yields a non-optically active product with an achiral oxidant. However, allylic alcohols give mainly one enantiomer of a pair when epoxidized with *t*-butyl hydroperoxide in the presence of $\text{Ti}(\text{OCHMe}_2)_4$ and a chiral dihydroxy compound such as (2*R*,3*R*)-diethyl tartrate, e.g.

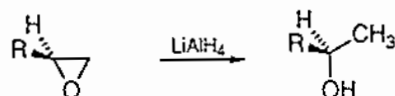


The (2*S*,3*S*)-tartrate gives the same enantiomers but in reversed proportions. It is thought that oxygen is transferred from the hydroperoxide to the alkene

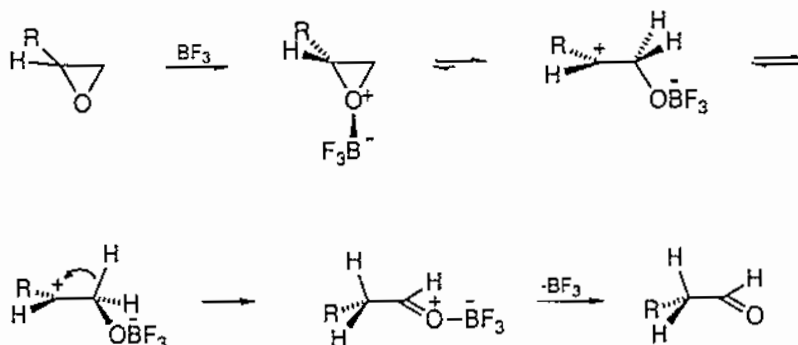
within a complex in which the configurations of the two ester groups determine whether the oxygen is delivered mainly from above the plane of the alkene or from below it.

Reactions of epoxides. The value of epoxidation lies in the variety of synthetic pathways readily available by cleavage of the strained ring:

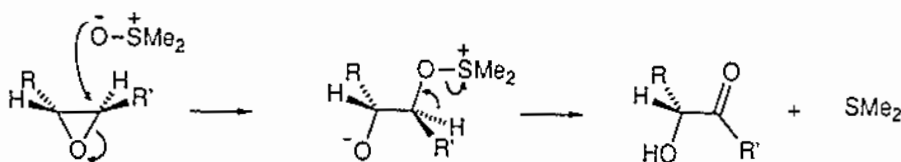
- (1) Reduction (lithium aluminium hydride) gives an alcohol (p. 656)



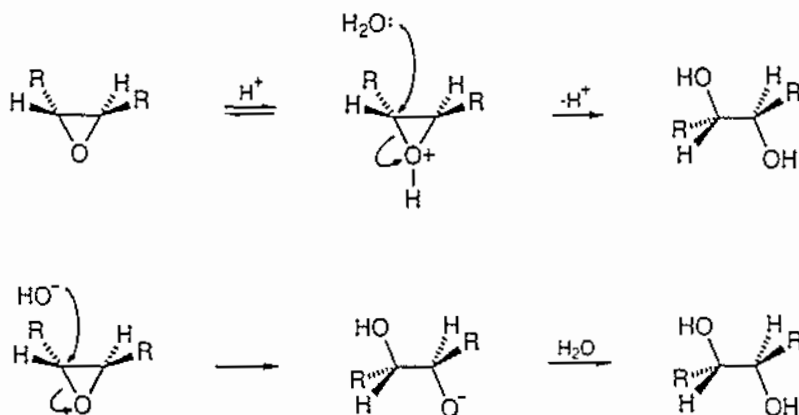
- (2) Treatment with a Lewis acid gives a carbonyl compound, e.g.



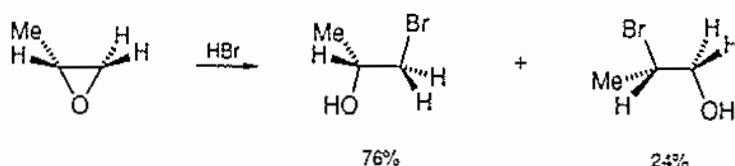
- (3) Treatment with dimethyl sulfoxide gives an α -ketol (cf. p. 610):



- (4) Hydrolysis gives a 1,2-diol. Either acidic or basic conditions can be used; in the former, the reactivity of the epoxide to a weak nucleophile (water) is enhanced by protonation of the ring oxygen. The product has *trans* stereochemistry as a result of the stereospecificity of $\text{S}_{\text{N}}2$ displacements (p. 160):



Other weak nucleophiles such as alcohols and bromide ion are also effective when the epoxide's reactivity is increased by protonation. As in other S_N2 reactions, the less alkylated carbon atom is the more reactive, e.g.



However, if ring opening of the protonated epoxide gives a carbocation that is strongly stabilized (e.g. by a phenyl substituent), the S_N1 mechanism is favoured. Consequently the stereospecificity is lost, and the nucleophile attacks at the substituted carbon.

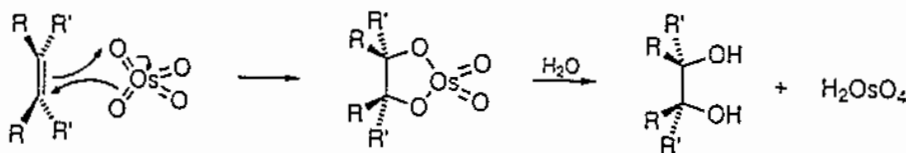
The ring opening of epoxides of rigid cyclohexene systems gives *trans*-diaxial products:



Industrially, ethylene oxide is made by the oxidation of ethylene with atmospheric oxygen at pressures up to 300 p.s.i. over a silver catalyst at 200–300°C. Its most important use is in the production of ethylene glycol, required, for example, for the manufacture of terylene.

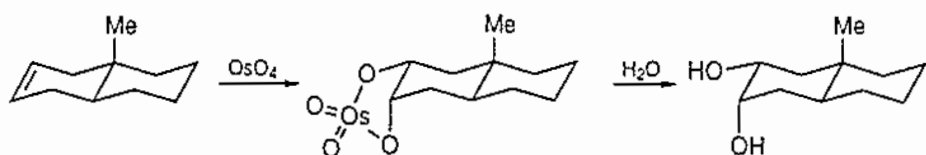
(ii) *Diol formation.* As well as the method for obtaining *trans*-1,2-diols via epoxides, there are three methods for converting alkenes directly into diols.

1. *Osmium tetroxide.* The addition of an alkene to osmium tetroxide in ether causes the rapid precipitation of a cyclic osmate ester. Pyridine, which complexes the osmium atom in the ester, is often added as a catalyst. The ester is then hydrolyzed, commonly with aqueous sodium sulfite, to give a *cis*-1,2-diol.

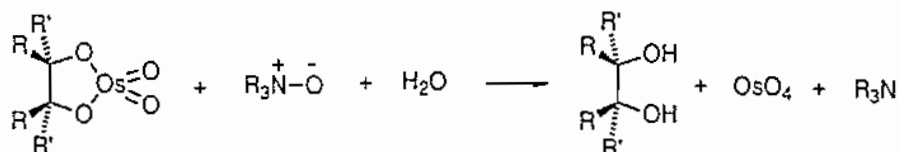


As well as its *syn*-stereospecificity, its other characteristics are the same as those of epoxidation:

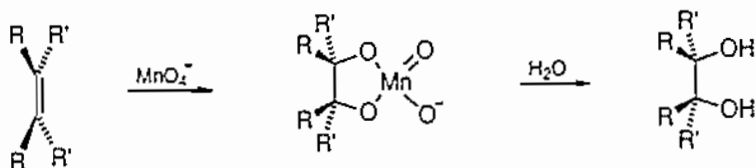
- It reacts with the most electron-rich double bond when more than one is present, so that it can be used regioselectively.
- It attacks rigid cyclic systems from the less hindered side, thereby yielding the more stable of the two possible *cis*-diols (compare the iodine-silver acetate method below), e.g.



Because of both its expense and its toxicity it is best to use osmium tetroxide in catalytic quantities by carrying out the reaction in the presence of a tertiary amine oxide or alkaline *t*-butyl hydroperoxide; the intermediate osmate ester regenerates the tetroxide, e.g.

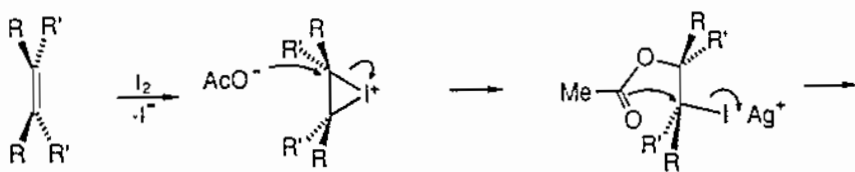


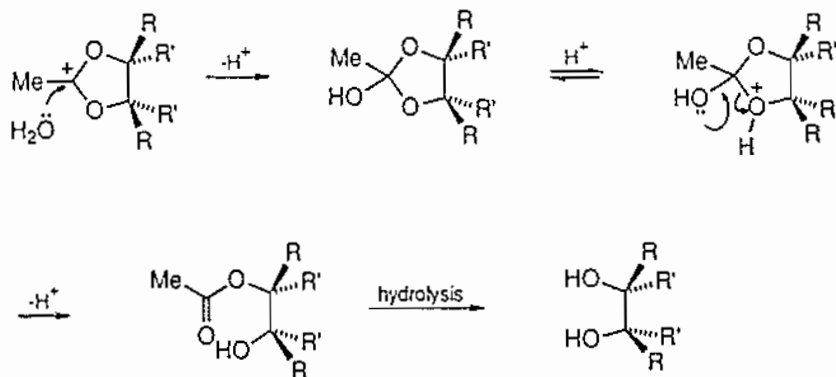
2. *Potassium permanganate acts similarly to osmium tetroxide:*



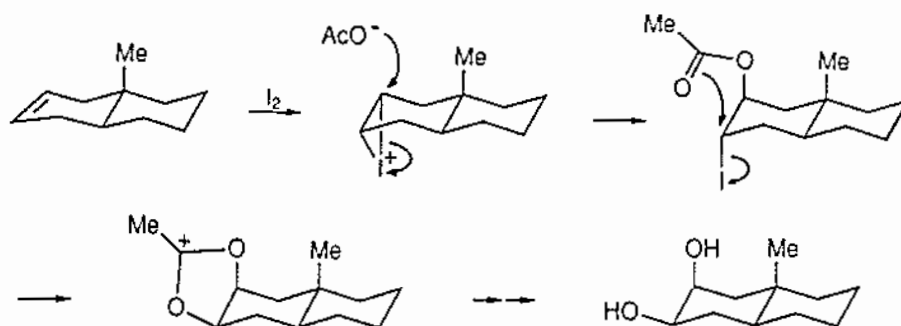
It is normally used in an aqueous solution in which the organic compound is dissolved or suspended; a co-solvent (usually *t*-butanol or acetic acid) is sometimes employed. In order to prevent further oxidation it is necessary to work in alkaline conditions, for otherwise the *cis*-diol is oxidized further with formation of an α -hydroxyketone or by cleavage of the C—C bond. In general, permanganate is less selective, and therefore less satisfactory, than osmium tetroxide, but it has the advantage of being less hazardous to use and very much cheaper.

3. *Iodine-silver acetate ('wet').* This method also yields *cis*-diols. The alkene is treated with iodine in aqueous acetic acid in the presence of silver acetate. Iodine reacts with the double bond to give an iodonium ion which undergoes displacement by acetate in the S_N2 manner, giving a *trans*-iodoacetate. Anchimeric assistance by the acetate group, together with the powerful bonding capacity of silver ion for iodide, then lead to the formation of a cyclic acetoxonium ion which in turn reacts with water to give a *cis*-hydroxyacetate. Final hydrolysis gives the *cis*-diol:

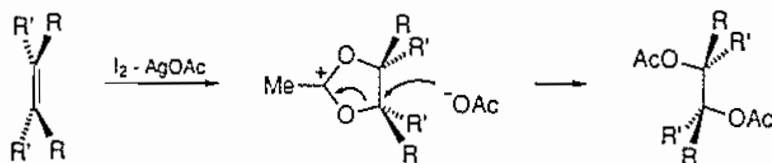




In a rigid cyclic alkene, approach of the iodine from the less hindered side leads ultimately to the *less* stable *cis*-diol, e.g.

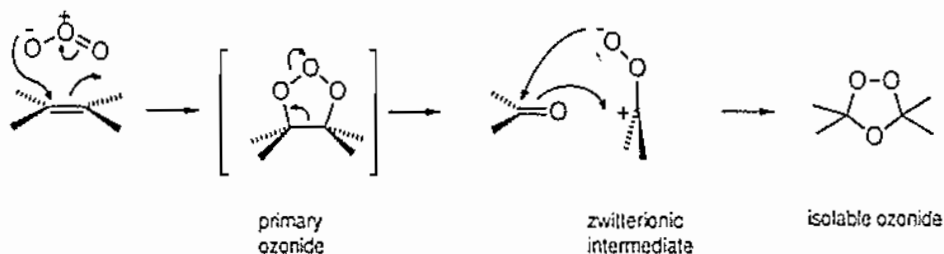


4. *Iodine-silver acetate ('dry').* The same reaction carried out in the absence of water leads to a *trans*-1,2-diacetate from which the *trans*-diol is obtained by hydrolysis (*Prévost reaction*):

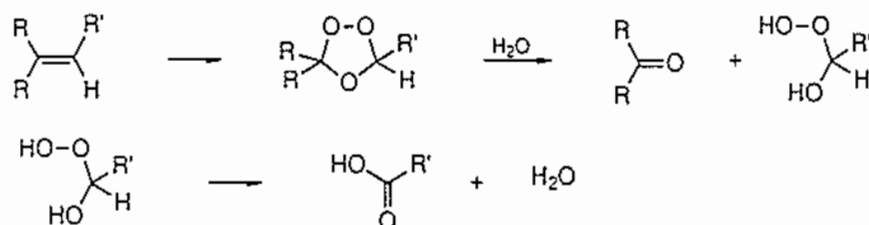


(iii) Cleavage of the double bond.

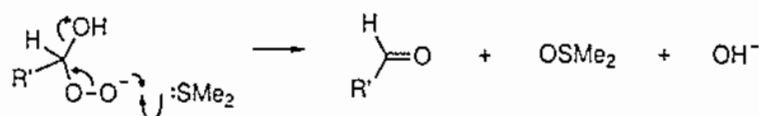
1. *Ozonolysis.* Ozone reacts as an electrophile with alkenes forming a primary ozonide which rearranges through a zwitterionic intermediate to an isolable ozonide:



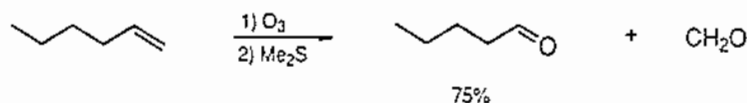
Ozonides are dangerously explosive, however, and are usually converted at once into products. Direct solvolysis gives ketones and/or acids, depending on the structure of the alkene e.g.



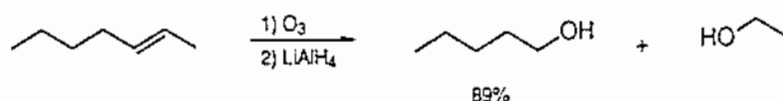
It is often more useful to carry out the solvolysis in the presence of a reducing agent. With zinc and acetic acid or dimethyl sulfide, only carbonyl compounds are formed; dimethyl sulfide has the advantage that other reducible groups in the compound are not affected and its suitability stems from its ability to reduce hydroperoxy compounds:



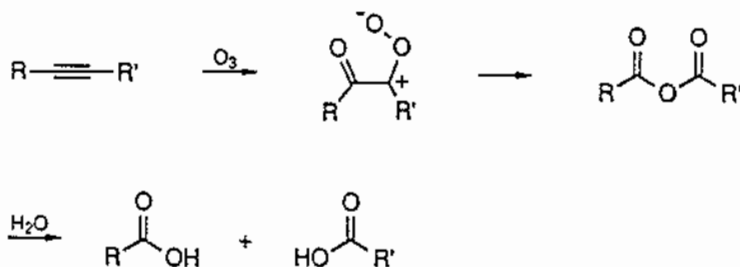
For example,



With lithium aluminium hydride, further reduction occurs to give alcohols, e.g.

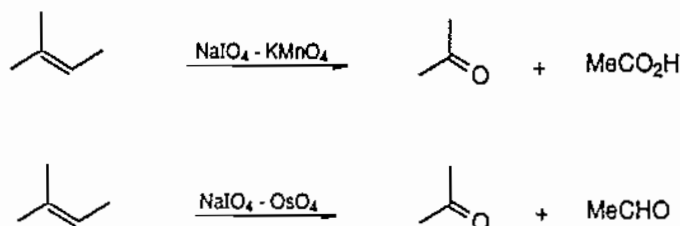


Alkynes are also oxidized by ozone but they generally react at only about one-thousandth the rate of alkenes just as they are less reactive than alkenes towards other electrophiles (p. 87). Selective oxidation of $\text{C}=\text{C}$ in the presence of $\text{C}\equiv\text{C}$ can therefore be achieved. Reaction with alkynes gives carboxylic acids, together with small amounts of α -dicarbonyl compounds, probably by a mechanism analogous to that for alkenes:

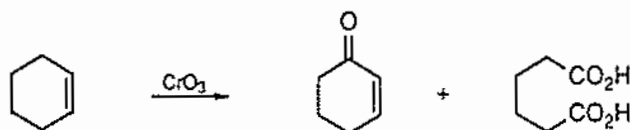


Aromatic rings also undergo ozonolysis (p. 596).

2. *The Lemieux reagents.* Ozone is unpleasant to handle and is not selective for alkenes, e.g. secondary alcohols are oxidized to ketones and tertiary C—H bonds to alcohols. Ozone has therefore largely been displaced by the Lemieux reagents which consist of dilute aqueous solutions of sodium periodate with a catalytic quantity of potassium permanganate and of osmium tetroxide, respectively. In each case, the alkene is first oxidized to the *cis*-diol which is then cleaved by the periodate (p. 616) to give aldehydes and/or ketones. The permanganate reagent then oxidizes aldehydic products to carboxylic acids. The low-valent states of manganese and osmium generated during the reaction are re-oxidized by the periodate to their original state so that only catalytic quantities are required. The reactions are rapid at room temperature and are selective for alkenes, e.g.



3. *Chromium(vi) oxide.* The cleavage of C=C bonds with chromic oxide is competitive with oxidation of allylic C—H bonds, e.g. cyclohexene gives a mixture of 3-cyclohexenone and adipic acid:



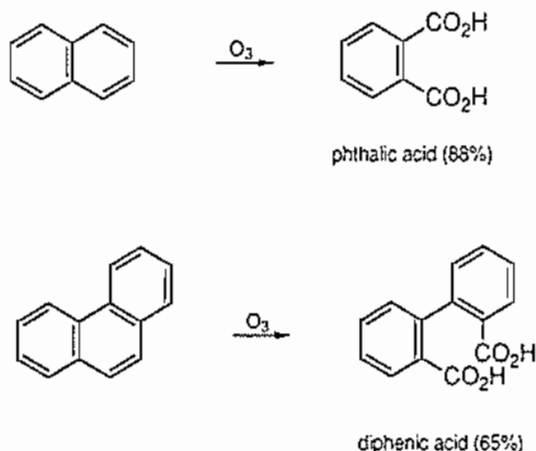
The use of a partially aqueous medium favours the cleavage process, whereas an anhydrous medium such as glacial acetic acid favours allylic oxidation. In addition, cleavage is promoted by the presence of phenyl substituents, evidently because the first step involves the formation of a carbocation which is stabilized by adjacent aromatic rings:



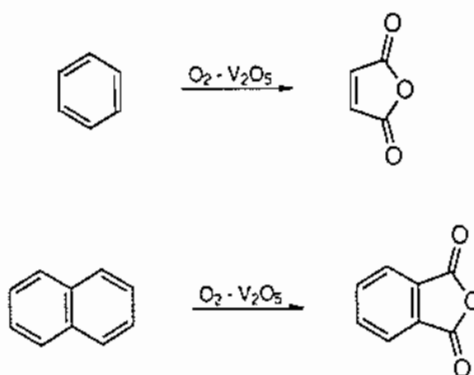
(b) Aromatic rings

The oxidation of unsubstituted aromatic rings which results in the loss of the associated stabilization energy requires vigorous conditions. Reaction can result either in the cleavage of the ring or in the formation of quinones.

Ozone effects cleavage, e.g.

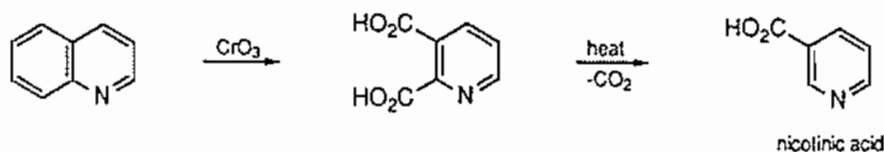


An industrial method, which is far cheaper than ozonolysis, employs aerial oxidation over a vanadium pentoxide catalyst at 400–500°C. For example, maleic anhydride and phthalic anhydride are cheaply available from benzene and naphthalene:



The latter process is particularly important since some of the derived diesters (e.g. dinonyl phthalate) are used as plasticizers for cellulose acetate and polyvinyl chloride. Polyester resins and phthalocyanine dyes are also manufactured from phthalic anhydride.

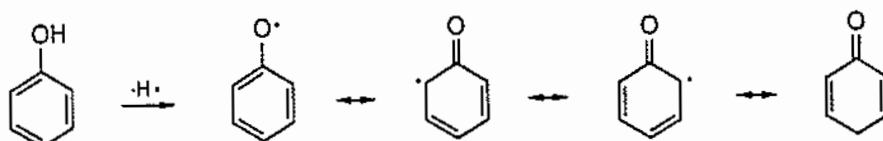
Chromium(vi) oxide can also be used: for example, quinoline is oxidized to pyridine-2,3-dicarboxylic acid which readily loses the 2-carboxy-substituent on being heated (p. 389), providing an easy route to nicotinic acid:



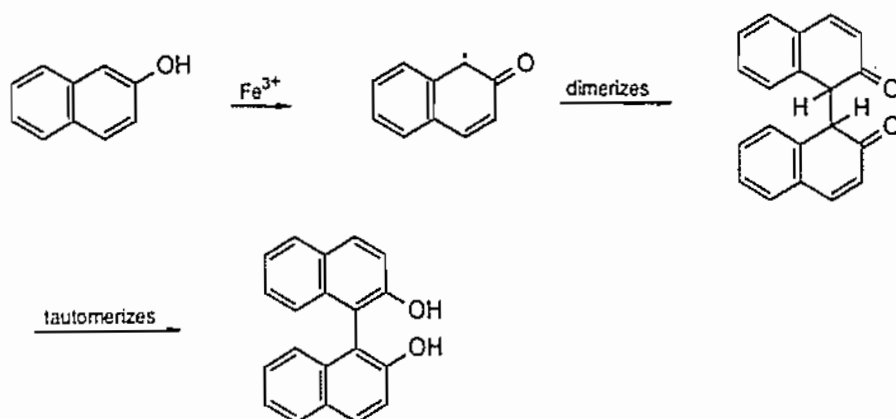
Chromium(vi) oxide does not invariably lead to ring fission. For example, it reacts with naphthalene in acetic acid solution at room temperature to give 1,4-naphthoquinone in about 20% yield [4]:



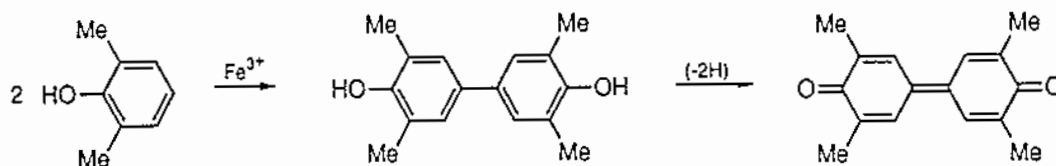
The aromatic rings of phenols are very susceptible to oxidation by one-electron oxidants, for the removal of a hydrogen atom gives a delocalized aryloxy radical,



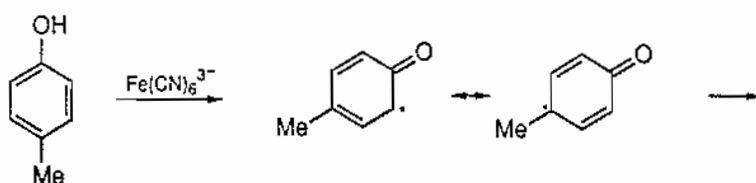
The fate of the radical depends on the structure of the phenol. Diphenols can be formed, as in the oxidation of 2-naphthol by iron(III) ion,

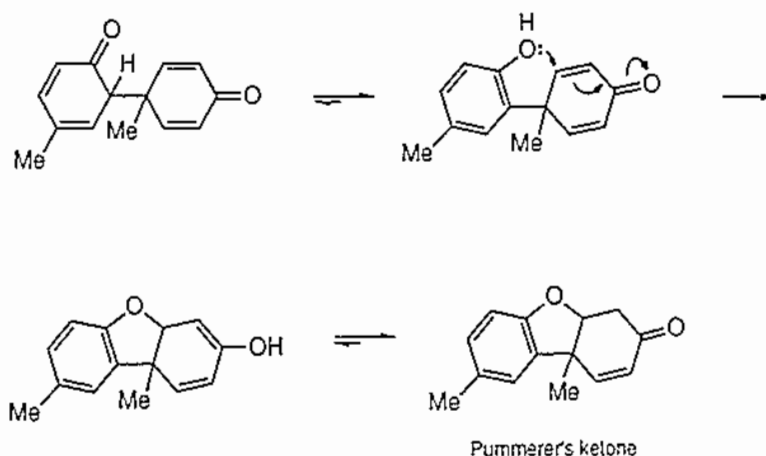


further oxidation to quinones can occur, e.g.

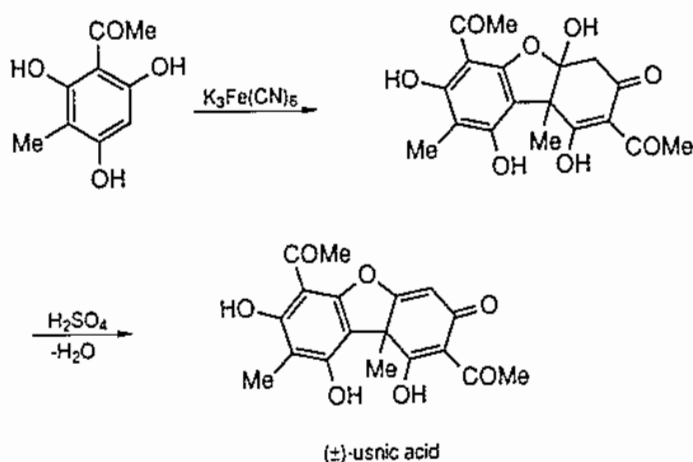


and cyclization to derivatives of dibenzofuran can arise through intramolecular nucleophilic attack on an intermediate quinone. For example, the oxidation of *p*-cresol by hexacyanoferrate(III) ion gives, in addition to carbon-carbon dimers, Pummerer's ketone:

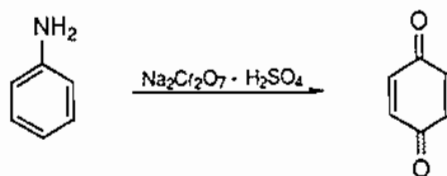




A synthesis of usnic acid is based on this mode of coupling. Hexacyanoferrate(III) oxidation of methylphloracetophenone gives a dimer which is dehydrated by concentrated sulfuric acid to ($\pm$)-usnic acid:

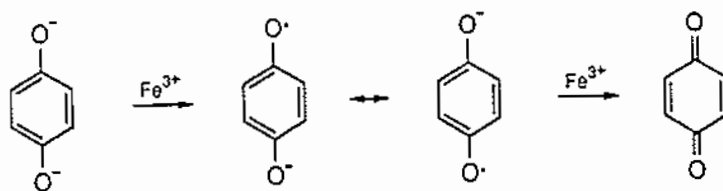


Aromatic amines are also sensitive to oxidation and discolour in air. The products are complex and the processes are of little use except in two cases: first, specific methods for oxidizing the amino group rather than the aromatic ring are available (p. 621) and second, aniline is oxidized to *p*-benzoquinone in 60% yield by dichromate:

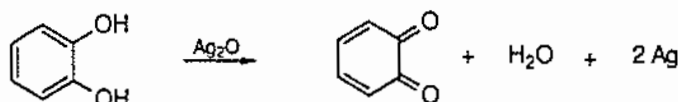


Industrially, aniline is oxidized to *p*-benzoquinone with manganese dioxide and sulfuric acid.

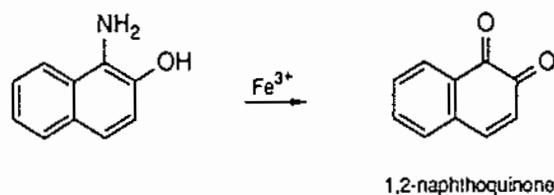
The *ortho* and *para* dihydric phenols are readily oxidized to the corresponding quinones by one-electron oxidants. Reaction occurs through delocalized semi-quinone radicals, e.g.



o-Benzoquinone may be prepared by the oxidation of catechol (from salicylaldehyde by Dakin's reaction; p. 447) with silver oxide suspended in ether; sodium sulfate is added as a dehydrating agent since the quinone is rapidly attacked by water (addition to the $\alpha\beta$ -unsaturated carbonyl system):

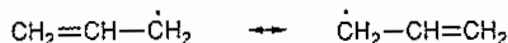


The corresponding aminophenols react similarly. For example, 1-amino-2-naphthol (readily available from 2-naphthol by nitrosation or diazo-coupling followed by reduction, p. 380) is oxidized by iron(III) chloride to 1,2-naphthoquinone in over 90% yield [2]:



(c) Saturated C—H groups

(i) *Allylic and benzylic systems.* The comparative stability of allyl and benzyl radicals, e.g.

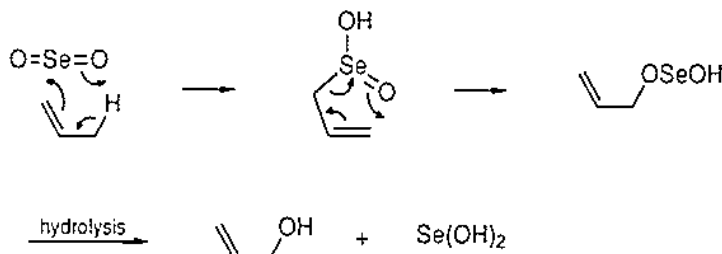


renders allylic and benzylic systems susceptible to oxidation *via* free-radical reactions. Autoxidation has been described (section 17.5). An alternative approach is through radical-catalyzed halogenation (p. 531). For example, the chlorination of toluene in the vapour phase or under reflux gives benzyl chloride, benzal chloride, and benzotrichloride, from which benzyl alcohol, benzaldehyde, and benzoic acid, respectively, are available by hydrolysis: the alcohol and aldehyde are produced in this way on an industrial scale. However, the difficulty of separating the halogenated products efficiently in the laboratory necessitates the use of more specific methods.

Oxidation to alcohols. Toluene can be oxidized to benzyl alcohol with sulfuryl chloride in the presence of a radical initiator such as a peroxide followed by hydrolysis of benzyl chloride,

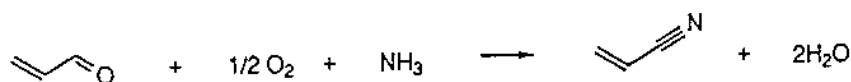


A method of wide application for allyl systems is the use of selenium dioxide. It is thought to act by an 'ene' reaction (p. 295) followed by a [3,2]-sigmatropic rearrangement and hydrolysis of a selenium ester:



By including *t*-butyl hydroperoxide to reoxidize $\text{Se}(\text{OH})_2$, selenium dioxide can be used in catalytic amount.

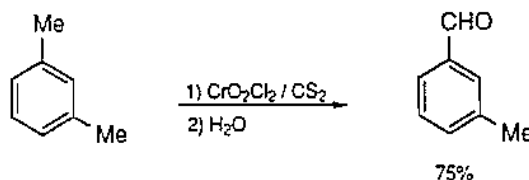
Oxidation to aldehydes. Commercially, a particularly important example is the oxidation of propylene to propenal $\text{CH}_2=\text{CH}-\text{CHO}$, over copper(II) oxide at $300-400^\circ\text{C}$. The product is largely converted into glycerol (required for nitroglycerine), by reduction to allyl alcohol followed by hydroxylation with hydrogen peroxide. A development involves the air oxidation of propylene in the presence of ammonia over a catalyst (e.g. bismuth molybdate); propenal is probably formed first and reacts further:



This process for acrylonitrile has superseded that from acetylene and hydrogen cyanide.

Three methods are available for the selective oxidation of ArCH_3 to ArCHO .

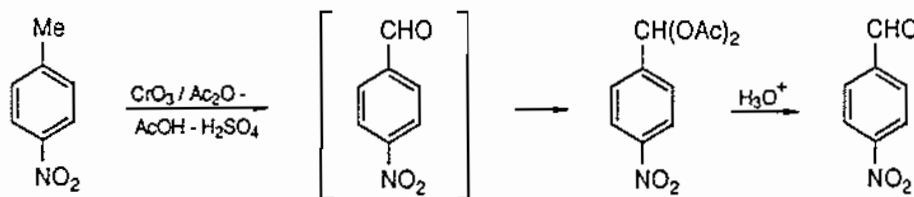
(1) *With chromyl chloride (Etard reaction).* A solution of chromyl chloride in carbon disulfide is added cautiously to the benzylic compound at $25-45^\circ\text{C}$ and the brown complex which separates is decomposed by water to give the aldehyde and chromic(VI) acid, e.g.



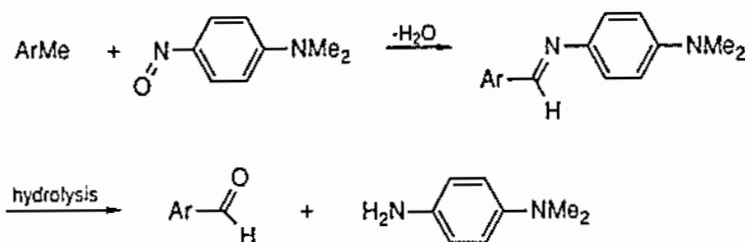
The aldehyde must be removed rapidly by distillation or extraction to prevent its further oxidation.

(2) *With chromium(VI) oxide in acetic anhydride.* Oxidation is carried out with chromium(VI) oxide in a mixture of acetic anhydride, acetic acid, and sulfuric acid

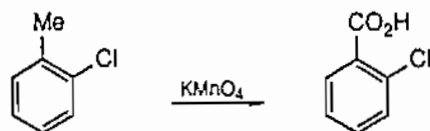
at low temperature. As it is formed, the aldehyde is converted into its 1,1-diacetate which is stable to oxidation; this is isolated and reconverted into the aldehyde by acid hydrolysis. For example, *p*-nitrotoluene gives *p*-nitrobenzaldehyde in about 45% overall yield [2]:



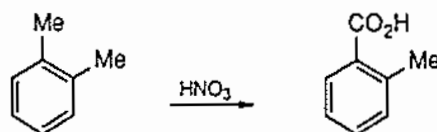
(3) *With p-nitrosodimethylaniline.* The benzylic compound is treated with *p*-nitrosodimethylaniline and the resulting imine is hydrolyzed. The method is applicable only to those compounds whose methyl groups are strongly activated, e.g. 2,4-dinitrotoluene



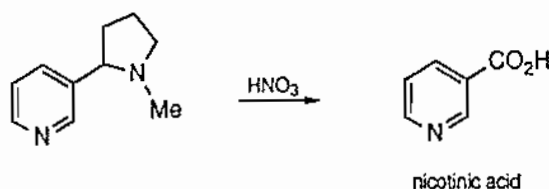
Oxidation to carboxylic acids. Methyl groups on aromatic rings can be oxidized with chromium(vi) oxide, permanganate, or nitric acid. For example, dilute permanganate oxidizes *o*-chlorotoluene to *o*-chlorobenzoic acid in about 65% yield [2],



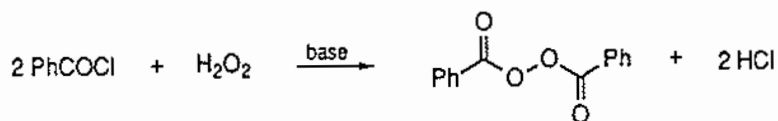
and concentrated nitric acid oxidizes *o*-xylene to *o*-toluic acid in 54% yield [3]:



Substituents on the methyl group are also removed in these conditions, e.g. nicotine is oxidized to nicotinic acid by concentrated nitric acid at 70°C :

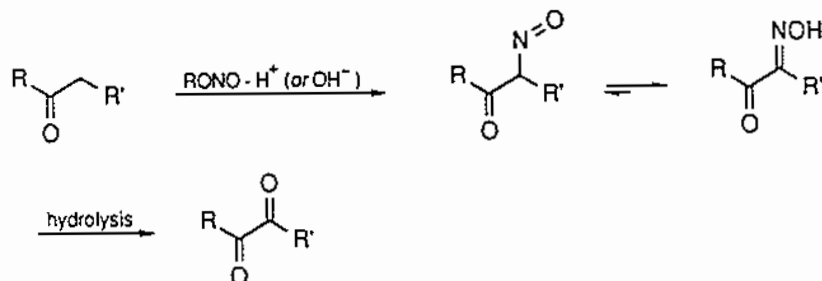


Industrially, benzoic acid is manufactured by the catalytic air oxidation of toluene. Its main use is in the production of sodium benzoate which is employed as a preservative for foodstuffs and pharmaceuticals. Benzoyl chloride is used to prepare dibenzoyl peroxide, used for initiating addition polymerization (and, on the laboratory scale, for other radical-catalyzed reactions) and as a bleaching agent.



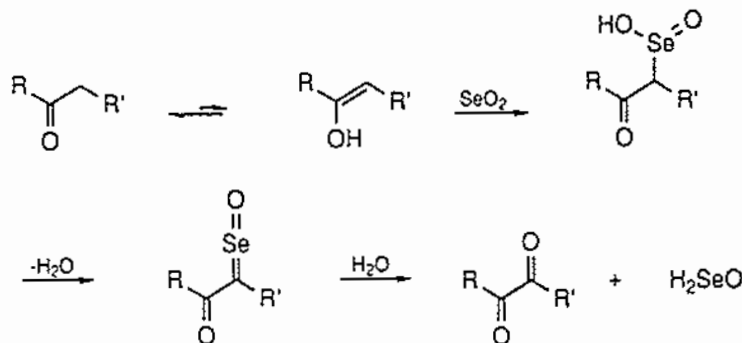
(ii) *The —CH₂—CO— system.* Methylene groups adjacent to carbonyl may themselves be oxidized to carbonyl in two ways.

1. *Through the oxime.* The methylene group is activated by the carbonyl group towards reaction with organic nitrites in the presence of acid or base (p. 319). The resulting nitroso compound tautomerizes to the oxime which may be hydrolyzed to the α -dicarbonyl compound:

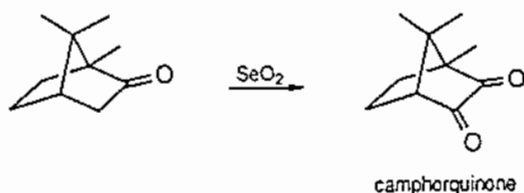


Methylene groups are oxidized in preference to methyl. For example, butanone gives $\text{CH}_3\text{COCOCH}_3$.

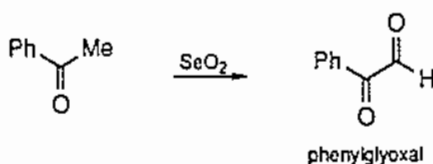
2. *By selenium dioxide (Riley reaction).* The reaction is thought to occur as follows:



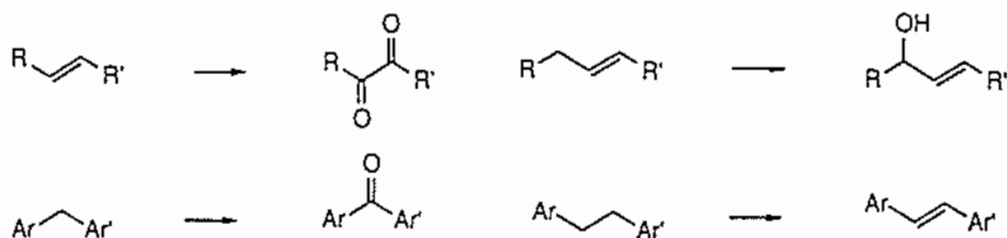
For example, camphor reacts in refluxing acetic anhydride to give camphorquinone in 95% yield,



and acetophenone reacts in dioxan to give phenylglyoxal in about 70% yield [2]:

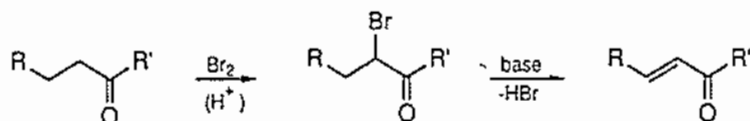


In contrast to the oxime method, selenium dioxide preferentially oxidizes methyl rather than methylene groups. For example, butanone gives mainly $\text{CH}_3\text{CH}_2\text{COCHO}$. Selenium dioxide is of limited scope because it is unselective: it can bring about the following oxidations:



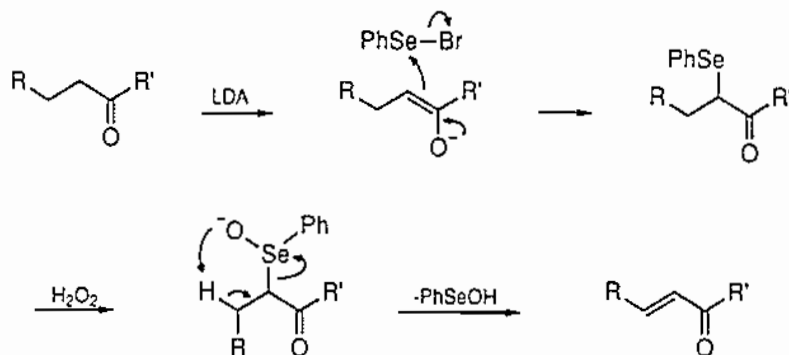
Oxidation of $-\text{CH}_2-\text{CO}-$ and $\text{>CH}-\text{CO}-$ to $-\text{CH}(\text{OH})-\text{CO}-$ and $\text{>C}(\text{OH})-\text{CO}-$ can also be effected. The enolate is formed in the presence of a strong base such as potassium *t*-butoxide and reacts with oxygen to give a hydroperoxide (p. 557). This is reduced to the corresponding alcohol with a trialkyl phosphite.

(iii) *The $-\text{CH}_2-\text{CH}_2-\text{CO}-$ and related systems.* Dehydrogenation to give an $\alpha\beta$ -unsaturated carbonyl compound can be brought about by acid-catalyzed bromination followed by base-catalyzed dehydrobromination:

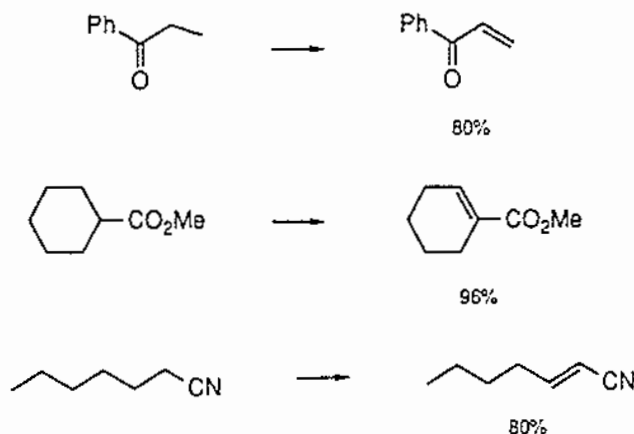


However, a more recent development has two advantages: acidic conditions are not required, and no base is present when the product is formed, so that possible base-catalyzed reactions are avoided. It consists of making either a sulfide or a selenide derivative and oxidizing with hydrogen peroxide to the sulfoxide or selenoxide which undergoes a ready thermal *syn*-elimination. The procedure *via* a

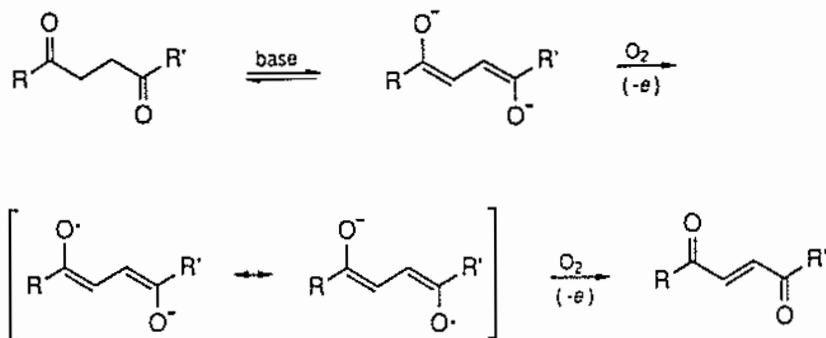
sulfide has been described (p. 474). That *via* a selenide is usually carried out by generation of the enolate with LDA followed by treatment with benzeneselenenyl bromide and then hydrogen peroxide:



The carbonyl group may be ketonic or part of an ester function, and $\alpha\beta$ -unsaturated nitriles also react. Yields are mostly very good, e.g.

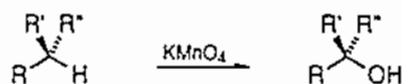


Finally, the γ -dicarbonyl system undergoes dehydrogenation by autoxidation in the presence of base; reaction occurs through the dienediolate dianion which donates two electrons successively to oxygen to form first a relatively stable anion-radical and then the enedione (cf. p. 599):

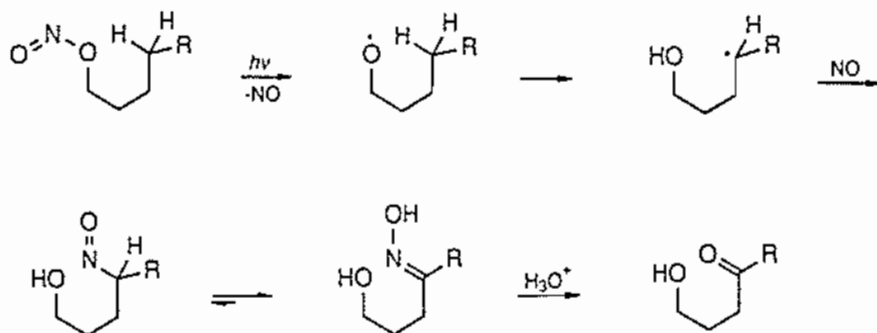


(iv) *Unactivated C—H*. The selective oxidation of one unactivated C—H group in a molecule possessing alternative centres for attack is attended by the

difficulty that the reagents with which C—H bonds react – essentially only free radicals – are relatively unspecific. However, selectivity is obtained in two circumstances. First, a relatively unreactive free radical, such as the bromine atom (p. 532), discriminates quite sharply in favour of tertiary C—H compared with primary or secondary C—H, and the oxidation of tertiary C—H to C—OH can sometimes be achieved directly with alkaline permanganate; reaction occurs with retention of configuration:



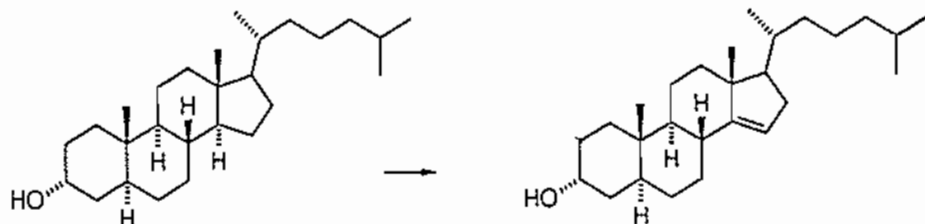
Second, intramolecular free-radical abstraction reactions occur specifically through six-membered cyclic transition states, so that it is possible selectively to oxidize δ -CH bonds, e.g.



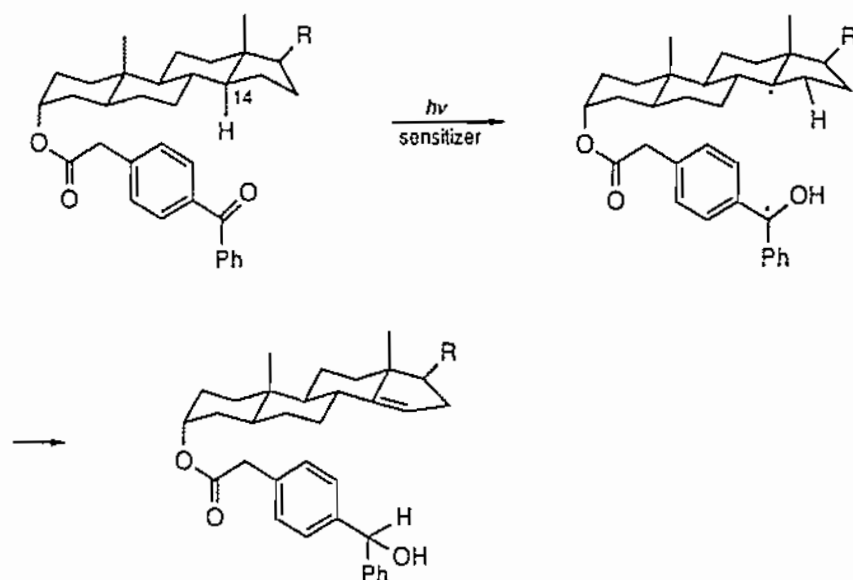
Examples of the application of this principle have been described earlier (p. 554).

The very vigorous oxidation of hydrocarbon chains by chromium(vi) oxide in concentrated sulfuric acid oxidizes most substances to carbon dioxide and water, but C-methyl groups give mainly acetic acid. This procedure has often been applied in determining the number of C-methyl groups in a compound of unknown structure (Kuhn–Roth method).

Regioselective dehydrogenation. A method for introducing a C=C bond in a steroid, with good regioselectivity, can be illustrated by the following conversion of 3 α -cholestanol:



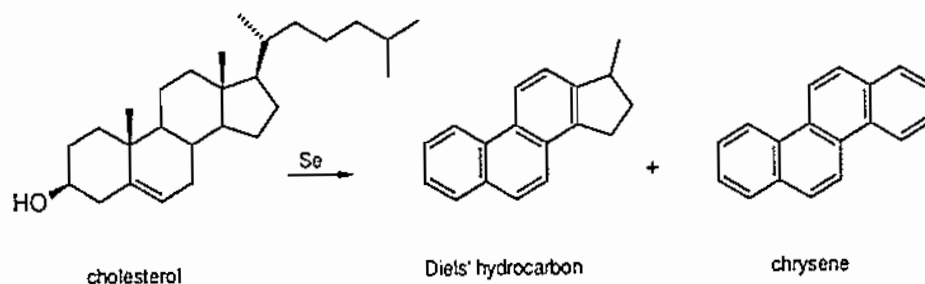
It is accomplished by irradiation of the ester shown; the geometry is such that the ketone's triplet can come to within bonding distance only of the hydrogen atom at C₁₄, and abstraction of that hydrogen followed by a second hydrogen gives the unsaturated ester from which the product is obtained by hydrolysis.



The method is capable of adaptation to other regioselectivities by varying the nature of the esterifying acid.

(v) *Aromatization.* Alicyclic compounds which are reduction products of aromatic systems can be dehydrogenated to the respective aromatics in several ways.

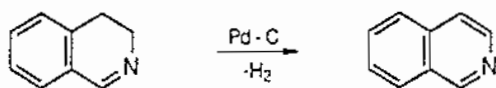
1. *With sulfur or selenium.* Reaction occurs with sulfur at about 200°C and with selenium at about 250°C, the hydrogen being removed as hydrogen sulfide or hydrogen selenide. Skeletal rearrangements may occur and carbon atoms may also be removed; in particular, angular methyl groups are degraded, e.g.



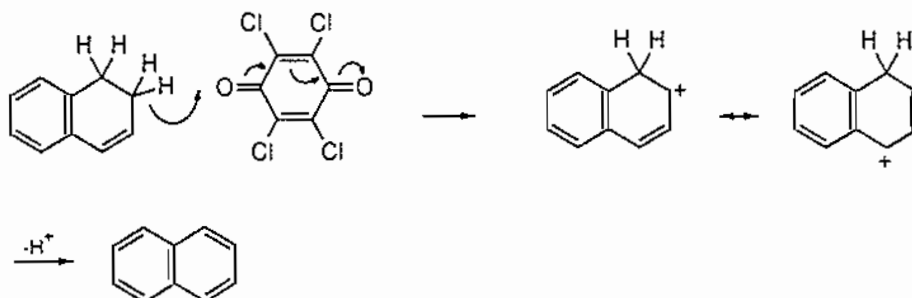
Selenium is less destructive than sulfur and is usually preferred but the methods are of more value in degradation than in synthesis. Distillation with zinc dust has essentially the same effect but is more destructive.

2. *Catalytically.* Alicyclic rings which contain some unsaturation can be dehydrogenated on those catalysts which are successful for hydrogenation (section 19.2): palladium on charcoal or asbestos is the most commonly used. The conditions are far milder than those using selenium and the procedure is widely

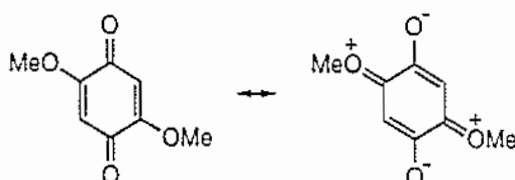
applied, e.g. the reduced isoquinolines obtained by the Bischler–Napieralski synthesis are usually oxidized in this way (p. 706):



3. *With quinones.* Partially unsaturated alicyclic rings are oxidized by quinones through hydride-ion transfer, e.g.

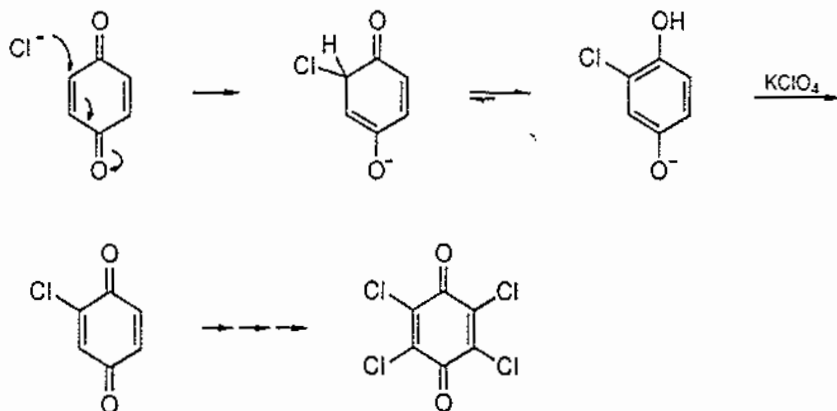


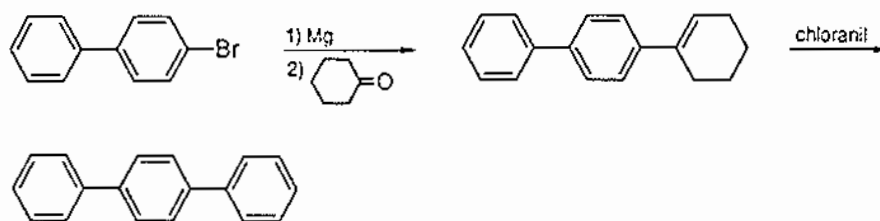
The driving force results from the conversion of both the quinone and the alicyclic system into aromatic compounds. Quinones which contain electron-releasing substituents are stabilized relative to their quinols, e.g.



and are less powerful oxidants than those containing electron-attracting substituents. Chloranil* is frequently used, as in a synthesis of *p*-terphenyl,

*Chloranil is obtained by heating benzoquinone with potassium perchlorate and hydrochloric acid. Nucleophilic attack by chloride ion on the quinone gives a chloroquinol which is oxidized by the perchlorate to the chloroquinone. Successive reactions of this type give chloranil:



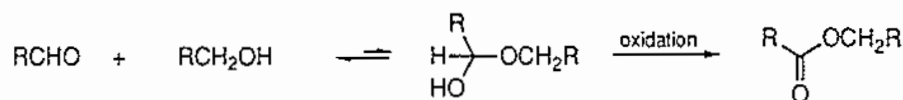


and 2,3-dichloro-5,6-dicyanobenzoquinone is a still more powerful reagent.

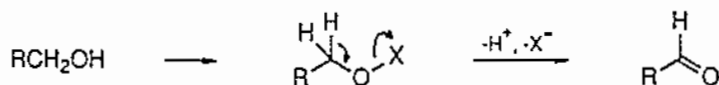
19.3 Systems containing oxygen

(a) Primary alcohols

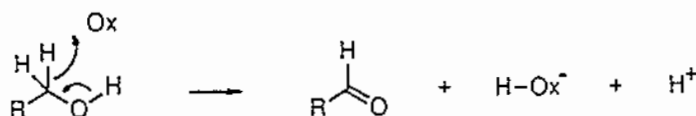
(i) *To aldehydes.* The oxidation of a primary alcohol to an aldehyde is attended by the difficulties that, first, aldehydes are readily oxidized to acids by many reagents and, second, under acidic conditions the aldehyde reacts with unchanged alcohol to give, in an equilibrium mixture, some of the hemiacetal which is readily oxidized to an ester:



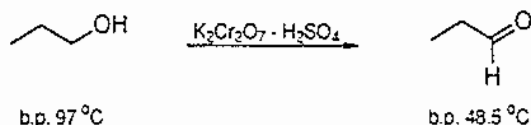
Mechanistically, one commonly employed principle is to attach a strongly electrophilic group, X, to the oxygen of the alcohol, so that H^+ and X^- can be eliminated,



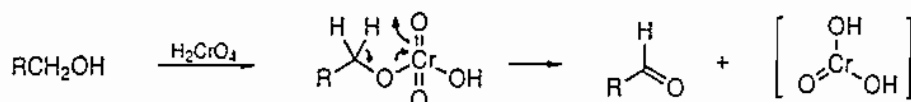
A second approach is to employ a hydride-accepting oxidant,



1. *Chromium(vi) oxide.* For the lower-boiling aldehydes, the simplest oxidation procedure is to add an acid solution of potassium dichromate slowly to the alcohol. Use is made of the fact that aldehydes boil at lower temperatures than the corresponding alcohols so that by maintaining the temperature above the boiling-point of the aldehyde but below that of the alcohol, the aldehyde distils as it is formed, e.g. propionaldehyde can be obtained in about 47% yield [2]:

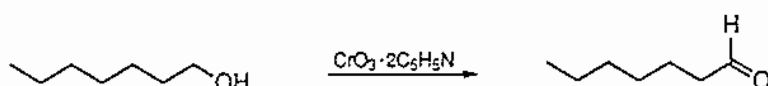


Reaction occurs through the chromate ester, *via* a cyclic transition state:



The chromium(IV) then disproportionates to give chromium(III) and chromium(VI).

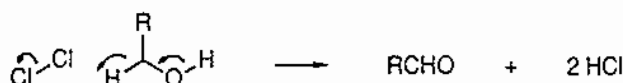
The yields of aldehydes can often be increased by use of the complex, $\text{CrO}_3 \cdot 2\text{C}_5\text{H}_5\text{N}$, formed by chromium(VI) oxide with pyridine (cf. the use of pyridine-sulfur trioxide as a mild sulfonating agent, p. 381). Addition of the chromium(VI) oxide to pyridine is followed either by addition of the alcohol or, better, by isolation of the complex followed by oxidation in dichloromethane. For example, with the latter variant, 1-heptanol gives heptanal in 70–80% yield [6]:



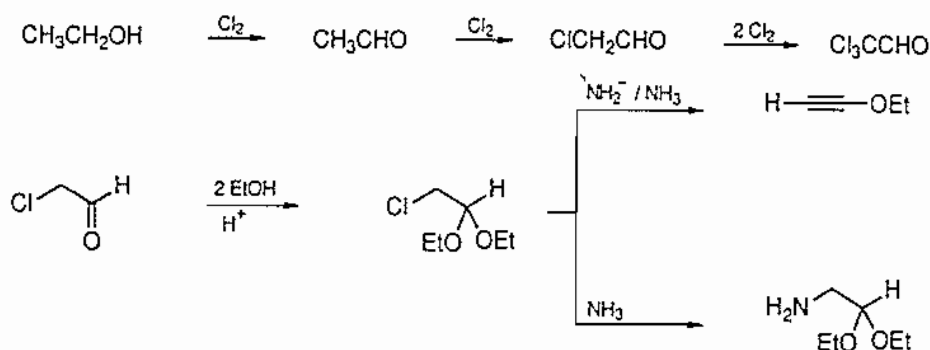
This method is particularly useful for compounds which contain acid-sensitive groups (e.g. acetals) or other easily oxidized groups (e.g. $\text{C}=\text{C}$ bonds).

In a variant of this method, pyridinium chlorochromate, $\text{C}_5\text{H}_5\text{NH}^+ \text{ClCrO}_3^-$, which is precipitated when pyridine is added to a solution of chromic oxide in hydrochloric acid, is used in dichloromethane solution.

2. *Chlorine.* Chlorine oxidizes by the acceptance of hydride ion from the alcohol:



The hydrogen chloride produced can catalyze chlorination of the $\text{C}-\text{H}$ bonds adjacent to the carbonyl group, so that the method is not in general suitable for the formation of simple aldehydes. However, a number of important compounds are available through the oxidation of ethanol with chlorine:

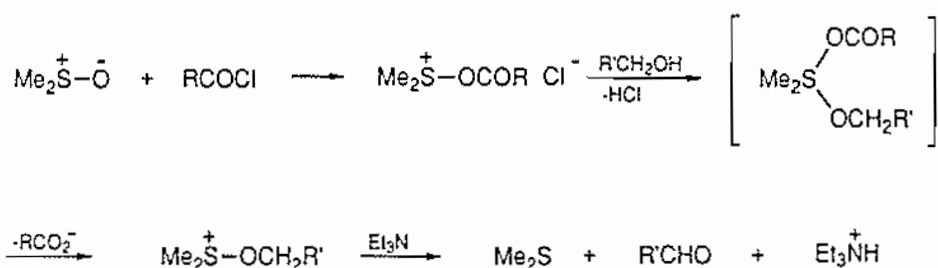




By passing chlorine into ethanol until the specific gravity reaches 1.025, monochloroacetaldehyde is obtained in good yield. Addition of more ethanol to the now acidic solution gives the corresponding acetal from which ethoxyacetylene and the amino-acetal (for the Pomeranz-Fritsch synthesis of isoquinolines, see pp. 707, 708) can be obtained. Further chlorination of monochloroacetaldehyde gives chloral and thence chloroform.

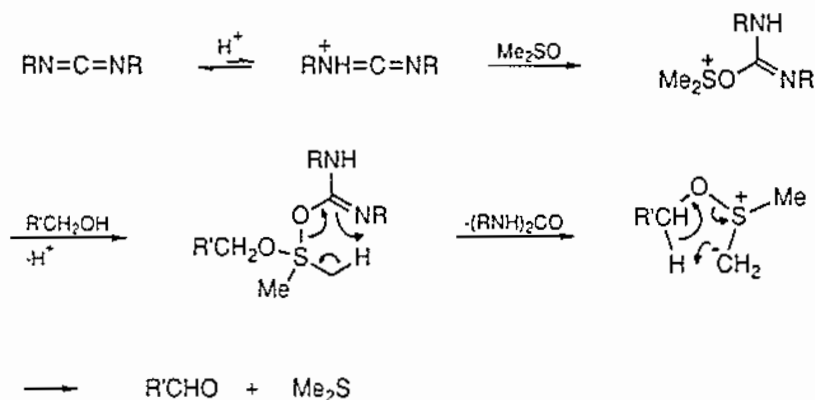
3. *Dimethyl sulfoxide.* There are two methods of using dimethyl sulfoxide. Either the sulfoxide is treated with an electrophile to form a species which is activated towards addition of the alcohol to the sulfur atom and which also possesses a good leaving group, or the alcohol is activated by conversion into its tosylate which reacts readily with the sulfoxide. In each case an alkoxysulfonium ion is formed which undergoes base-catalyzed elimination to give the carbonyl compound.

The first method can be illustrated by reaction with an acid chloride:

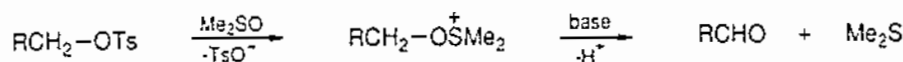


Oxalyl chloride, $\text{ClCO}-\text{COCl}$, is a particularly effective electrophile (*Swern's method*); reaction occurs at temperatures as low as -60°C .

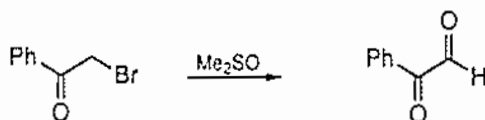
Dicyclohexylcarbodiimide is also a very effective reagent. An acid is added to catalyze the first step, and the addition of base to complete the reaction is unnecessary because an intramolecular hydrogen shift occurs spontaneously:



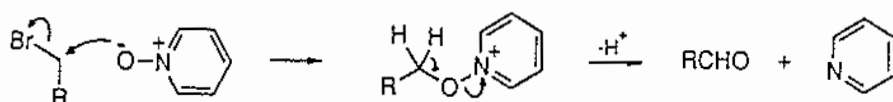
In the second method (*Kornblum's method*), the tosylate is treated with dimethyl sulfoxide in the presence of sodium hydrogen carbonate for a few minutes at 150°C :



Since the first, and key, step is an $\text{S}_{\text{N}}2$ substitution, benzylic compounds, which are more reactive than alkyl derivatives towards nucleophiles, react at a lower temperature (100°C) and the very reactive α -bromo-ketones undergo oxidation directly, even though bromide ion is not as good a leaving group as toluene-*p*-sulfonate, e.g.



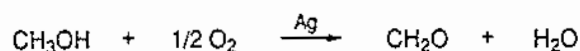
Amine oxides act similarly to dimethyl sulfoxide, and pyridine *N*-oxide in particular has been used to prepare aldehydes from halides and sulfonates:



4. *Catalytic dehydrogenation.* Dehydrogenation over copper or copper chromite occurs at about 300°C . Industrially, silver is employed as the catalyst for the production of formaldehyde and acetaldehyde. The dehydrogenation step, e.g.



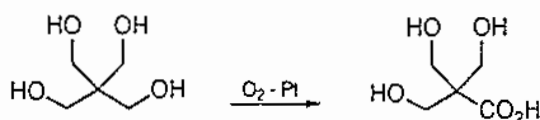
is endothermic ($\Delta H > +100 \text{ kJ mol}^{-1}$), but in the presence of air a second type of oxidation occurs,



This is strongly exothermic ($\Delta H = -150 \text{ kJ mol}^{-1}$), so that by carefully controlling the air supply, a steady temperature is maintained and the reaction becomes self-supporting. Acetaldehyde is also manufactured from ethylene (p. 578).

(ii) *To carboxylic acids.* Primary alcohols can be oxidized directly to acids by reagents such as chromic(VI) acid, nitric acid, and potassium permanganate but in each case side reactions occur and yields are usually not high. For example, chromic(VI) acid degrades carboxylic acids to smaller molecules, ultimately giving acetic acid (from *C*-methyl groups) and carbon dioxide (see Kuhn-Roth oxidation; p. 605).

A selective method is, however, available: namely, the use of molecular oxygen on a platinum catalyst. For example, pentaerythritol gives trihydroxymethylacetic acid in 50% yield in carefully controlled conditions (sodium hydrogen carbonate buffer, 35°C):



An example of the selectivity of this system as between primary and secondary alcohols occurs in a synthesis of ascorbic acid (p. 587). The reagent has the added advantage that double bonds are not attacked, but halides and amines are degraded.

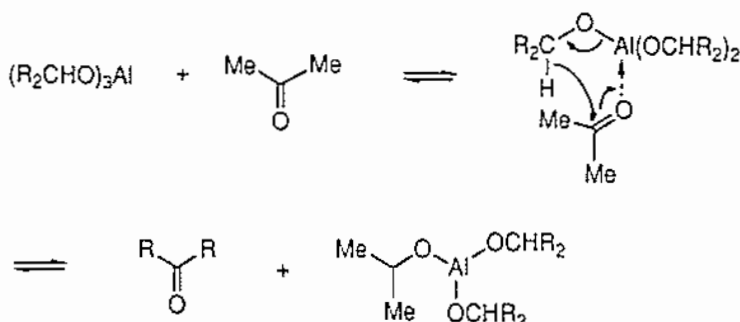
(b) *Secondary alcohols*

Since ketones are only oxidized (with C—C bond-breakage) under vigorous conditions, the oxidation of secondary alcohols to ketones is not attended by the difficulty which applies to the oxidation of primary alcohols to aldehydes. Three methods are widely employed.

(1) *Chromic oxide.* Large-scale oxidations of alcohols which can withstand vigorous conditions can be carried out with an inexpensive reagent such as sodium dichromate in hot aqueous acetic acid. However, milder conditions, at room temperature or below, are often necessary: either pyridinium chlorochromate in dichloromethane or chromium trioxide in a solution of sulfuric acid in aqueous acetone (*Jones' reagent*) are used. For the latter, it is usual to add a solution of chromic oxide in aqueous sulfuric acid from a burette to a cooled solution of the secondary alcohol in acetone. Addition is stopped when a permanent yellow colour, indicating a slight excess of Cr(vi), is obtained. In this way, over-oxidation is prevented and yields of over 90% can be obtained.

Alternative techniques are available. For example, the oxidant power is increased by using acetic acid as the solvent and milder conditions can be obtained by using chromium(vi) oxide in pyridine (see also primary alcohols).

(2) *Oppenauer method.* The secondary alcohol and a ketone are equilibrated with the corresponding ketone and secondary alcohol by heating in the presence of aluminium t-butoxide. By using the added ketone, usually acetone, in large excess, the equilibrium is forced to the right. The reaction, which is the reverse of Meerwein–Ponndorf–Verley reduction (p. 654), involves the transfer of hydrogen within a cyclic complex:

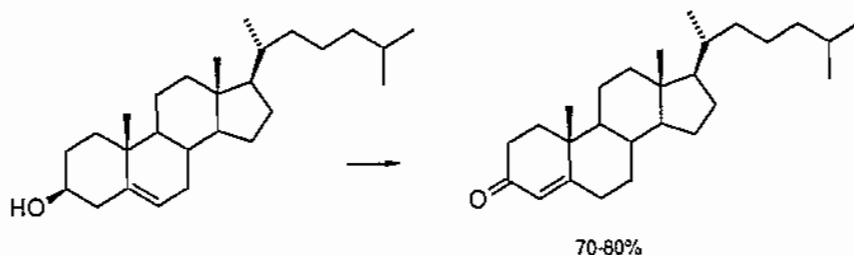


Benzene or toluene is often added as a co-solvent to raise the temperature of the reaction, or alternatively cyclohexanone is used instead of acetone.

The method is specific for alcohols and is therefore suitable, for example, for compounds containing C=C bonds or phenolic groups, e.g.



One disadvantage of the method is that the aluminium compounds are basic and can bring about prototropic shifts within the product, e.g. the oxidation of cholesterol is accompanied by migration of the C=C bond to give the $\alpha\beta$ -unsaturated ketone:

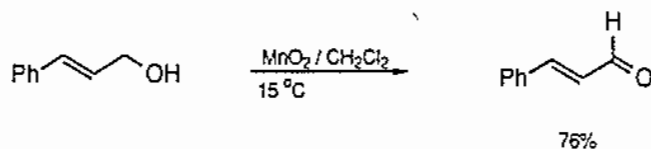


Aldehydes are not satisfactorily prepared by Oppenauer oxidation because the basic medium induces reactions between the aldehyde and the ketone.

(3) *Catalytic dehydrogenation.* Like primary alcohols, secondary alcohols are dehydrogenated when passed over certain heated catalysts. For example, acetone is obtained industrially by the dehydrogenation of isopropanol over copper or zinc oxide at about 350°C. (Acetone is also produced by the cumene hydroperoxide process, p. 446).

(c) *Allylic alcohols*

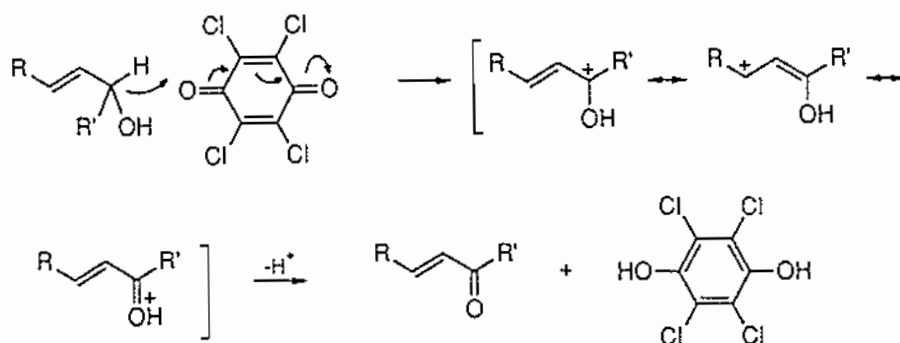
In addition to the methods described above, allylic alcohols are oxidized to $\alpha\beta$ -unsaturated carbonyl compounds on the surface of manganese dioxide suspended in an inert solvent such as dichloromethane. The nature of the manganese dioxide affects the yields: it is best prepared as a non-stoichiometric compound by reducing permanganate ion with manganese(II) ion in alkaline solution. Yields are then high, e.g.



The method is suitable also for benzylic alcohols but not for primary and secondary alcohols which are oxidized only very slowly.

The high potential quinones (p. 607) are suitable for the oxidation of allylic,

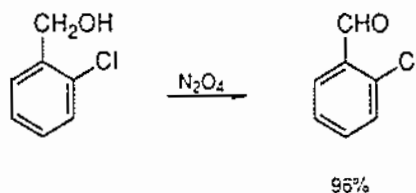
benzylic, and propargylic ($-\text{C}\equiv\text{C}-\text{CH}(\text{OH})-$) alcohols: reaction occurs through the relatively stable carbocations formed by the loss of hydride ion:



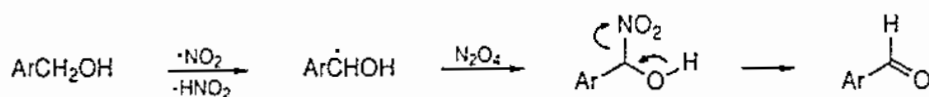
(d) *Benzylic alcohols*

Several specific processes have been developed.

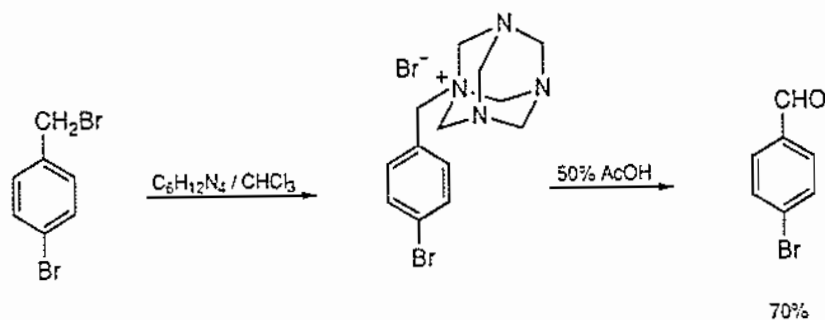
(1) *Nitrogen dioxide (Field's method)*. Primary and secondary benzyl alcohols react with dinitrogen tetroxide in chloroform at 0°C , e.g.



Reaction occurs through (the radical) nitrogen dioxide with the formation and decomposition of a hydroxynitro compound:



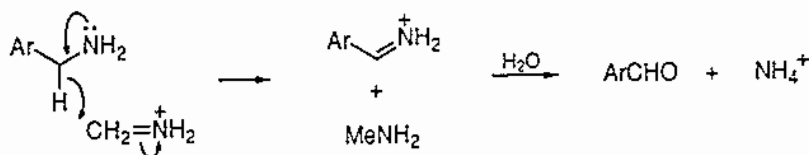
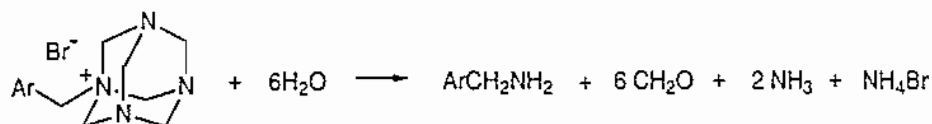
(2) *Hexamethylenetetramine (Sommelet reaction)*. The halide from the benzyl alcohol is treated with hexamethylenetetramine (p. 307) and the resulting salt is hydrolyzed in the presence of more hexamethylenetetramine, usually with aqueous acetic acid, to the aldehyde, e.g.



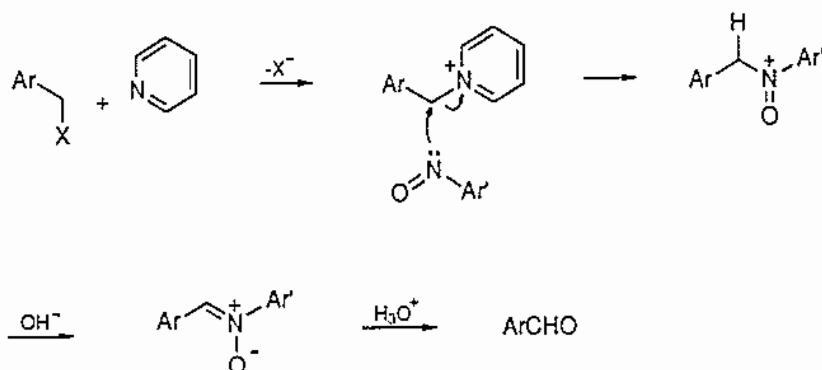
It is not necessary to isolate the intermediate salt.

Electron-withdrawing substituents decrease the yield and *ortho*-substituents hinder the reaction, e.g. neither 2,4-dinitro- nor 2,6-dimethylbenzaldehyde can be prepared in this way.

The reaction involves hydride-ion transfer. At the acidity employed, the quaternary benzyl salt is hydrolyzed to the benzylamine, ammonia, and formaldehyde. The benzylamine transfers hydride ion to methyleneimine (from formaldehyde and ammonia) giving an imine which is hydrolyzed to the aromatic aldehyde:



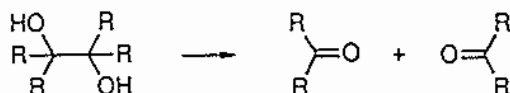
(3) *Kröhnke reaction*. A benzyl halide is converted into its pyridinium salt and thence, with *p*-nitrosodimethylaniline, into a nitrone. Acid hydrolysis gives the aromatic aldehyde.



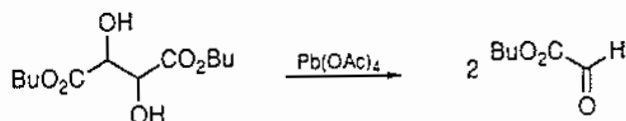
Reaction occurs under mild conditions so that it is suitable for the preparation of sensitive aldehydes. In addition, it is facilitated by electron-attracting substituents, e.g. 2,4-dinitrobenzaldehyde can be prepared in this way (cf. the Sommelet reaction).

(e) 1,2-diols

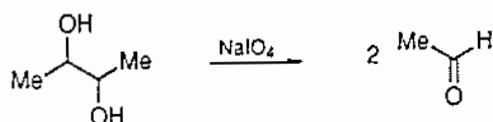
1,2-Diols are cleaved by lead tetraacetate, phenyliodoso acetate, and periodic acid or sodium metaperiodate:



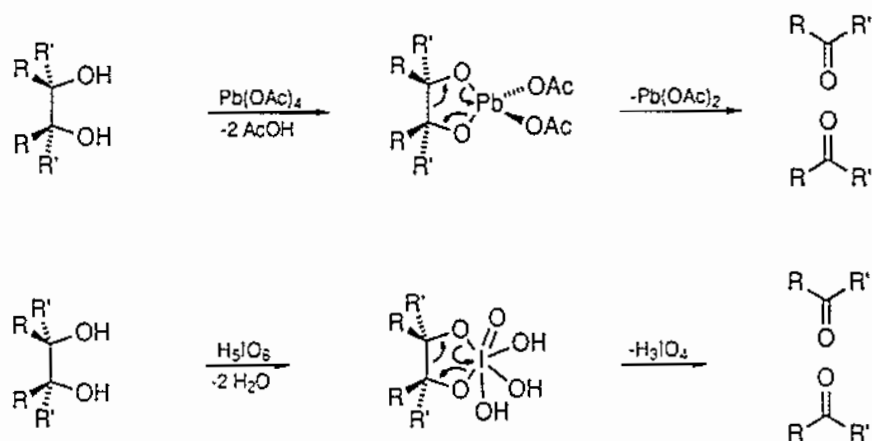
The first two reagents are used in an organic medium, commonly glacial acetic acid, whereas periodate is used in aqueous solution. Phenyliodoso acetate is the least powerful and is not often employed, but all three are entirely specific: the aldehydes formed are not oxidized further. Yields are usually excellent, e.g. dibutyl tartrate and lead tetraacetate give butyl glyoxylate in up to 87% yield [4],



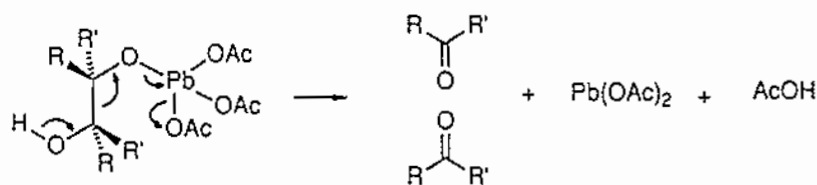
and 2,3-butane diol and sodium periodate give acetaldehyde quantitatively,



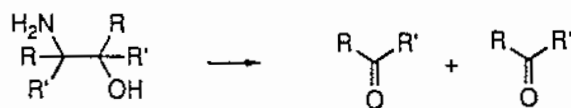
Reaction normally occurs by two-electron oxidation (Pb(IV) to Pb(II) and I(VII) to I(V)) within a cyclic intermediate:

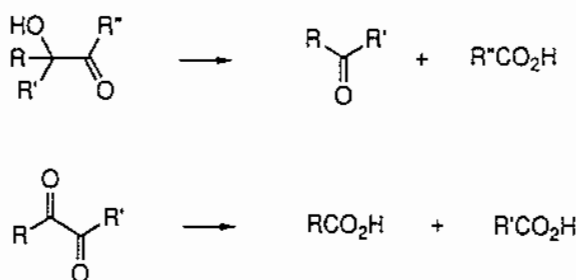


so that *cis*-diols are oxidized faster than *trans*-diols, e.g. *cis*-cyclohexane-1,2-diol reacts about 25 times faster than its *trans*-isomer. However, an acyclic path is also followed, e.g.

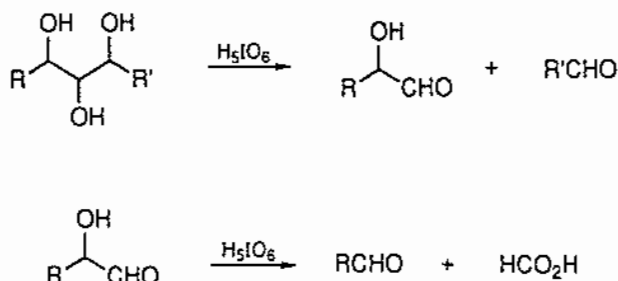


In addition to diols, α -amino-alcohols, α -ketols, and α -dicarbonyl compounds are cleaved:





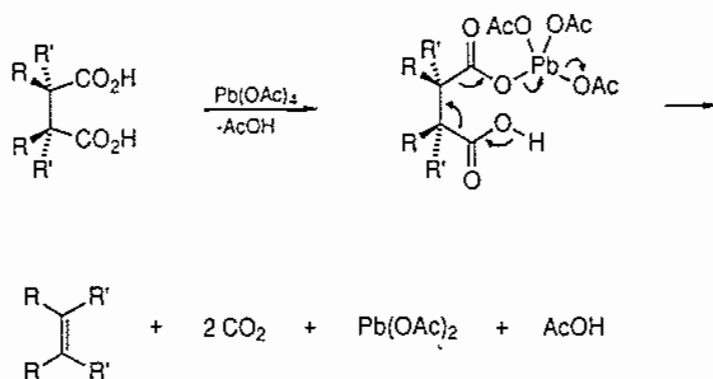
The grouping $\text{—CH(OH)—CH(OH)—CH(OH)—}$ is oxidized to a mixture of aldehydes and formic acid,



Periodate has proved a valuable degradative agent in carbohydrate chemistry, for it reacts nearly quantitatively. Since the grouping $\text{—CH(OH)—CH}_2\text{OH}$ is oxidized to formaldehyde and the grouping $\text{—CH(OH)—CH(OH)—CH(OH)—}$ gives one equivalent of formic acid, the estimation of formaldehyde and formic acid provides information about the occurrence of these groups and the total periodate consumed indicates the total number of 1,2-diol groups present.

An interesting example of the synthetic use of periodate, in the oxidation of the system $\text{—CO—CH(OH)—CH(OH)—}$, occurs in the synthesis of reserpine (section 22.1). Other uses are in the Lemieux reagents (p. 595).

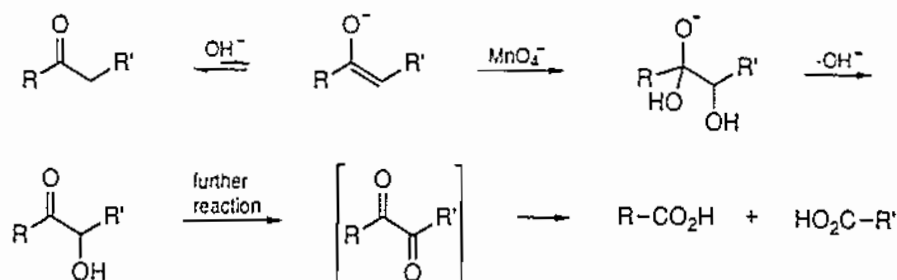
Lead tetraacetate also effects the bisdecarboxylation of succinic acids:



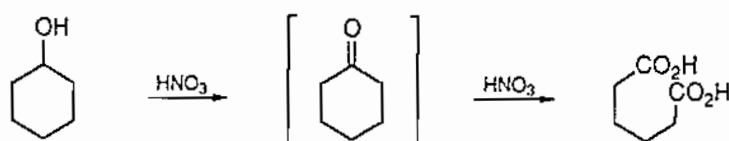
(f) Aldehydes

Aldehydes can be oxidized to acids by vigorous reagents such as chromic(vi) acid and permanganate. This can be satisfactory for compounds which do not possess

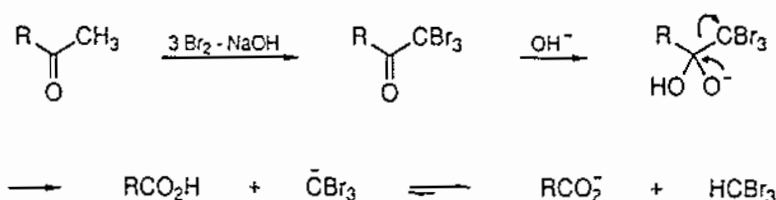
ditions give carboxylic acids, reaction occurring through the enol (acid solution) or the enolate (basic solution), e.g.



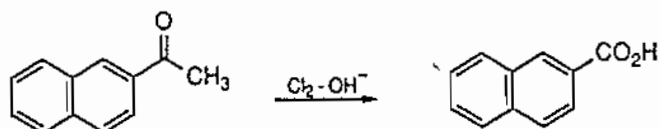
Attack can occur on both sides of the carbonyl group so that, unless the ketone is cyclic, a mixture of products is obtained. With cyclic ketones, however, reasonable yields can be achieved, e.g. cyclohexanol is oxidized by hot 50% nitric acid, *via* cyclohexanone, to give adipic acid in 60% yield [1]:



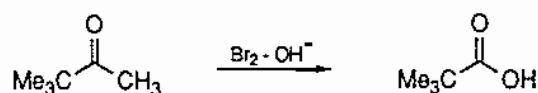
(2) *By the halogens in alkali.* Methyl ketones are oxidized by chlorine, bromine, or iodine in alkaline solution to give acids and the corresponding haloform. Reaction occurs by base-catalyzed halogenation followed by elimination of the conjugate base of the haloform:



This provides a particularly useful method for the synthesis of aromatic acids, for the corresponding methyl ketones are often readily available through the Friedel-Crafts reaction (p. 365). For example, the acetylation of naphthalene in nitrobenzene solution gives 2-acetylnaphthalene (p. 368) from which 2-naphthoic acid can be obtained in 97% yield with chlorine in sodium hydroxide solution at 55°C:



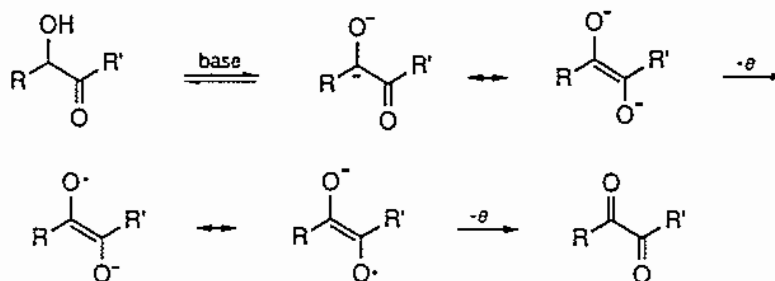
There are also applications in the aliphatic series. For example, pinacolone, available from acetone by reduction and rearrangement (p. 433), reacts with bromine in sodium hydroxide solution at below 10°C to give trimethylacetic acid in over 70% yield [1]:



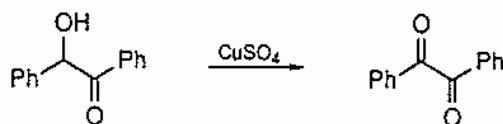
(3) *By peroxyacids.* Ketones undergo oxidative rearrangement with peroxyacids to give esters or lactones, $\text{RCOR} \rightarrow \text{RCO}_2\text{R}$ (Baeyer–Villiger reaction, p. 445).

(h) *α -Ketols*

These systems are very readily oxidized to α -dicarbonyl compounds. One-electron oxidants in basic solution are effective, for the enediolate formed by base can donate one electron to the oxidant to give a delocalized radical (cf. semiquinone radicals, p. 598). Loss of a second electron completes the oxidation.



For example, copper sulfate in pyridine at 95°C oxidizes benzoin to benzil in 86% yield [1]:

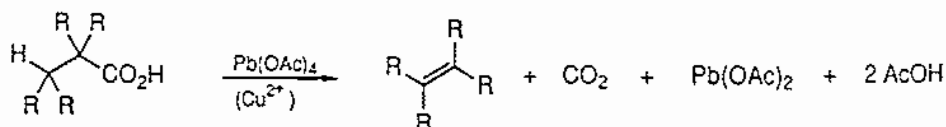


(i) *α -Dicarbonyl compounds*

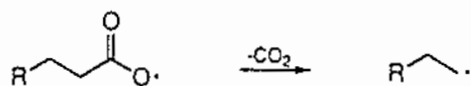
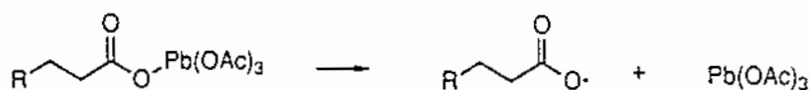
Oxidation can be brought about in three ways: peroxyacids in inert solvents give anhydrides, $\text{RCO}-\text{COR} \rightarrow \text{RCO}-\text{O}-\text{COR}$ (Baeyer–Villiger reaction, p. 445); and both warm hydrogen peroxide in acetic acid and the glycol-cleavage reagents give carboxylic acids, $\text{RCO}-\text{COR} \rightarrow 2\text{RCO}_2\text{H}$.

(j) *Acids: oxidative decarboxylation*

Carboxylic acids undergo oxidative decarboxylation when heated with lead tetraacetate in the presence of a catalytic amount of a copper(II) salt:



Reaction is thought to occur by homolysis of the lead carboxylate, followed by oxidation of the resulting radical by copper(II), e.g.



The copper(I) ion is then oxidized by lead(IV) to regenerate copper(II).

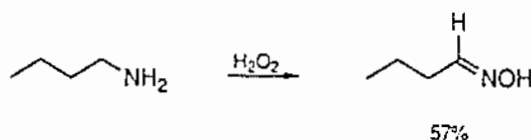
19.4 Systems containing nitrogen

(a) Primary amines

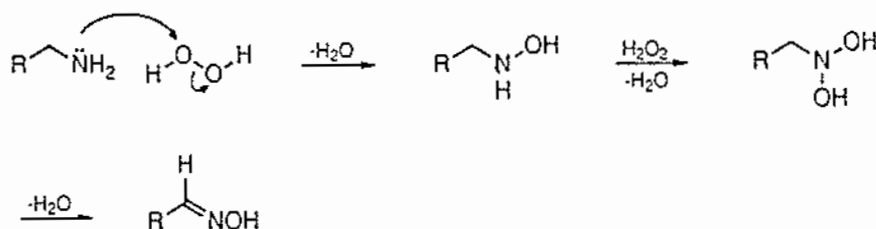
These are very sensitive to oxidation and generally darken on exposure to air, through autoxidation at the surface, to give mixtures of complex products. This is particularly true of aromatic amines: a number of products arise by oxidation to nitroso and nitro compounds followed by condensations, e.g. $\text{ArNH}_2 \rightarrow \text{ArNO}$; $\text{ArNH}_2 + \text{ArNO} \rightarrow \text{ArN}=\text{NAr}$. Repeated reactions give aniline blacks.

Synthetically useful methods have therefore to be highly selective. The most successful reagents are hydrogen peroxide and peroxyacids.

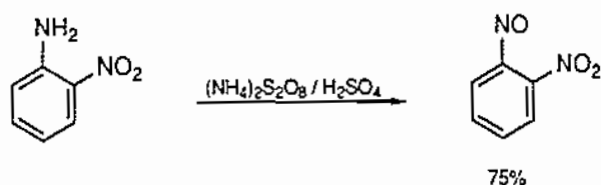
(1) *Hydrogen peroxide* converts primary aliphatic amines into aldoximes, e.g.



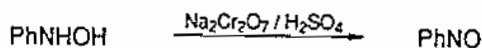
by nucleophilic displacement by the amine on the peroxide:



Oxidation of aromatic amines to the corresponding level (nitroso compounds) is generally brought about with peroxydisulfuric acid, $\text{HO}_3\text{S}-\text{O}-\text{O}-\text{SO}_3\text{H}$, e.g.

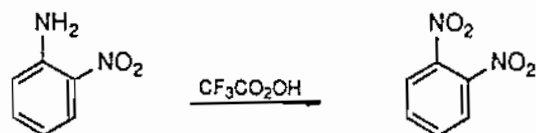


Aromatic nitroso compounds can also be prepared from the corresponding hydroxylamines by oxidation with dichromate at low temperatures (to prevent further oxidation). For example, phenylhydroxylamine, sodium dichromate, and 50% sulfuric acid at 0°C give nitrosobenzene in about 50% yield [3]:



Since arylhydroxylamines are readily obtained by the reduction of nitro compounds in neutral solution (p. 667), this method is convenient for introducing the nitroso group *via* nitration.

(2) *Trifluoroperoxyacetic acid*, a more powerful oxidant than hydrogen peroxide, converts primary amines directly into nitro-compounds. The yields with aromatic amines are generally high, e.g. *o*-nitroaniline is oxidized in refluxing dichloromethane to *o*-dinitrobenzene in 92% yield:



With aliphatic amines, however, the yields are low. Other oxidants have been used with moderate success. For example, 1-hexylamine is oxidized in 33% yield to the nitro compound by peroxyacetic acid.

Aromatic nitro-compounds can also be obtained by the oxidation of nitroso-compounds with dichromate or dilute nitric acid, but it is normally more convenient to oxidize the amine.

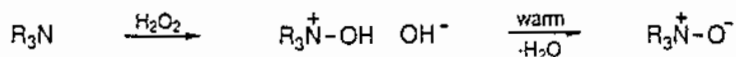
(b) Secondary amines

Oxidation with hydrogen peroxide gives hydroxylamines (cf. primary amines):



(c) Tertiary amines

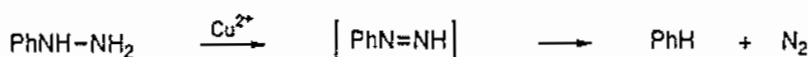
Hydrogen peroxide converts tertiary amines into their *N*-oxide hydrates, by nucleophilic displacement analogous to the reaction of primary amines. The *N*-oxide is obtained by warming the hydrate *in vacuo*.



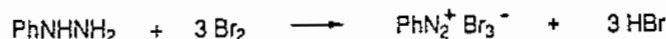
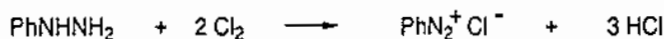
(d) *Hydrazines*

Hydrazine itself is readily oxidized to nitrogen. Some at least of the oxidants, e.g. copper(II) ion, yield diimide, $HN=NH$, as an intermediate and this is employed for the reduction of alkenes (p. 634).

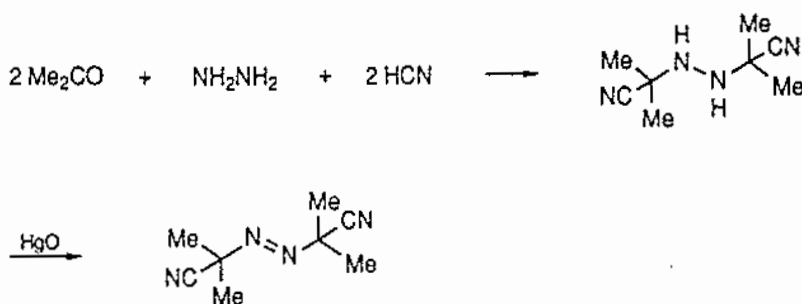
Monosubstituted hydrazines react with one-electron oxidants such as copper(II) and iron(III) ion to give unstable azo compounds which decompose with loss of nitrogen to hydrocarbons, e.g.



Arylhydrazines are oxidized by the two-electron oxidants chlorine and bromine giving diazonium salts, e.g.

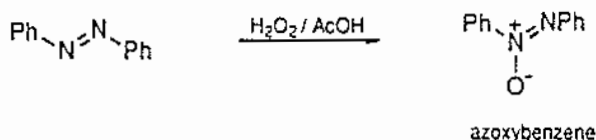


N,N'-Disubstituted hydrazines give azo-compounds readily. For example, azobisisobutyronitrile (a useful reagent for initiating free-radical reactions) can be obtained from acetone, hydrazine, and cyanide (compare the Strecker synthesis of α -amino acids, p. 329) followed by oxidation, e.g. with mercury(II) oxide:



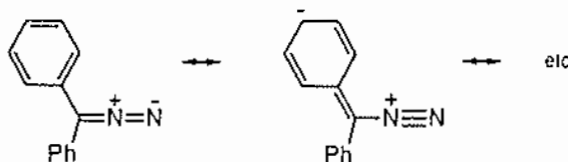
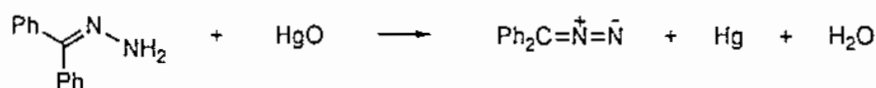
Diarylhydrazines oxidize with exceptional ease since a conjugated system is formed. For example, hydrazobenzene gives azobenzene even on standing in air for some time.

Azo-compounds may be further oxidized to azoxy-compounds in more vigorous conditions, e.g.

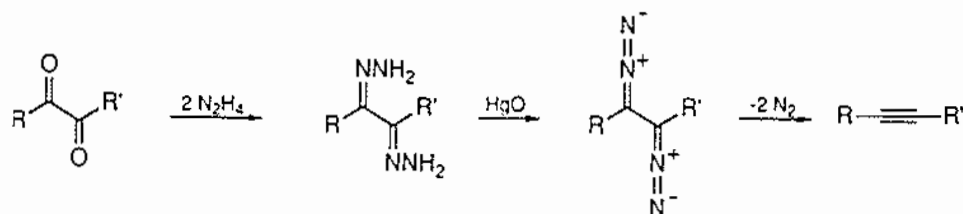


(e) Hydrazones

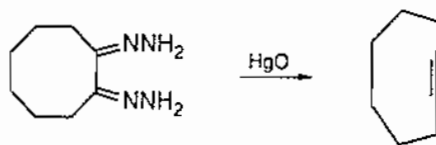
Hydrazones are oxidized to diazoalkanes by mercury(II) oxide. Those containing aryl-substituents give moderately stable (conjugated) products which can be isolated, e.g. benzophenone hydrazone gives diphenyldiazomethane nearly quantitatively [3]:



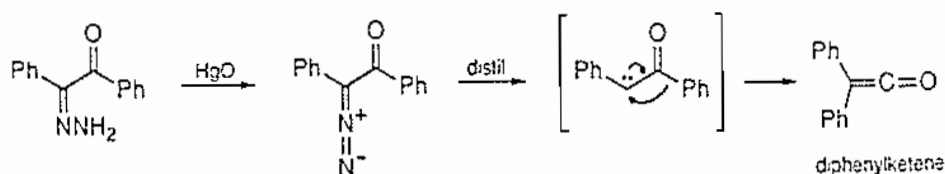
Diazoalkanes containing only saturated groups decompose rapidly to nitrogen and products derived from carbenes ($\text{R}_2\text{CN}_2 \rightarrow \text{R}_2\text{C} + \text{N}_2$). Use is made of this in a synthetic procedure for alkynes: the bishydrazone of an α -diketone is oxidized with mercury(II) oxide and the unstable bisdiazo-compound decomposes to the alkyne:



For example, benzil bishydrazone gives diphenylacetylene in about 70% yield [4] and the bishydrazone from 1,2-cyclooctanedione gives cyclooctyne in 9% yield, the low yield doubtless resulting from the extreme ring strain in the product (which is the smallest cycloalkyne to have been made).



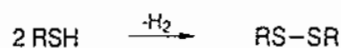
The monohydrazones of α -diketones give ketenes in these conditions, e.g. the treatment of benzil monohydrazone with mercury(II) oxide gives a diazo-compound which rearranges on distillation, with loss of nitrogen, to diphenylketene (58%) [3]:



(a) *Thiols*19.5
Systems
containing
sulfur

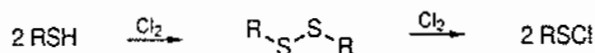
Whereas hydroxyl-containing compounds are oxidized at carbon, thiol-containing compounds are oxidized at sulfur, largely because of the comparatively low bond energy of S—H compared with O—H (p. 46).

(1) *Oxidation to disulfides.* A variety of relatively weak oxidants, such as hydrogen peroxide, iron(III) ion, and iodine, oxidize thiols to disulfides,

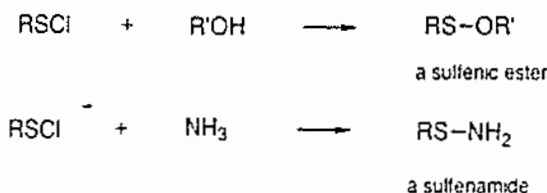


For example, cysteine is readily oxidized to cystine, and the disulfide ring in thioctic acid is readily formed from the dithiol precursor (p. 255).

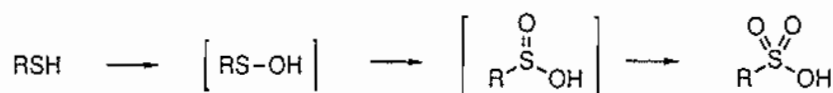
(2) *Oxidation to sulfenyl chlorides.* Thiols are oxidized by chlorine, through disulfides, to sulfenyl chlorides,



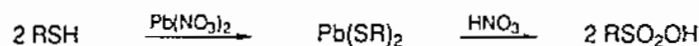
from which derivatives of sulfenic acids may be obtained, e.g.



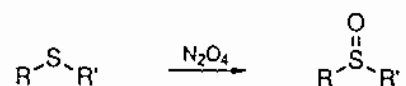
(3) *Oxidation to sulfonic acids.* Vigorous reagents such as nitric acid and permanganate give sulfonic acids, probably through sulfenic and sulfinic acids, which are too easily oxidized to be isolated:



Sulfonic acids are also formed by the treatment of lead thiolates with nitric acid:

(b) *Sulfides*

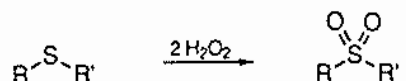
Sulfides can be oxidized to both sulfoxides and sulfones. The former are obtained in high yield (c. 90%) with liquid dinitrogen tetroxide in ethanol cooled with solid carbon dioxide. Further oxidation to the sulfone does not occur, and the product is anhydrous (wet sulfoxides are very difficult to dry).



The oxidation can also be carried out with a slight excess of sodium metaperiodate at 0°C and with hydrogen peroxide but these methods cause some further oxidation to the sulfone (see below).

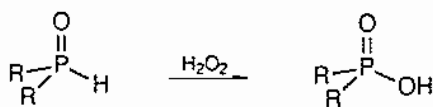
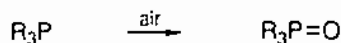
Very sensitive sulfides require more delicate treatment. For example, manganese dioxide is the only reagent which oxidizes diallyl sulfide to diallyl sulfoxide in reasonable yield.

Sulfones are normally obtained from sulfides by oxidation with hydrogen peroxide in aqueous or acetic acid solution:



19.6 Systems containing phosphorus

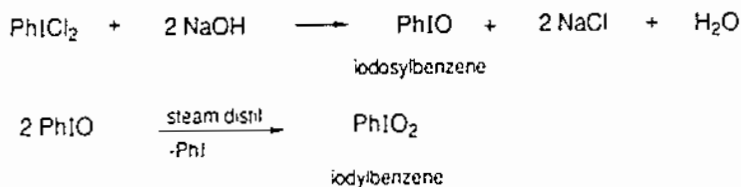
The characteristics of phosphorus chemistry compared with that of nitrogen are that three-valent phosphorus is readily oxidized to the five-valent state and that P—O bonds are more stable than N—O bonds. Oxidations at phosphorus occur under mild conditions, e.g.



19.7 Systems containing iodine

The iodine atom in aryl iodides can be oxidized to both the three- and the five-valent states. For example, iodobenzene reacts with chlorine in dry chloroform to give dichloriodobenzene in about 90% yield [3]; this may be hydrolyzed to iodosylbenzene in about 60% yield [3] and iodylbenzene may be obtained in over 90% yield by steam-distilling iodosobenzene to remove the iodobenzene formed by disproportionation.



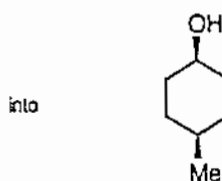
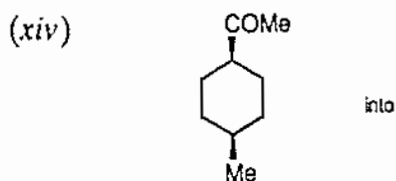
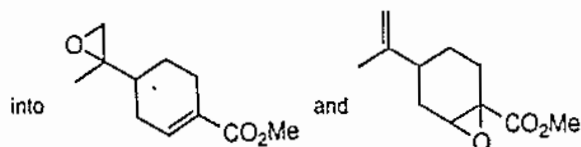
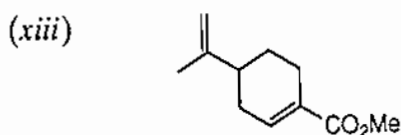
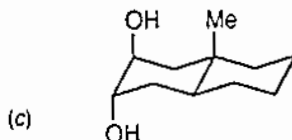
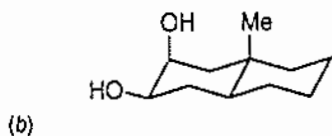
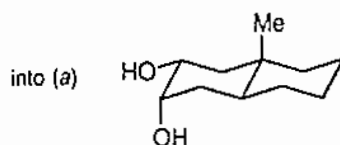
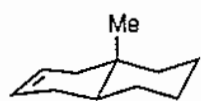


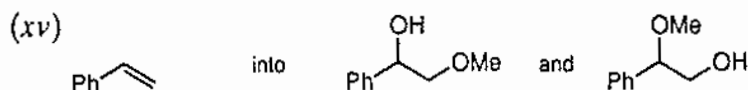
The aliphatic analogues of these higher-valent iodine compounds are unstable.

1. How would you carry out the following transformations?

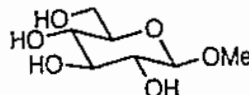
Problems

- (i) $\text{R}-\text{CH}=\text{CH}_2$ into (a) $\text{R}-\text{CHO}$, (b) $\text{R}-\text{CH}_2\text{OH}$, (c) $\text{R}-\text{CH}_2-\text{CHO}$, (d) $\text{R}-\text{CH}(\text{OH})-\text{CHO}$, (e) $\text{R}-\text{CH}(\text{OH})-\text{CH}_3$.
- (ii) $\text{R}-\text{CH}=\text{CH}-\text{CH}_2\text{OH}$ into (a) $\text{R}-\text{CH}(\text{OH})-\text{CH}(\text{OH})-\text{CH}_2\text{OH}$, (b) $\text{R}-\text{CH}=\text{CH}-\text{CHO}$.
- (iii) $\text{R}-\text{CH}_2-\text{CO}-\text{CH}_3$ into (a) $\text{R}-\text{CH}_2-\text{CO}-\text{CHO}$, (b) $\text{R}-\text{CO}-\text{CO}-\text{CH}_3$.
- (iv) PhCHO into (a) $\text{PhCH}(\text{OH})-\text{COPh}$, (b) $\text{PhCO}-\text{COPh}$.
- (v) $\text{PhCO}-\text{COPh}$ into (a) $\text{Ph}_2\text{C}=\text{C}=\text{O}$, (b) $\text{PhC}\equiv\text{CPh}$.
- (vi) PhCH_2Br into PhCHO .
- (vii) PhCOCH_2Br into PhCOCHO .
- (viii) PhNH_2 into (a) *p*-benzoquinone, (b) azoxybenzene, (c) *o*-dinitrobenzene.
- (ix) PhOH into (a) *o*-benzoquinone, (b) chloranil (tetrachloro-*p*-benzoquinone).
- (x) PhSH into (a) $\text{PhS}-\text{SPh}$, (b) PhSCl , (c) PhSO_2OH .
- (xi) *trans*-2-Butene into (a) ($\pm$)-2,3-butanediol, (b) *meso*-2,3-butanediol.
- (xii)



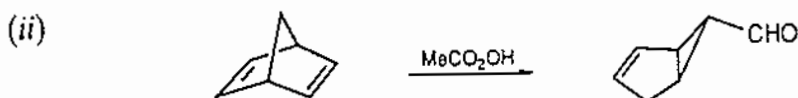
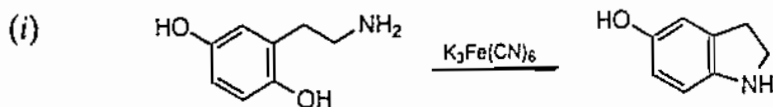


- Summarize the reagents which may be used to oxidize the methyl group in a compound $X-CH_3$ according to the nature of the group X (i.e. X = acetyl, phenyl, etc.).
- A methylated derivative of D-glucose is thought to have the structure (I). How could an oxidative method be used to provide evidence that the ring is six-membered?

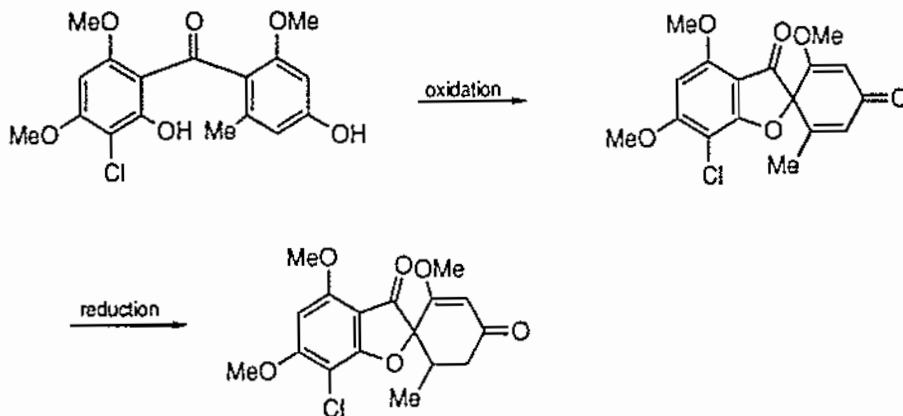


(I)

- Rationalize the following reactions:



- Griseofulvin (a fungal metabolite which is an important antibiotic) has been made by the following sequence:



How would you attempt to effect the oxidative step?

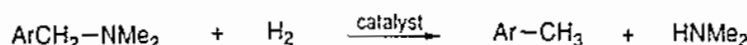
Reduction 20

20.1 Introduction

The reductive processes described in this chapter fall into three categories: the removal of oxygen, the addition of hydrogen, and the gain of electrons. The addition of hydrogen may be subdivided into *hydrogenation*, the addition of hydrogen to an unsaturated system, e.g.

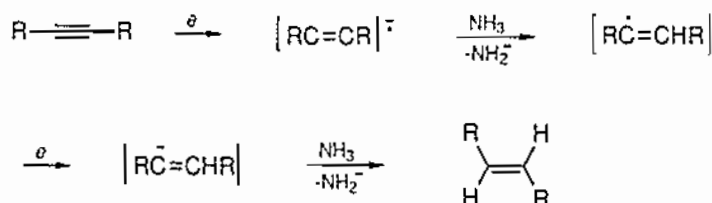


and *hydrogenolysis*, the addition of hydrogen with concomitant bond rupture, e.g.

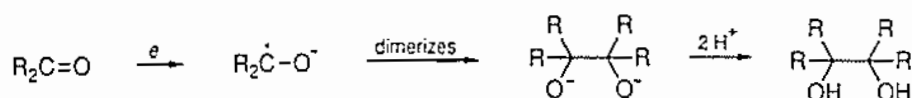


Mechanistically, there are three main pathways for reduction:

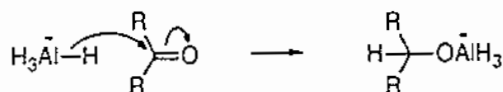
(1) *By the addition of electrons*, followed either by the uptake of protons, as in the reduction of an alkyne by sodium in liquid ammonia (p. 639),



or by coupling, as in the reduction of ketones to pinacols (p. 656),



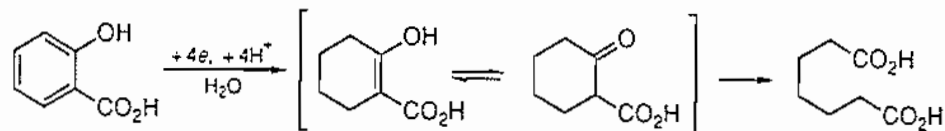
(2) *By the transfer of hydride ion*, as in the reduction of the carbonyl group by lithium aluminium hydride (p. 652):



Such transference may also occur intramolecularly, as in Meerwein-Ponndorf-Verley reduction (p. 654).

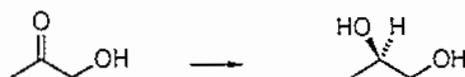
(3) *By the catalyzed addition of molecular hydrogen*, as in the reduction of alkenes on metals (p. 631).

Subsidiary reactions, such as the breaking of C—C bonds, are relatively uncommon in reductive processes, but some are usefully applied. For example, pimelic acid (heptanedioic acid) may be obtained in 45% yield from salicylic acid by refluxing for 8 hours with sodium in isoamyl alcohol (3-methyl-1-butanol) [2]; reaction occurs through a β -keto-acid which undergoes a reverse Claisen reaction



There have been enormous developments in reductive methods since the 1950s, with respect both to the types of bond which may be reduced and to the selectivity of the processes. The older methods involving electron-transfer, such as sodium and alcohol, and zinc and acetic acid (in which the metal acts as the electron-source and the hydroxylic-compound as the proton-donor), are now supplemented by the metal-ammonia and metal-amine systems which have increased the scope of these reductions. Hydride-transfer agents such as formic acid (Leuckart reduction, p. 670) are supplemented by the complex hydrides some of which are of remarkable selectivity. Catalytic methods have been improved by procedures for obtaining more active catalysts.

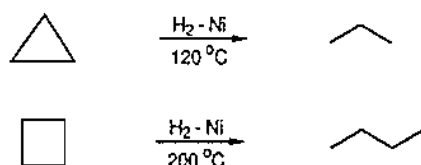
Enzymic systems are as yet of little general importance but are likely to achieve wider use as methods for isolating enzymes are developed. Their particular merit is that many (being themselves optically active) are enantioselective so that optically active compounds can be obtained from inactive reactants. For example, hydroxyacetone is reduced specifically to (*R*)-propylene glycol (50%) by incubation with yeast reductase at 32°C for 3 days [2]:



This chapter, like that on oxidation, is classified according to the type of group to be reduced. Each major class of reducing agent is described in the context of the system for which it has been most used.

20.2 Hydrocarbons (a) Alkanes

Alkanes can be reduced only by rupturing carbon-carbon bonds. Reducing agents are not normally sufficiently powerful to bring this about, in contrast to oxidizing agents of which the strongest cleave aliphatic chains ultimately to carbon dioxide. There is, however, one circumstance in which reduction can be effected catalytically, namely, with strained cyclic compounds, since C—C cleavage relieves the strain, e.g.



The nature of the catalyst for the reduction is described in the following section.

(b) *Alkenes*

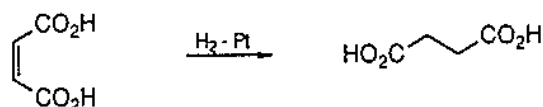
(i) *Catalytic hydrogenation.* Almost all alkenes can be saturated in very high yield by treatment with hydrogen and a metal catalyst. The most active catalysts are specially prepared platinum and palladium.

1. *Palladium.* An active form of palladium can be obtained from palladium chloride in a similar way to that used for the Adam's catalyst, but more commonly the palladium chloride is reduced in the presence of a suspension of charcoal or other solid support on which the metal is deposited in a very finely divided state.

2. *Adams' catalyst.* Chloroplatinic acid is fused with sodium nitrate to give a brown platinum oxide (PtO_2) which can be stored. When required, it is treated with hydrogen to give a very finely divided black suspension of the metal. This is the reagent usually chosen in the laboratory, reaction being carried out in solvents such as acetic acid, ethyl acetate, and ethanol. About 0.2 g of platinum oxide is usually employed per 10 g of reactant.

A particularly reactive catalyst is obtained by the reduction of platinum oxide *in situ* with sodium borohydride in the presence of carbon. The molecular hydrogen required for hydrogenation is then generated by the addition of acid to the excess of borohydride.

Most $\text{C}=\text{C}$ bonds are reduced on these catalysts at temperatures below 100°C and at atmospheric or slightly increased pressure. For example, maleic acid gives succinic acid in 98% yield in 30 minutes on platinum at 20°C and 1 atmosphere [1],



and this particular reaction is often used for testing the apparatus and assessing the catalyst.

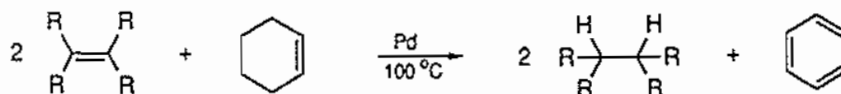
3. *Raney nickel.* This is a slightly less active catalyst than platinum or palladium. It is prepared by treating a nickel-aluminium alloy with sodium hydroxide,



and washing away the sodium aluminate to leave the nickel as a black suspension saturated with hydrogen (c. 50–100 cm³ per g) which is pyrophoric when dry. Most alkenes are hydrogenated over Raney nickel at about 100°C and pressures up to 3 atmospheres. Since the Raney alloy is fairly cheap, it is usually convenient to employ relatively large quantities of the catalyst (c. 2 g per 10 g) so as to minimize the effects of catalyst-poisons. This is particularly important with sulfur-containing compounds (p. 672).

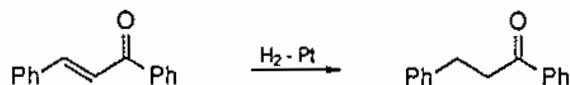
4. '*Copper chromite*'. The early catalytic methods developed by Sabatier and Senderens employed metals such as iron, cobalt, nickel, and copper at temperatures of about 300°C. This approach has been considerably improved by the use of high pressures (up to 300 atmospheres of hydrogen) and more active catalysts. The best, *Adkins' catalyst*, the so-called copper chromite, is made from copper nitrate and sodium dichromate and corresponds to CuO·CuCr₂O₄. This is a far cheaper catalyst than palladium or platinum and is therefore more attractive industrially.

5. *Transfer hydrogenation*. The hydrogen is supplied by a donor such as cyclohexene or hydrazine, e.g.



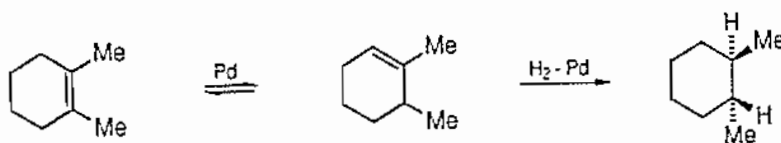
The driving force in the case of cyclohexene is the gain in aromatic stabilization energy when benzene is formed; with hydrazine, the strongly bonded N₂ molecule is formed. The advantage of this method is that no special apparatus for handling and measuring gaseous hydrogen is required.

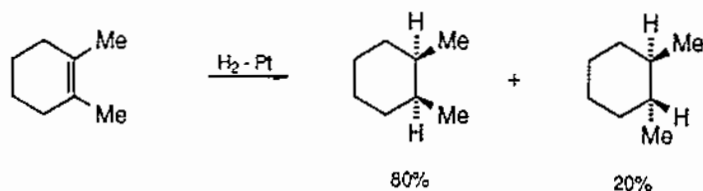
Selectivity. The C≡C bond is reduced more readily than C=C but other unsaturated groupings, with the exception of nitro groups and acid chlorides, are reduced less readily. Catalytic hydrogenation can therefore be used for the selective reduction of C=C in the presence of aromatic rings and carbonyl groups, whether or not the unsaturated functions are conjugated. For example, benzylideneacetophenone is reduced over platinum at 20°C in 90% yield [1]:



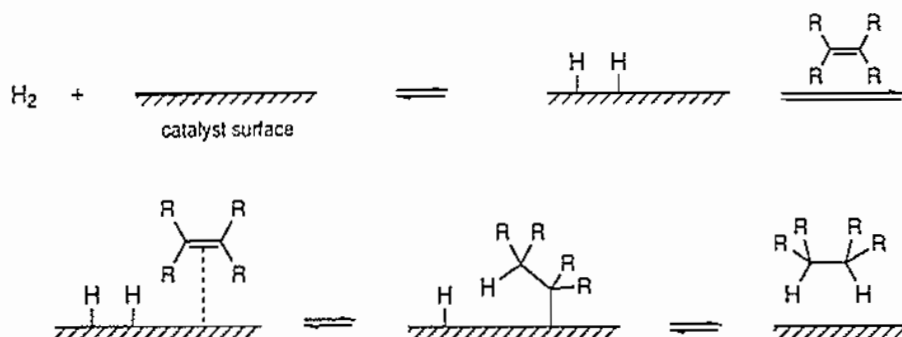
Stereochemistry. Reduction is highly stereoselective, giving predominantly *cis*-alkene,* e.g.

* With palladium, however, the *trans*-product predominates. This is because Pd catalyzes the isomerization of the alkene:

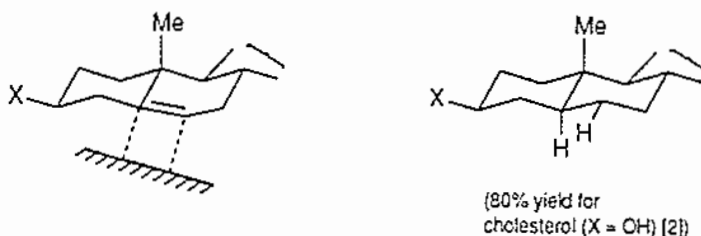




This can be understood in terms of the following simplified representation:



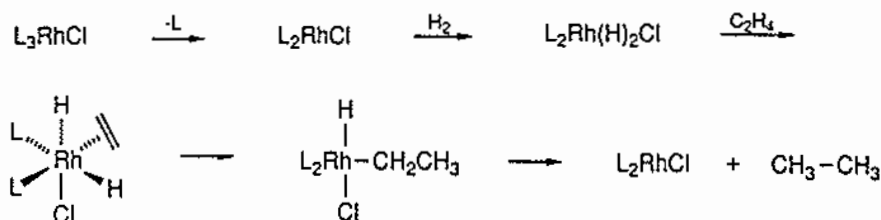
In rigid ring systems, the direction of addition is from the less hindered side. For example, octalins of the type shown below (e.g. cholesterol; $\text{X}=\text{OH}$) give mainly *trans* ring-junctions on reduction, since the angular methyl group hinders the fit of the catalyst on the opposite side of the double bond:



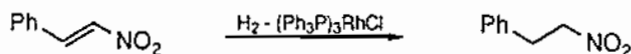
On the other hand, when the substituent X is axial, the fit to the catalyst is hindered on both sides and reduction gives a mixture of *cis* and *trans* decalins in comparable amounts.

6. *Homogeneous catalytic hydrogenation.* The use of complexes of rhodium or ruthenium as catalysts enables hydrogenation to be carried out in homogeneous solution.

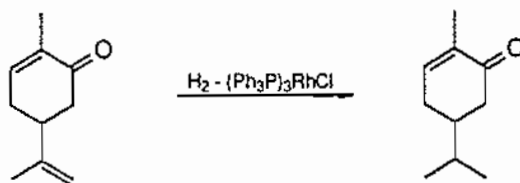
The rhodium reagent is Wilkinson's catalyst, $(\text{Ph}_3\text{P})_3\text{RhCl}$ (p. 575). Alkenes and alkynes are reduced at room temperature and atmospheric pressure, usually in benzene solution. The mechanism, illustrated for ethylene ($\text{L} = \text{PPh}_3$), is:



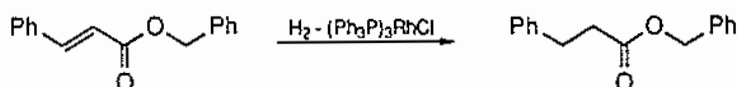
The method has several advantages. First, only alkenes and alkynes are reduced; other common groups such as $\text{C}=\text{O}$, $\text{C}\equiv\text{N}$, and NO_2 are unaffected. This makes possible selective reductions such as



Second, mono- and disubstituted alkenes are reduced much more rapidly than tri- or tetrasubstituted ones. This provides a further degree of selectivity, illustrated by the reduction of carvone to dihydrocarvone – leaving the trisubstituted double bond intact – in over 90% yield [6]:



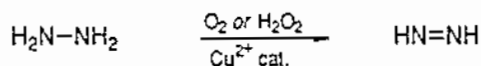
Third, hydrogenolysis does not occur. For example, benzyl cinnamate gives the dihydro derivative,

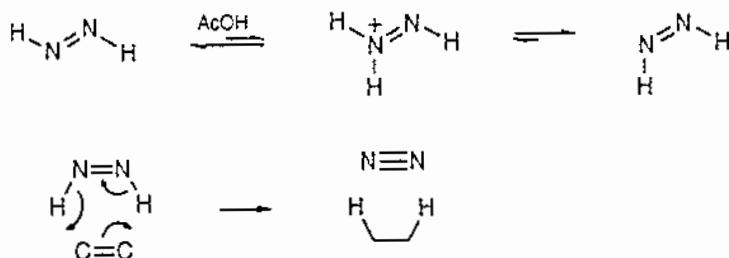


whereas hydrogenation on a metal catalyst results also in cleavage of the *O*-benzyl bond to give $\text{PhCH}_2\text{CH}_2\text{CO}_2\text{H}$ (cf. p. 643).

The major disadvantage of the method is that, because of the strong affinity of the rhodium complex for carbon monoxide, aldehydes are degraded, for example, $\text{PhCH}=\text{CHCHO} \rightarrow \text{PhCH}=\text{CH}_2$. The ruthenium complex used is $(\text{Ph}_3\text{P})_3\text{RuClH}$, which is formed *in situ* from $(\text{Ph}_3\text{P})_3\text{RuCl}_2$ and molecular hydrogen in the presence of a base such as triethylamine. It is specific for the hydrogenation of monosubstituted alkenes and disubstituted acetylenes, which yield *cis*-alkenes.

(ii) *The diimide method.* Diimide is an unstable compound which is usually made by the copper(II)-catalyzed oxidation of hydrazine with air or hydrogen peroxide. In the absence of a species which traps it, it decomposes to nitrogen and hydrogen but when it is generated in the presence of an alkene and a small quantity of acetic acid, rapid *syn*-stereospecific reduction occurs, the driving force being the great stability of the nitrogen molecule compared with the $-\text{N}=\text{N}-$ system. The role of the acetic acid is almost certainly to catalyze the formation of the *cis*-isomer of diimide from the more stable *trans*-isomer, the former then reacting with the alkene *via* a cyclic transition state:



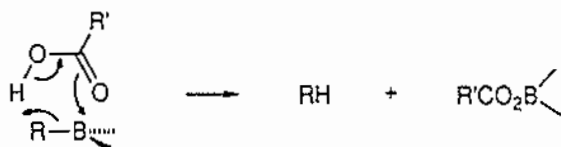


Alkynes and azo-compounds are also reduced, but carbonyl-containing groups, nitro groups, sulfoxides, and S—S bonds are not affected.

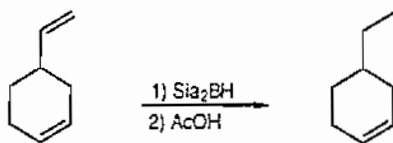
(iii) *Hydroboration*. The organoboranes formed by alkenes with diborane or other boronating agents (p. 483) are solvolyzed by organic acids (but not by mineral acids) to alkanes:



It is thought that reaction occurs as follows:



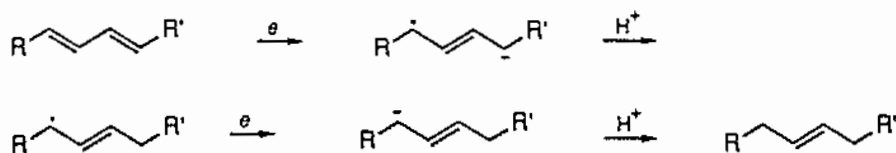
In the case of dienes, it is possible to reduce selectively the less substituted alkene by using a hindered borane such as that from trimethylethylene, Si_2BH (cf. p. 485), e.g.

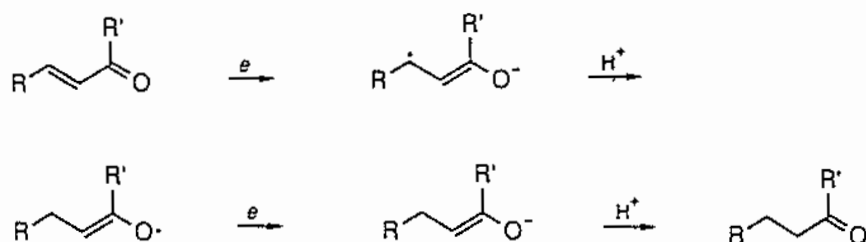


Aldehydes, ketones, acids, esters, nitriles, and epoxides are also reduced.

(c) Conjugated alkenes

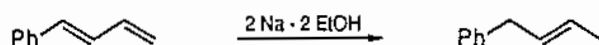
(i) *Electron transfer*. In addition to being reducible by catalytic methods, conjugated dienes and polyenes, and compounds containing C=C bonds conjugated with carbonyl-containing groups, are reducible with electron-transfer agents. The reason is that the uptake of electrons by these systems gives delocalized intermediates whereas the addition of electrons to alkenes does not:



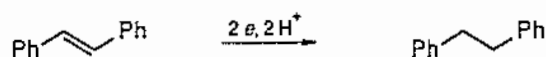


The common reducing agents in this category are sodium and an alcohol, sodium amalgam, zinc and acetic acid, and the metal-ammonia and metal-amine systems. The last two groups are discussed more fully below.

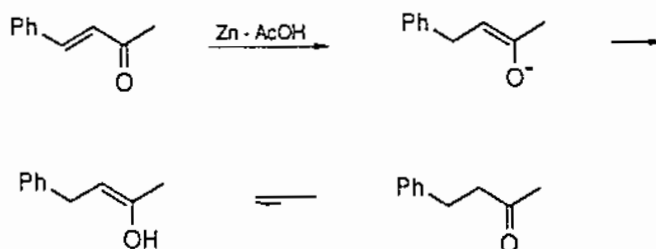
The reduction of a conjugated diene occurs by 1,4-addition, e.g.



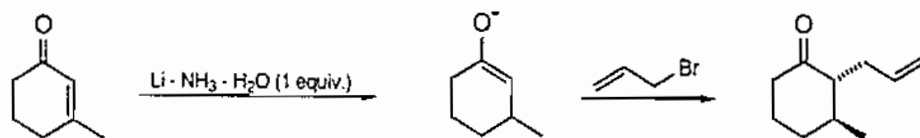
A C=C bond conjugated with an aromatic ring is also reduced, for the aromatic π -system is available to delocalize the charge on the anion, but the 1,2-adduct is formed since formation of the 1,4-adduct would be accompanied by the loss of aromatic stabilization energy. For example, stilbene is reduced to bibenzyl,



The reduction of an $\alpha\beta$ -unsaturated carbonyl compound gives an enolate ion as the intermediate. Although this is protonated more rapidly on oxygen than carbon, the enol rapidly tautomerizes to the more stable carbonyl compound, e.g.



When only one equivalent of the proton-donor is used, the enolate remains as such and can be used for generating new C—C bonds, e.g.



Metal-ammonia and metal-amine solutions. Strongly electropositive metals dissolve in liquid ammonia and in some amines to give characteristic blue solutions which contain metal cations and solvated electrons. Reduction to give amide ions is slow, but can be accelerated by the addition of catalytic amounts of transition-metal ions, e.g.



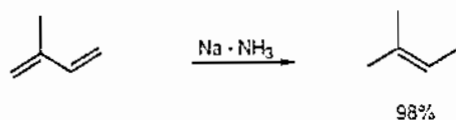
The use of sodamide in liquid ammonia as a powerful catalyst for condensations has already been described (p. 242). The solution before amide formation acts as a powerful source of electrons for the reduction of organic compounds. A proton source is usually added to complete the reaction, for ammonia itself is a very weak proton donor. The donor should not be a strong enough acid to liberate hydrogen rapidly with sodium; ethanol is commonly used, for it reacts with sodium only very slowly at the boiling point of liquid ammonia. The reducing power of the system increases with the strength of the proton donor: the stronger this is, the more rapidly is the initial equilibrium displaced to the right:



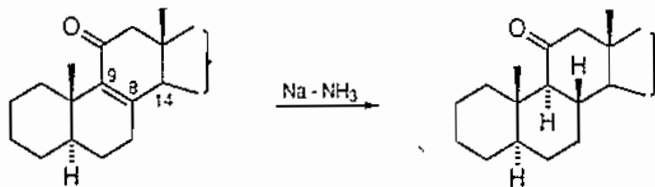
As the affinity of R for electrons is reduced, increasingly powerful proton donors are needed to effect reduction.

The use of ammonia is limited by the low solubility in it of most organic compounds at its boiling-point (-33°C). Co-solvents such as ethers can be added, but they depress the solubility of the metals: high mutual solubility is rare. Primary amines such as ethylamine are more powerful solvents and lithium is the most soluble of the alkali metals in such solvents. Since the reducing power of the system increases with the concentration of the metal, lithium and ethylamine provide about the most vigorous available conditions.

In general, these systems are of less value for the reduction of $\text{C}=\text{C}$ bonds than in the aromatic series (p. 641) partly because so many effective alternatives are available for the former and partly because the conditions are often too vigorous. However, yields in simple cases are high, e.g.

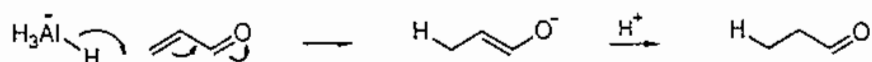


and in certain situations these methods succeed where catalytic methods fail. For example, the C_8-C_9 and especially the C_8-C_{14} double bonds in steroid systems are almost impossible to reduce catalytically but are reduced efficiently with sodium in liquid ammonia, e.g.

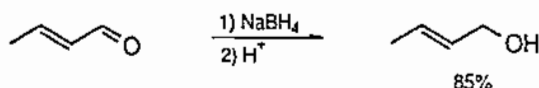


The main product is usually the thermodynamically most stable stereoisomer, as in the above example. The first protonation, at C-8, preferentially occurs from above the molecular plane, giving an axial C—H bond and therefore equatorial C—C bonds and protonation of the ensuing enolate gives the more stable *trans* adduct.

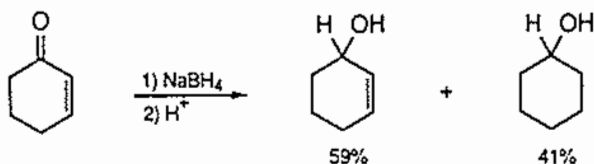
(ii) *Hydride-transfer.* Compounds in which $C=C$ is conjugated to substituents of $-M$ type are susceptible to nucleophiles (p. 91), so that reductions of the type,



might be expected with $\alpha\beta$ -unsaturated carbonyl compounds. When the $-M$ group is $-CHO$, this does not normally occur to a significant extent and reduction of the carbonyl occurs instead, e.g.

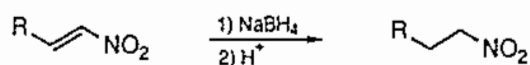


When conjugation is to a keto group (which is less reactive than $-CHO$ to nucleophiles), mixtures of carbonyl-reduced and fully reduced products are formed, e.g.



Lithium aluminium hydride has a relatively stronger tendency for reaction at the carbonyl group: in the above case, it gives 94% of 2-cyclohexenol and only 2% of cyclohexanol.

However, if the $-M$ substituent is not itself reducible by the hydride, selective reduction of the $C=C$ bond can be effected. For example, reduction of an $\alpha\beta$ -unsaturated nitro-compound by sodium borohydride,



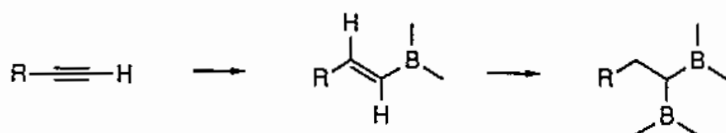
was employed in the synthesis of chlorophyll (section 22.4). The more powerful hydride-donor lithium aluminium hydride reduces both the $C=C$ bond and the nitro group in this situation and the sequence, $ArCH=CHNO_2 \rightarrow ArCH_2CH_2NH_2$, is employed in the building up of isoquinoline rings from aromatic aldehydes (p. 706).

(d) Alkynes

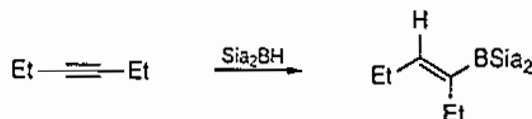
(i) *Catalytic reduction.* The $C\equiv C$ bond can be fully reduced on the catalysts used for the reduction of alkenes, but this is seldom needed in synthesis. It is much more useful to exploit the fact that the catalytic reduction of $C\equiv C$ to $C=C$ is faster than the reduction of $C=C$ to $C-C$. However, it is not satisfactory simply to use the conventional catalysts and to inject the amount of hydrogen needed for reduction to the alkene level, since this gives mixtures of

partially and fully reduced compounds. The technique is to deactivate ('poison') the catalyst so that reduction of $C\equiv C$ to $C-C$ is inhibited, and this is usually done by using *Lindlar's catalyst*, in which palladium is deposited on a calcium carbonate support to which lead acetate and a small quantity of quinoline are added as the 'poison'. The reduction product is the *cis*-alkene.

(ii) *Hydroboration*. The hydroboration of alkynes is carried out in the same way as that of alkenes. With terminal alkynes, dihydroboration occurs to some extent,

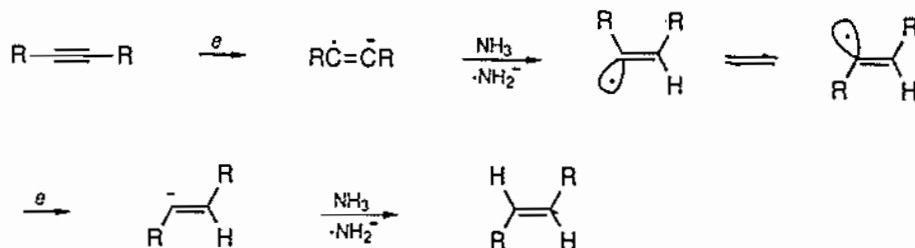


so that, on hydrolysis, complete reduction occurs. To prevent this, it is best to use the hindered and more selective disiamyl borane (p. 485) which allows quantitative monohydroboration of both mono- and disubstituted acetylenes, e.g.

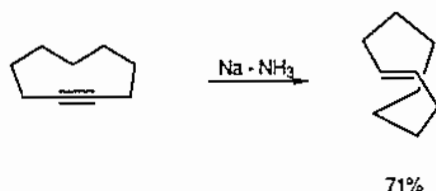


The products can be hydrolyzed with acetic acid at room temperature to give specifically *cis*-alkenes.

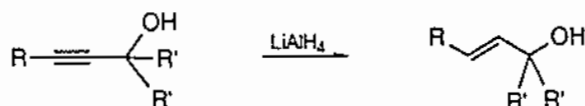
(iii) *Electron transfer*. Electrons are transferred to alkynes more readily than to alkenes (cf. the greater reactivity of alkynes towards nucleophiles). The best reagents are the metal-amine and metal-ammonia systems; no added proton donor is needed. Reaction is thought to occur by alternating electron and proton transfers:



The formation of the *trans*-product is thought to be dictated by the occurrence of a rapid equilibrium between the *cis* and *trans* vinyl radicals in which the latter, being the less congested, predominates; the *trans* anion is thereby formed the faster. This method is therefore complementary to the catalytic methods. For example, *trans*-cyclononene can be obtained from cyclononyne:



(iv) *Hydride-transfer.* Diisobutyl aluminium hydride (p. 659) reduces alkynes to alkenes, but lithium aluminium hydride is effective only when the alkyne has a hydroxyl group in the α -position: the hydroxyl group facilitates reaction through the formation of a complex with aluminium, and the product is the *trans*-alkene.



(e) *Aromatic rings*

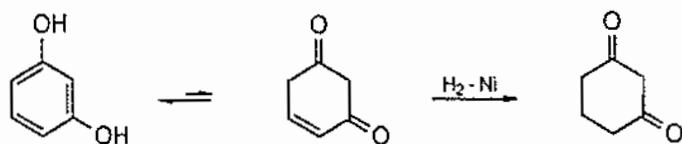
(i) *Catalytic hydrogenation.* Aromatic rings can be reduced on the catalysts suitable for alkenes but more vigorous conditions are required because aromatic stabilization energy is lost in the process. The conditions vary according to the amount of stabilization energy which is sacrificed, e.g. naphthalene is reduced to tetralin more easily than benzene is reduced to cyclohexane (cf. p. 38). Typical conditions on Adams' catalyst are 10 hours at 100–150°C and 100–150 atmospheres (i.e. in a steel bomb), compared with 1 hour at 20°C and atmospheric pressure for the reduction of simple alkenes.

If there is a danger of the catalyst becoming poisoned, Raney nickel is normally selected because it is cheaper than palladium or platinum and can be used in excess. This is particularly important with sulfur-containing molecules but it must be remembered that sulfur is removed from the system during reduction (p. 672). Pyridine also poisons catalysts, but its hydrochloride can be reduced efficiently over platinum in ethanol.

Reduction in mild conditions gives mainly the *cis* product, but if the *trans* product is the more stable it can be formed in vigorous conditions:* e.g. the reduction of naphthalene in solution over platinum gives mainly *cis*-decalin whereas over copper chromite in the gas phase *trans*-decalin predominates.

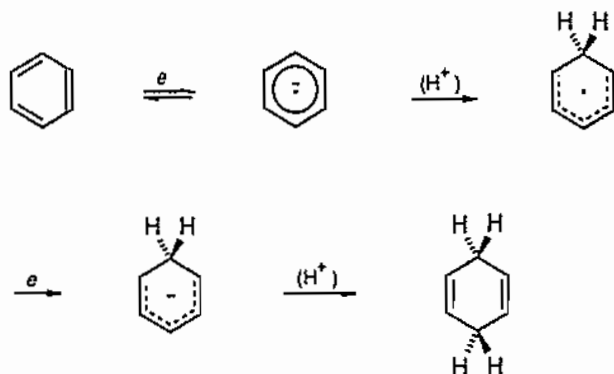
Those aromatic compounds which are in equilibrium with significant proportions of aliphatic tautomers can be reduced in conditions appropriate to alkenes. For example, *m*-dihydroxybenzene is reduced, *via* its diketo-tautomer (p. 13), at 50°C and 80 atmospheres over Raney nickel:

* This is an example of thermodynamic control (p. 68). In vigorous conditions, equilibrium is established with the *cis* product while the slower *trans* reduction gradually removes the reactant to give the more stable product.



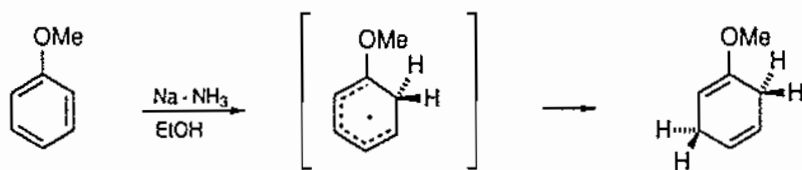
(ii) *Electron transfer.* Whereas catalytic methods normally fully reduce aromatic rings, electron-transfer reagents can be highly selective.

Reaction occurs by successive electron and proton transfers, e.g.

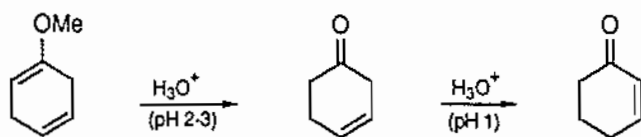


Sodium in liquid ammonia is commonly used as the reducing agent. Ammonia can act as the source of protons, but when an electron-releasing substituent such as alkyl, OMe, or NMe₂ is present, the first equilibrium lies well to the left and, because ammonia is a weak proton-transfer agent, reaction is very slow. It is therefore usual to include an alcohol to trap the anion radical rapidly.

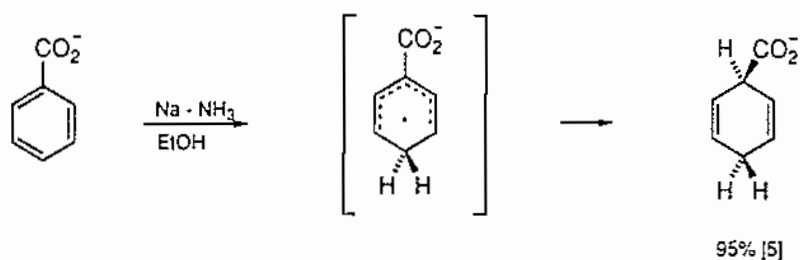
The structure of the product is determined by the site of the first protonation: the anion that is then formed reacts at the central carbon of the delocalized system to complete the 1,4-addition. Electron-releasing substituents direct the first protonation to the *ortho* position, e.g.



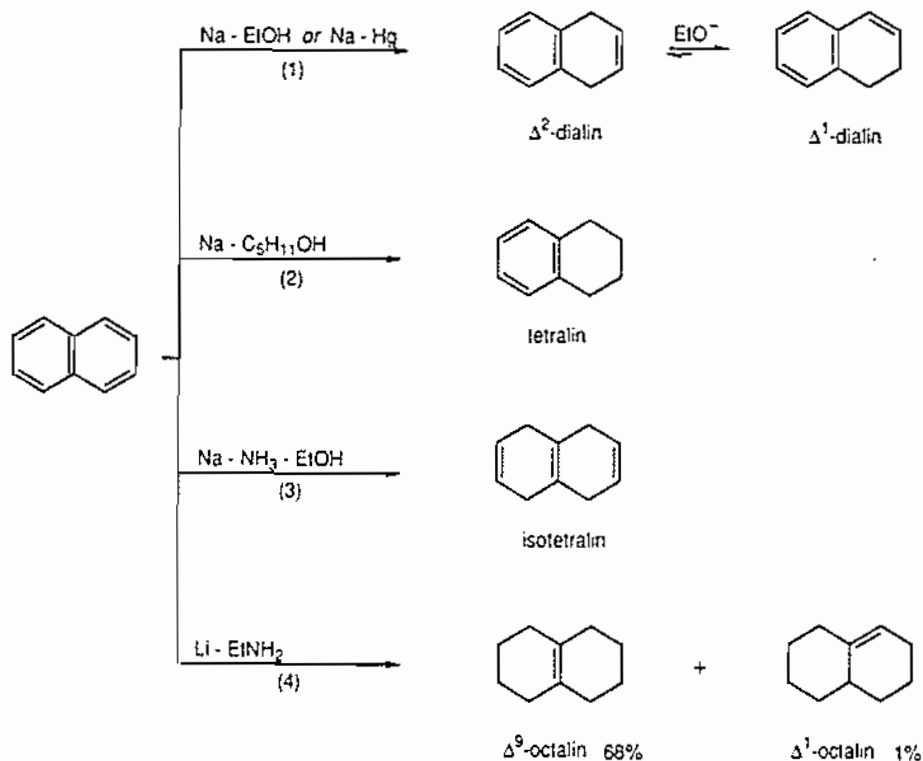
Acid-catalyzed hydrolysis (pH 2–3) of the product (an enol ether) gives an unsaturated ketone which, in slightly more acidic conditions (pH 1), tautomerizes to the conjugated isomer:



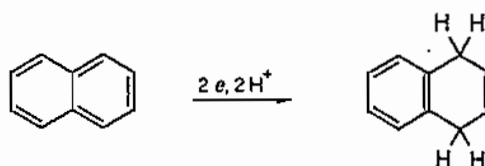
In contrast, the —CO₂[−] substituent directs protonation to the *para* position, e.g.



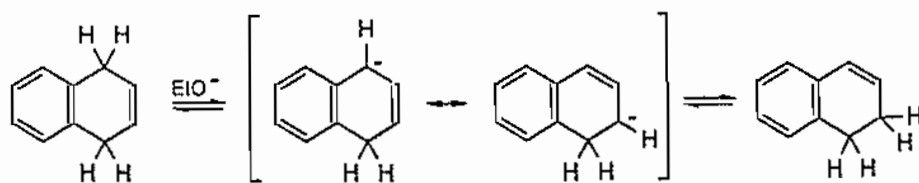
With naphthalene, a variety of products can be obtained, depending on the conditions:



(1) The mildest conditions give a dihydro compound:



The product is isomerized with base to the more stable (conjugated) Δ^1 -dialin, via a delocalized anion,

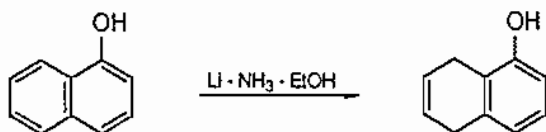


(2) Further reduction occurs in the higher-boiling 1-pentanol probably through formation of Δ^2 -dialin, isomerization as in (1), and reduction of the conjugated Δ^1 -dialin.

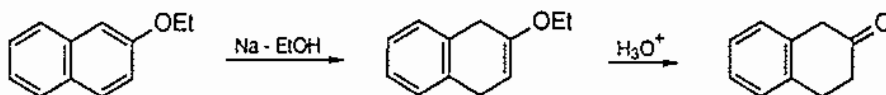
(3) The use of liquid ammonia with an added proton-donor enables both benzenoid rings to be reduced in the 1,4-manner.

(4) Lithium in ethylamine provides the most vigorous conditions (p. 637): initial reduction as in (3) is followed by isomerization and further reduction.

The reductions of 1-naphthol and ethyl 2-naphthyl ether provide an interesting contrast. The former is reduced by lithium in liquid ammonia containing ethanol in the *unsubstituted* ring in 98% yield [4],



whereas the latter is reduced by sodium and ethanol in the *substituted* ring to give 2-tetralone in 45% yield after hydrolysis [4]:



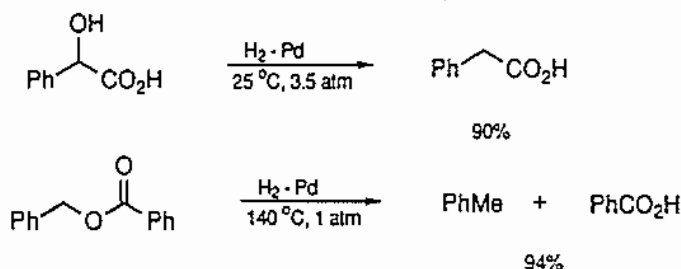
The difference is consistent with the mechanism described for the reduction of anisole. With 1-naphthol, protonation of the anion-radical at the 2-position and subsequent protonation at the 5-position would result in the complete loss of the aromatic stabilization energy and is less favourable than 1,4-addition in the unsubstituted ring. In contrast, protonation of the anion-radical from ethyl 2-naphthyl ether at the 1-position and subsequent protonation at the 4-position leaves a benzenoid system.

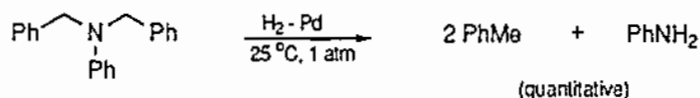
(a) Benzylic systems

20.3 Hydrogenolysis

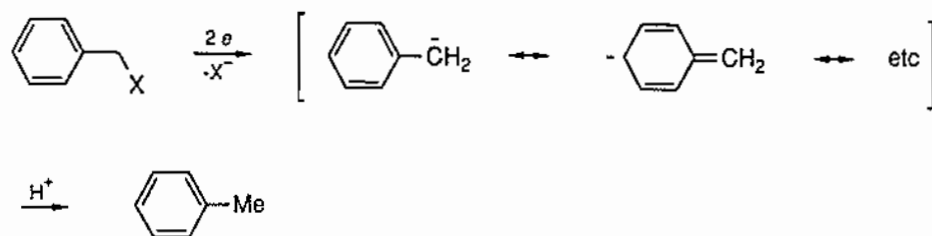
The benzylic group, when attached to OH, OR, OCOR, NR₂, SR, or halogen, is particularly susceptible to nucleophilic reducing agents, catalytic reduction, and electron-transfer reagents.

Catalytic reduction is widely employed, as in the following examples:

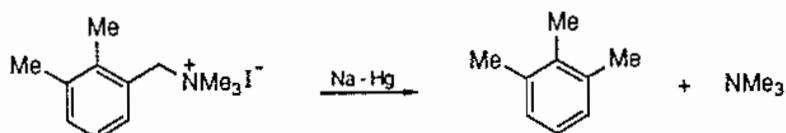
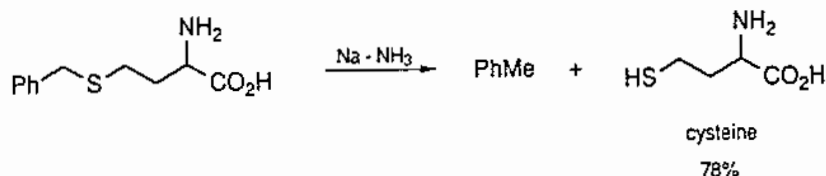
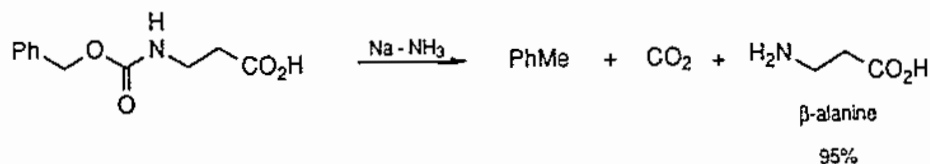




Electron-transfer reduction is probably facilitated by the ability of the benzenoid ring to delocalize the charge of an anion, i.e.

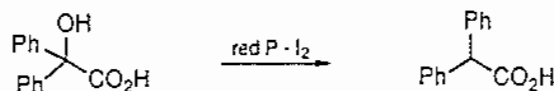


The following are examples:

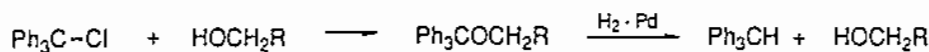


The first example illustrates the use of the benzyl group as part of a protecting agent in amino acid chemistry: the benzyloxycarbonyl group is used to protect the amino group in peptide synthesis and is afterwards removed as above or catalytically (p. 330). The second example represents a standard procedure for the protection of thiol groups: the catalytic method is unsuitable for the removal of the benzyl residue because sulfur is present. The final example illustrates a synthetic method for introducing methyl groups *ortho* to existing methyl groups in aromatic rings: the reactant is obtained from *o*-xylene via the Stevens rearrangement (p. 447) and is converted into 1,2,3-trimethylbenzene by sodium amalgam at 80°C in 87% yield [4].

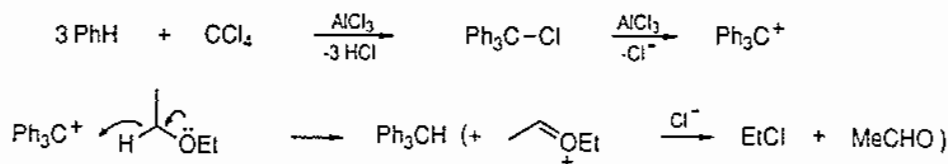
Diphenylmethyl systems are more easily hydrogenolyzed than benzyl systems. For example, benzoic acid is reduced in 95% yield to diphenylacetic acid by treatment with red phosphorus and iodine in refluxing acetic acid [1]:



Triphenylmethyl (trityl) systems are still more easily reduced. Advantage is taken of this, together with the selectivity of triphenylmethyl chloride for primary alcoholic groups as compared with secondary and tertiary, in protecting primary alcohols (e.g. in the synthesis of adenosine triphosphate, section 22.2):



Triphenylmethyl systems readily form the trityl cation and this is particularly susceptible to reduction by hydride-transfer agents. For example, triphenylmethane can be obtained in 70–80% yield by Friedel–Crafts reaction between benzene and carbon tetrachloride in the presence of aluminium trichloride (p. 361) followed by the addition of diethyl ether [1]:



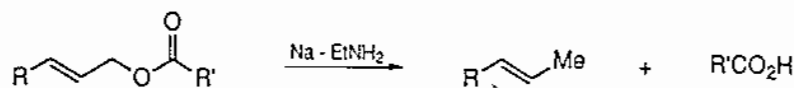
(b) Allylic systems

Allylic systems behave in most reactions analogously to benzylic systems. However, whereas the latter are normally reduced by hydrogenolysis rather than by hydrogenation of the aromatic ring, the former more readily undergo hydrogenation with those reducing agents which react with the C=C bond, so that catalytic reduction is in general unsuitable for the hydrogenolysis of allylic systems.

Fortunately, lithium aluminium hydride and sodium in an amine are suitable reagents since neither reacts with isolated C=C bonds. For example, allyl ethers are cleaved by lithium aluminium hydride in tetrahydrofuran,



and allyl esters are cleaved by sodium in ethylamine,

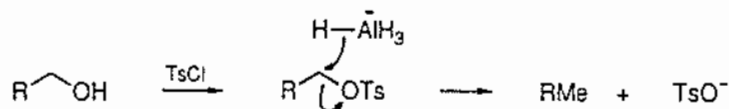


The corresponding alkyl compounds are unaffected in these conditions.

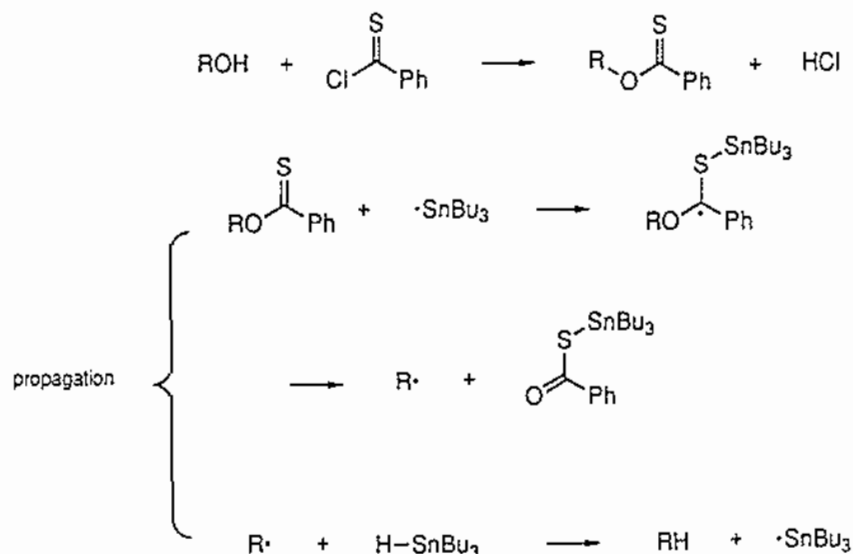
(c) Alkyl systems

Alkyl systems are generally much less easy to hydrogenolyse than the corresponding benzyl systems.

(i) *Alcohols* cannot be hydrogenolyzed directly but can be converted into their toluene-*p*-sulfonates which are reducible with sodium cyanoborohydride or with lithium aluminium hydride (i.e. by taking advantage of the fact that the toluene-*p*-sulfonate anion is so good a leaving group in S_N2 displacements, p. 103):

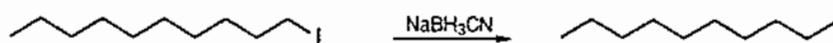


Alternatively, the thioester formed with $PhCSCl$ can be reduced with tributylstannane in a radical-chain reaction that is usually initiated with AIBN (cf. p. 538):

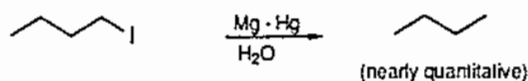


This method is of particular value compared with the tosylate method when the organic compound contains other groups, such as carbonyl, which would also be reduced by $LiAlH_4$. However, although good yields are obtained from secondary alcohols, they are generally low with primary alcohols.

(ii) *Halides* are much more reactive than alcohols in S_N2 displacements and are correspondingly more susceptible to hydride-transfer from lithium aluminium hydride or, better, sodium cyanoborohydride. For example, with the latter reagent in hexamethylphosphoramide, 1-iododecane gives decane in about 90% yield [6]:



Electron-transfer agents are also efficient. Since the highly electropositive metals like sodium induce Wurtz coupling, the metal-amine systems are unsuitable, but magnesium amalgam in water and a zinc-copper couple in ethanol can be used, e.g.



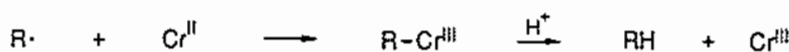
Alternatively, magnesium in ether is used to form the Grignard reagent which is subsequently decomposed by water,



Two radical-mediated processes are available. In one, the reducing agent is chromium(II) ion, usually complexed with ethylenediamine. It is thought to act by first removing the halide to form an alkyl radical,

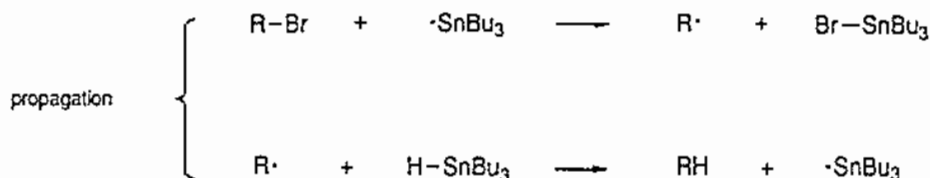


and then to complete reduction *via* an organochromium species:

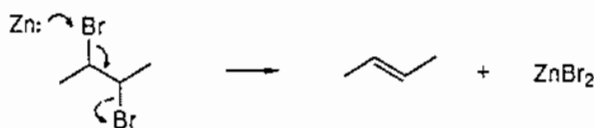


Alkenyl halides can also be reduced in this way.

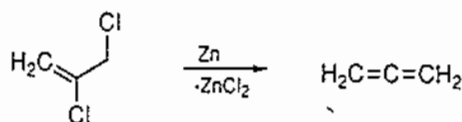
The second is a radical-chain reaction, initiated by AIBN, in which the reducing agent is tributylstannane:



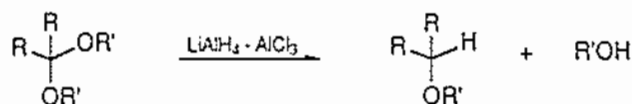
Electron-transfer agents react with 1,2-dihalides in a different manner: both halogen atoms are eliminated and a C=C bond is introduced. Zinc is normally used as the reducing agent:



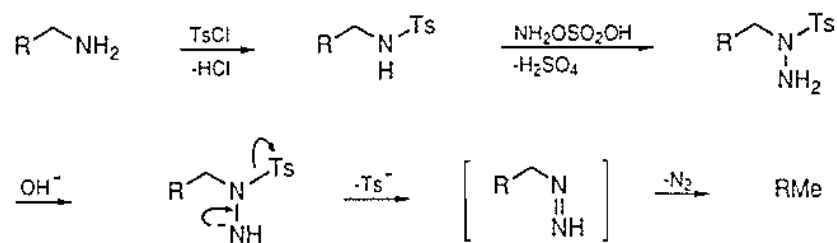
For example, allene may be obtained in 80% yield by the treatment of 2,3-dichloropropylene with zinc in aqueous ethanol:



(iii) *Acetals and ketals* are hydrogenolyzed to ethers in high yield by lithium aluminium hydride in the presence of aluminium trichloride:



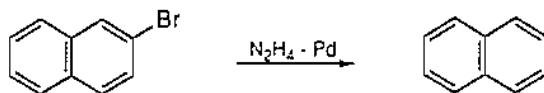
(iv) *Primary amines* can be reduced by treating the derived toluene-*p*-sulfonamide with a large excess of hydroxylamine-*O*-sulfonic acid in alkaline ethanol. Use is made of the strong leaving-group potential of sulfates and sulfinates and the instability of the —N=NH grouping (cf. p. 650).



For example, 1-hexylamine gives hexane in 41% yield.

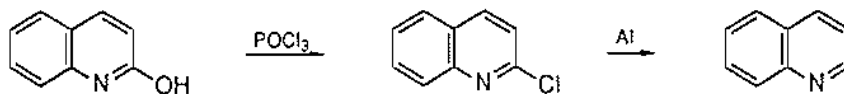
(d) *Aromatic systems*

Aryl halides can be reduced by the formation of the Grignard reagent (from bromides and iodides) or the lithium compound (from chlorides) followed by treatment with water, or with hydrazine over a palladium catalyst. For example, 2-bromonaphthalene is reduced to naphthalene in 95% yield in 5 minutes in boiling ethanol:

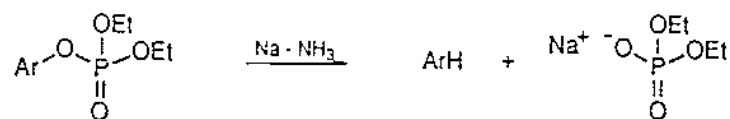


Reduction with chromium(II) ion is also efficient. For example, 1-bromonaphthalene with chromium(II) complexed with ethylenediamine in aqueous dimethylformamide gives 93–98% of naphthalene [6].

The halogen atoms at the 2- and 4-positions of pyridine and its derivatives can be efficiently removed both catalytically by hydrogen and by electron-transfer reducing agents (cf. the ease of nucleophilic substitution at these positions, p. 399). Since hydroxyl groups at these positions are convertible into chloro-substituents with phosphorus oxychloride, this provides a method for reducing phenols, as in the Conrad–Limpach synthesis (p. 705), e.g.



Phenols can be hydrogenolyzed *via* their phosphate esters by reduction with sodium in liquid ammonia, but yields are not usually above 50%.



Aromatic amines cannot be hydrogenolyzed directly; they are deaminated by reduction of the corresponding diazonium ions (p. 413).

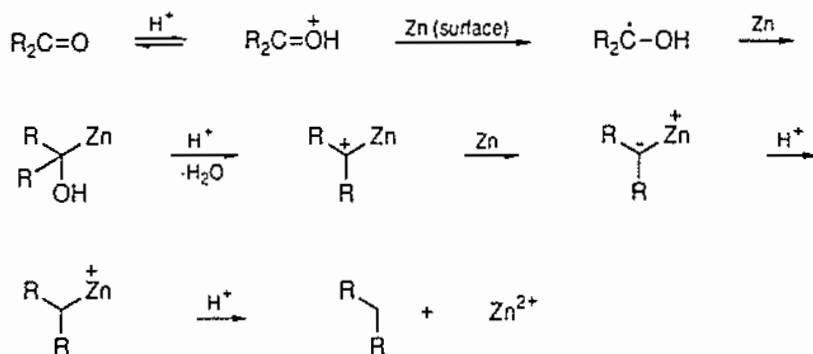
20.4 Aldehydes and ketones

Aldehydes and ketones can be reduced to hydrocarbons, alcohols, and 1,2-diols.

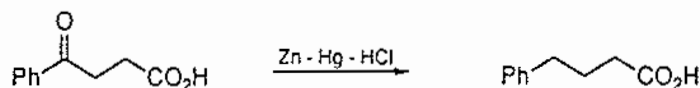
(a) To hydrocarbons

Five methods are available. The choice between them is based on the sensitivity of other functional centres in the reactant in the reducing conditions.

(1) *Clemmensen method*. The reducing agent is amalgamated zinc and concentrated hydrochloric acid. The mechanism is not certain but may be as follows:



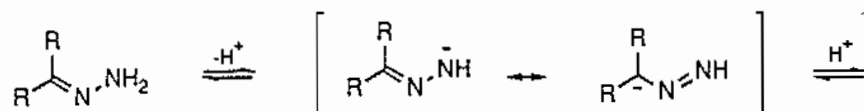
The method is particularly useful for ketones which contain phenolic or carboxylic groups. For example, the reduction of 3-benzoylpropionic acid in toluene gives 4-phenylbutyric acid in 85% yield [2]:

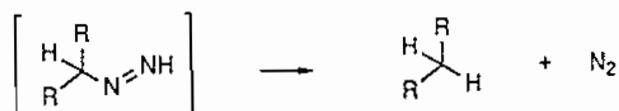


This type of reduction forms one step in the extension of benzenoid systems *via* Friedel-Crafts acylations (p. 368).

The reagent also reduces the C=C bond in $\alpha\beta$ -unsaturated ketones, acids, and esters. Benzyl halides and alcohols are hydrogenolyzed. Strongly hindered ketones give low yields and sometimes rearrangement products (e.g. $\text{Ph}_3\text{C}-\text{CO}-\text{Ph} \rightarrow \text{Ph}_2\text{C}=\text{CPh}_2$).

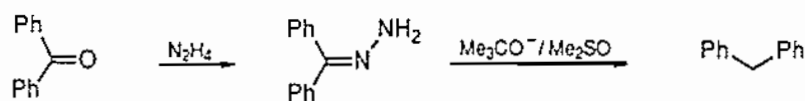
(2) *Wolff-Kishner method*. The hydrazones of aldehydes and ketones are reduced in vigorously basic conditions with the evolution of nitrogen, probably as follows:



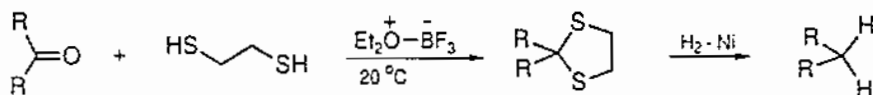


One procedure is the *Huang–Minlon* modification. The hydrazone is formed by heating the carbonyl compound with hydrazine hydrate and potassium hydroxide in $(\text{HOCH}_2\text{CH}_2)_2\text{O}$ under a water condenser. After completion of the formation of the hydrazone, the water condenser is removed so that the water liberated in the first reaction is distilled and the temperature rises to over 200°C , so bringing about decomposition of the hydrazone.

A newer modification employs potassium *t*-butoxide as the base and dimethyl sulfoxide as solvent. Alkoxide bases are very much more powerful in this solvent than in hydroxylic solvents (p. 64) and reaction occurs at room temperature in high yield. For example, benzophenone gives about 90% of diphenylmethane:



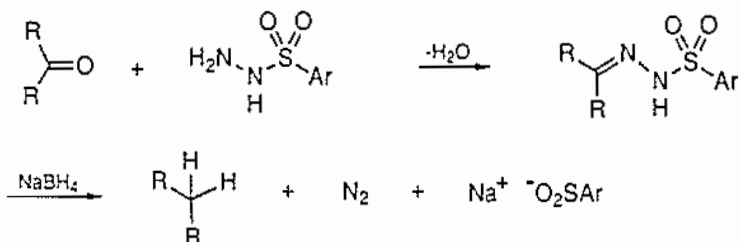
(3) *Mozingo method*. The carbonyl compound is converted with ethanedithiol in the presence of a Lewis acid into its dithio-acetal or ketal and this is hydrogenolyzed over Raney nickel:



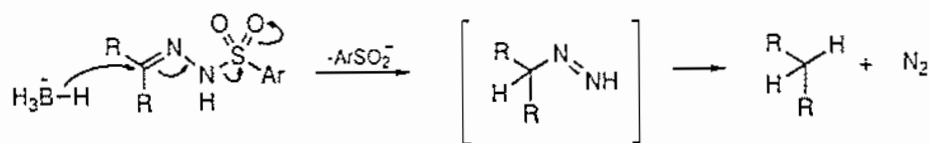
Alternatively, the cyclic dithio-compound is reduced by hydrogen-transfer from hydrazine at $100\text{--}200^\circ\text{C}$.

The Mozingo reaction is useful for reducing carbonyl compounds which are sensitive to mineral acid and bases, for the Clemmensen and Wolff–Kishner methods are then unsuitable.

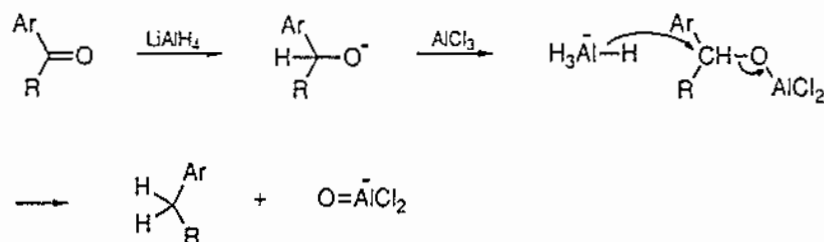
(4) *Tosylhydrazone method*. Reaction of the carbonyl compound with toluene-*p*-sulfonylhydrazine gives the tosylhydrazone which is efficiently reduced by sodium borohydride or cyanoborohydride ($\text{Ar} = p\text{-tolyl}$):



Reaction probably occurs as follows:



(5) *Lithium aluminium hydride*. Aromatic ketones are reduced by lithium aluminium hydride in the presence of aluminium trichloride. Reaction occurs by reduction to the alcohol (p. 652) followed by hydrogenolysis of the benzylic system, aided by the Lewis acid:

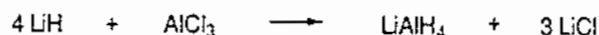


(b) To alcohols

Carbonyl compounds are reduced to alcohols by a variety of reagents. Of the three general classes of reductive process, catalytic hydrogenation is not normally chosen because it is slow, but both hydride-transfer and electron-transfer reagents are employed.

(i) *Hydride transfer*. The alkali-metal hydrides such as sodium hydride, NaH, are unsuitable as reducing agents because of their insolubility in organic solvents and, particularly, because of their powerful effects as catalysts for the base-catalyzed reactions which aldehydes and ketones undergo. Instead, lithium aluminium hydride, LiAlH₄, or sodium borohydride, NaBH₄, or derivatives in which some of the hydrogens have been replaced by other groups, are used; they are effective at transferring a hydride ion to a carbonyl group, but not as bases in condensations.

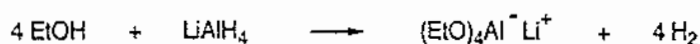
1. *Lithium aluminium hydride* was the first such compound to be developed. It is prepared by treating lithium hydride with aluminium trichloride,



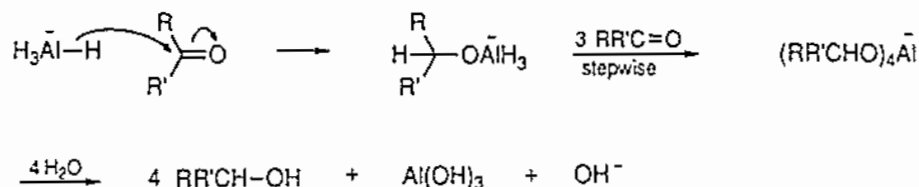
and is soluble in ethers such as diethyl ether and tetrahydrofuran, in which it is normally used. The solvent must be scrupulously dry, since the hydride is destroyed by water,



and it must also be handled as a solid with great care since it may inflame spontaneously while being crushed in a pestle and mortar and it explodes violently on strong heating. All hydroxyl-, amino-, and thiol-containing compounds liberate hydrogen quantitatively from it, e.g.



Each of the four hydrogen atoms in lithium aluminium hydride is available for transfer to carbonyl groups, e.g.

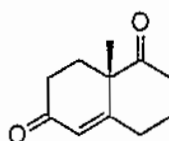


each step occurring less rapidly than the preceding one. (As a result, the replacement of two or three hydrogen atoms of the aluminium hydride anion by alkoxy groups gives less reactive and more selective reducing agents: examples are described later.) Finally, hydrolysis of the aluminium alkoxide gives the alcohol.

2. *Sodium borohydride* is much less reactive. It can be used in alcoholic solvents and even in water, for it decomposes only enough to make the solution alkaline, after which it is stable.



Chemoselectivity. The two reagents differ considerably in their reducing power. Lithium aluminium hydride reduces not only aldehydes and ketones but also acids, acid chlorides, esters, nitriles, imines, and nitro groups, whereas sodium borohydride reduces only aldehydes, ketones, acid chlorides, and imines. Neither reagent normally reduces $C=C$, $C\equiv C$, or $N=N$, although lithium aluminium hydride reduces alkynes containing α -hydroxy-substituents (p. 640) and reduces azo-compounds in the presence of a Lewis acid (p. 672). For the reduction of aldehydes and ketones, sodium borohydride is normally preferred, both because of its greater chemoselectivity and because of its greater ease of handling. Sodium borohydride is also selective between carbonyl groups in different environments. For example, with only one equivalent of reducing agent, the compound

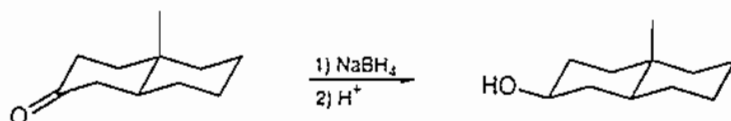
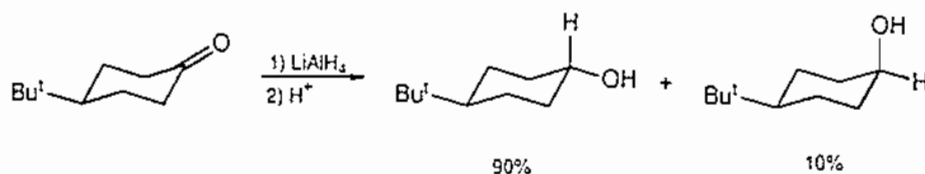


is reduced only at the non-conjugated carbonyl position; the other carbonyl group is less reactive, owing to the delocalization energy in the conjugated system,

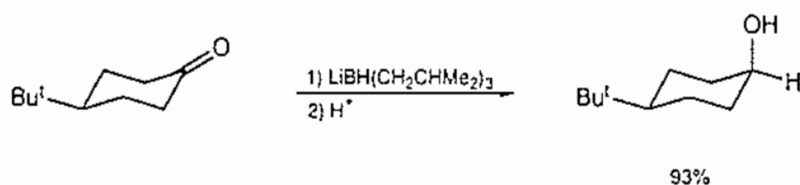


which is lost on addition of a nucleophile. Sodium cyanoborohydride, NaBH_3CN , is less reactive, and more selective, than the borohydride because the electron-attracting cyano-group stabilizes the negative charge on boron. It reduces aldehyde groups and imines but not the less reactive keto-groups and acid chlorides.

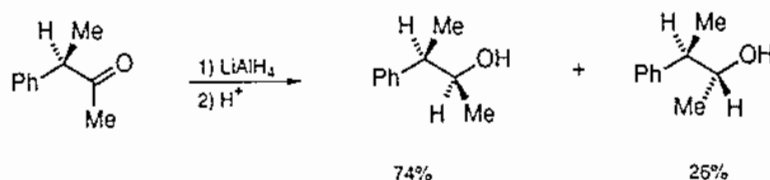
1. *Stereoselectivity with cyclohexanones.* With relatively small hydride-donors, such as the AlH_4^- and BH_4^- ions, steric hindrance to axial approach is less important than the stereoelectronic effect that favours it (p. 175), e.g.



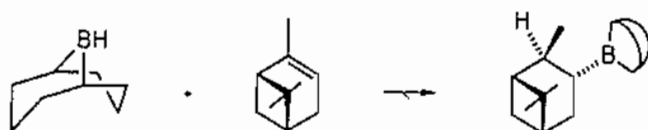
However, the opposite is the case with bulky hydrides, e.g.



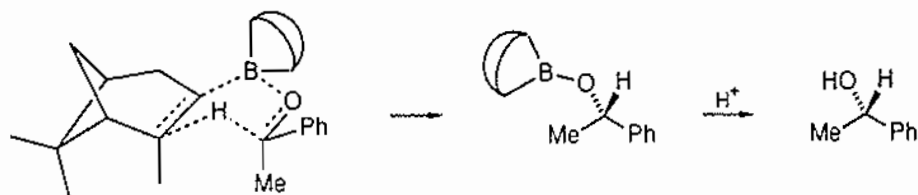
2. *Stereoselectivity with acyclic asymmetric ketones.* The major product can be predicted by Cram's rule (p. 150), e.g.



3. *Stereoselectivity with symmetric ketones and an asymmetric hydride donor.* (+)- α -Pinene reacts with the boron derivative 9-BBN (p. 485) to give an asymmetric adduct,



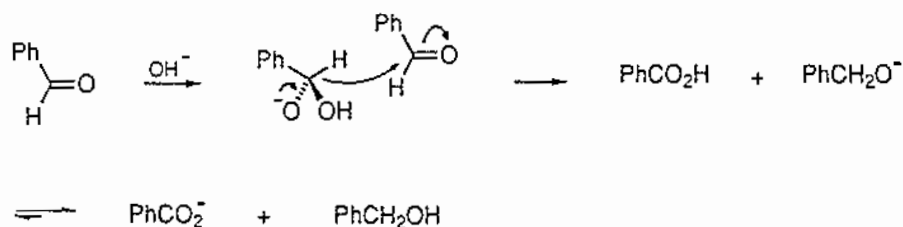
which reduces ketones with a very high degree of enantioselectivity, in some cases 100%. It is thought that the adduct transfers hydride to the ketone *via* a six-membered, boat-shaped cyclic transition state in which the larger group on the ketone preferentially lies away from the α -pinene residue to minimize steric congestion, e.g.



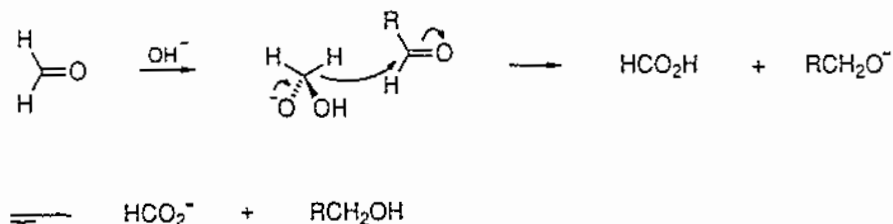
The adducts of 9-BBN with both (+)- and (-)- α -pinene are available commercially, so that either enantiomeric alcohol can be synthesized.

(ii) *Other hydride-transfer systems.*

(1) *Cannizzaro reaction.* Aldehydes which do not have α -CH groups cannot undergo base-catalyzed reactions. Instead they react with bases by disproportionation involving the transfer of hydride ion, e.g.

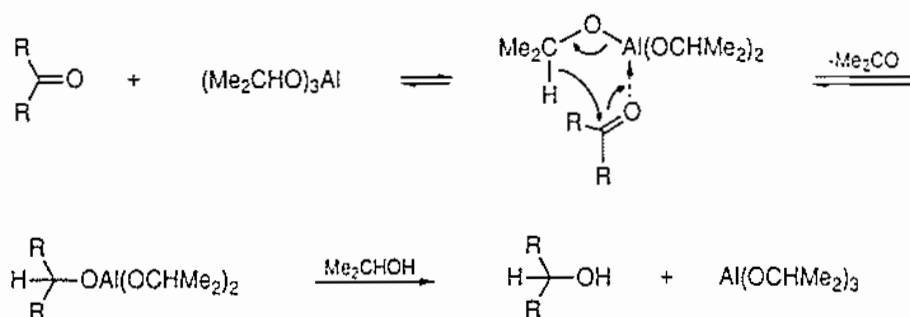


Crossed Cannizzaro reactions between one such aldehyde and formaldehyde result in the reduction of the former and the oxidation of the latter, for formaldehyde is more reactive than other aldehydes towards nucleophiles and rapidly gives a high concentration of the donor anion:



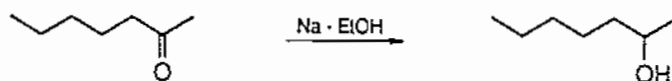
This fact can be exploited for reductions. For example, benzaldehyde is reduced by formaldehyde in the presence of potassium hydroxide in refluxing methanol to give 80% of benzyl alcohol [2]. The preparation of pentaerythritol from acetaldehyde and formaldehyde is also dependent on a crossed Cannizzaro reaction (p. 212).

(2) *Meerwein-Ponndorf-Verley reaction.* This is the reverse of Oppenauer oxidation (p. 612): equilibrium is established between the carbonyl group to be reduced and isopropanol on the one hand, and the required alcohol and acetone on the other, in the presence of aluminium isopropoxide. Since acetone is the lowest boiling constituent of the mixture, it can be continuously distilled so that the equilibrium is displaced to the right.



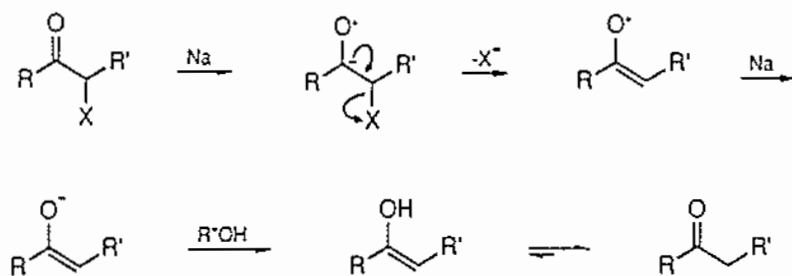
For example, trichloroacetaldehyde is reduced to trichloroethanol in about 80% yield. The reaction is specific to aldehydes and ketones; in particular, the C=C bond in $\alpha\beta$ -unsaturated aldehydes or ketones is not reduced (compare lithium aluminium hydride, p. 638). However, the basic conditions may bring about side reactions (as in the synthesis of reserpine, section 22.1). Very hindered Grignard reagents effect reduction in a similar way (p. 190).

(ii) *Electron-transfer reagents.* These reagents are less selective than sodium borohydride and the Meerwein-Ponndorf-Verley reagent, e.g. they also reduce the C=C bond in $\alpha\beta$ -unsaturated carbonyl compounds. Nevertheless, in simple cases they are rapid and efficient, e.g. 2-heptanone is reduced by sodium in ethanol to 2-heptanol in over 60% yield [2],

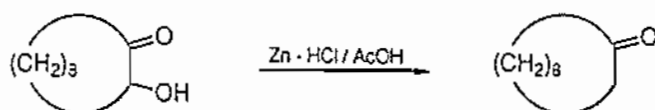


and heptanal is reduced by iron in aqueous acetic acid to 1-heptanol in 80% yield.

If the ketone contains at the α -position a substituent which is a good leaving group, the intermediate anion-radical undergoes elimination:



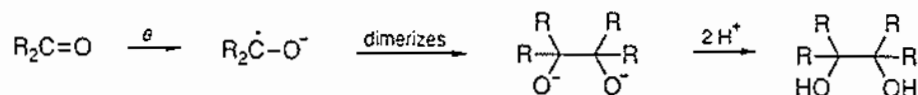
Zinc is often used as the electron source in these reductions. For example, α -hydroxycyclodecanone gives cyclodecanone in about 75% yield when treated with zinc in a mixture of hydrochloric and acetic acids at 75–80°C [4]:



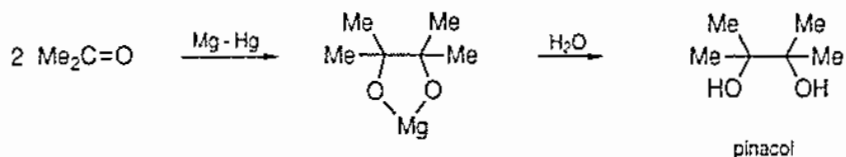
(iii) *The borane 9-BBN.* Whereas lithium aluminium hydride and sodium borohydride reduce $\alpha\beta$ -unsaturated aldehydes and ketones by both 1,2- and 1,4-addition, 9-BBN (p. 485) reduces only the carbonyl group.

(c) *To 1,2-diols*

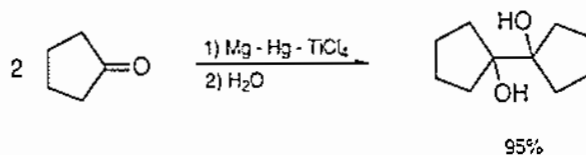
In the absence of a proton-donor, electropositive metals reduce ketones to 1,2-diols *via* the dimerization of anion-radicals:



The standard procedure employs amalgamated magnesium with benzene as solvent: the solid magnesium salt of the diol is formed and is hydrolyzed to the diol itself. For example, acetone gives pinacol in 45% yield after 2 hours in refluxing benzene [1]:



In an improved procedure, titanium tetrachloride is included; reaction is probably brought about by titanium metal formed *in situ*, e.g.

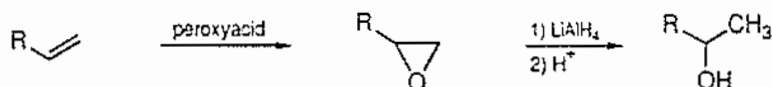


The reaction is generally ineffective for aldehydes because they are too readily reduced to alcohols. 1,2-Diols may also be formed by photochemical dimerization (p. 499).

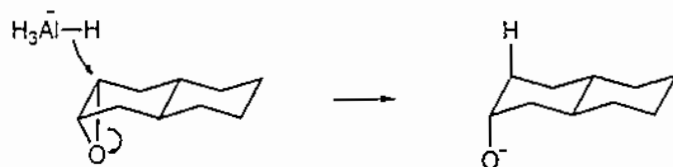
20.5

Epoxides (a) *Lithium aluminium hydride*

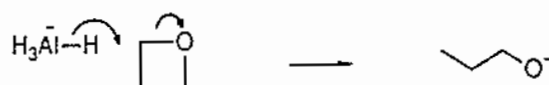
Epoxides are reduced to alcohols by lithium aluminium hydride. Since epoxides are readily obtained from alkenes (p. 588), the overall reaction serves to hydrate the alkene. The procedure is complementary to the hydroboration method (p. 657) since the hydride selectively attacks the less alkylated carbon of the epoxide ring, so giving the more highly alkylated alcohol (p. 590), e.g.



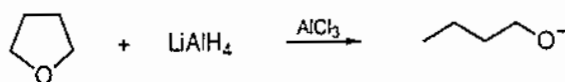
The reaction has the stereochemistry characteristic of S_N2 processes, for example in a rigid cyclic system the *axial* alcohol is formed,



The strained ring in four-membered cyclic ethers is also cleaved by lithium aluminium hydride, e.g.

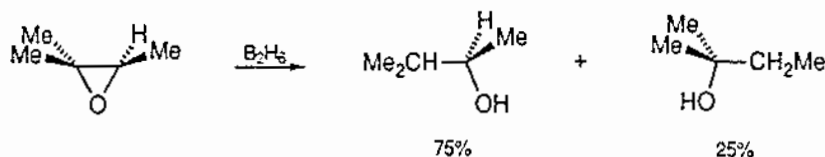


but the near-strainless five-membered ethers are resistant. They are, however, opened in the more vigorous conditions obtained by using lithium aluminium hydride in the presence of aluminium trichloride (cf. p. 651), e.g. tetrahydrofuran gives 1-butanol:



(b) Hydroboration

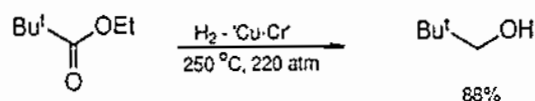
Epoxides are reduced by diborane to give mainly the *less* substituted alcohol, e.g.



The method is therefore complementary to the use of lithium aluminium hydride.

20.6 Acids and their derivatives

Until the advent of the complex metal hydrides (1947), the only methods for reducing acids and related compounds were relatively unselective. For example, esters could be reduced to alcohols over copper chromite in vigorous conditions, although not over other catalysts, e.g.

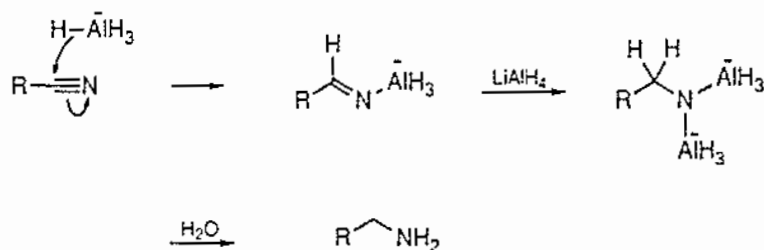
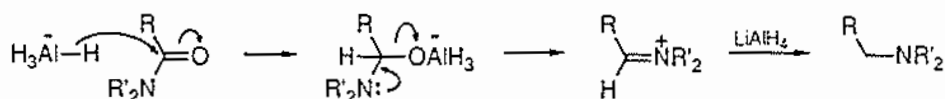
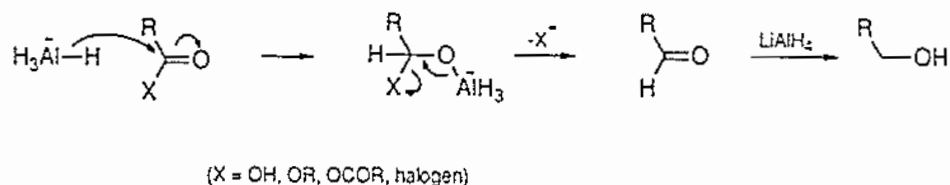


but other reducible groups in the molecule would also react (e.g. $C=C$, $C\equiv C$, $C=O$).

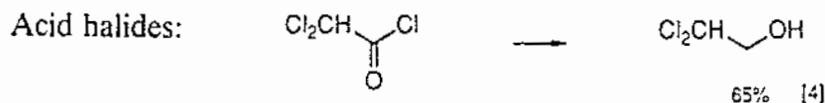
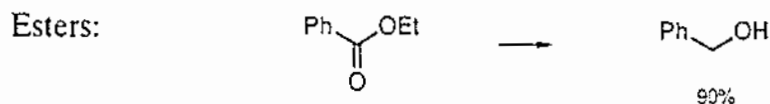
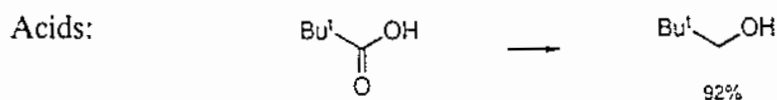
Many methods, some of unique and others of moderate selectivity, are now available.

(a) *Reduction to the alcohol or amine*

Lithium aluminium hydride reduces acids and all their derivatives to the alcoholic or amino level.* The general mechanisms are:



For example:



* Note the difference between the reaction with amides and those with other acid derivatives. It arises because $-\text{NR}_2$ is not as good a leaving group as, e.g. $-\text{Cl}$.

Anhydrides:



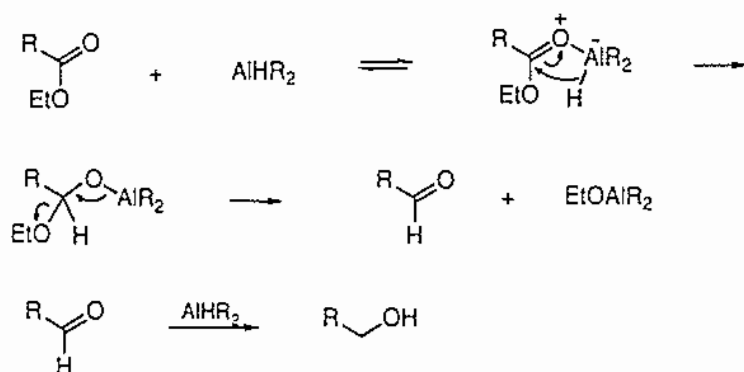
Amides:



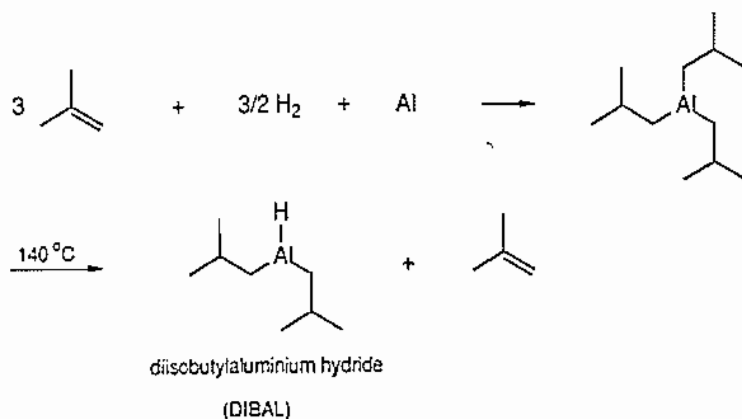
Nitriles:



The order of reactivity is the same as that in other nucleophilic displacements on acid derivatives: $\text{RCOCl} > \text{RCO}_2\text{R} > \text{RCONR}_2 > \text{RCN} > \text{RCO}_2\text{H}$. In contrast, with a *neutral* aluminium hydride (AlH_3) – an *alane* (cf. boranes) – the order of reactivity is almost the exact opposite: $\text{RCO}_2\text{H} > \text{RCONR}_2 > \text{RCN} > \text{RCO}_2\text{R} > \text{RCOCl}$. This is because the crucial step is the formation of a complex in which the aluminium hydride acts as an electrophile, so that it is the capacity for donation of an electron pair by the organic compound that is important, e.g.

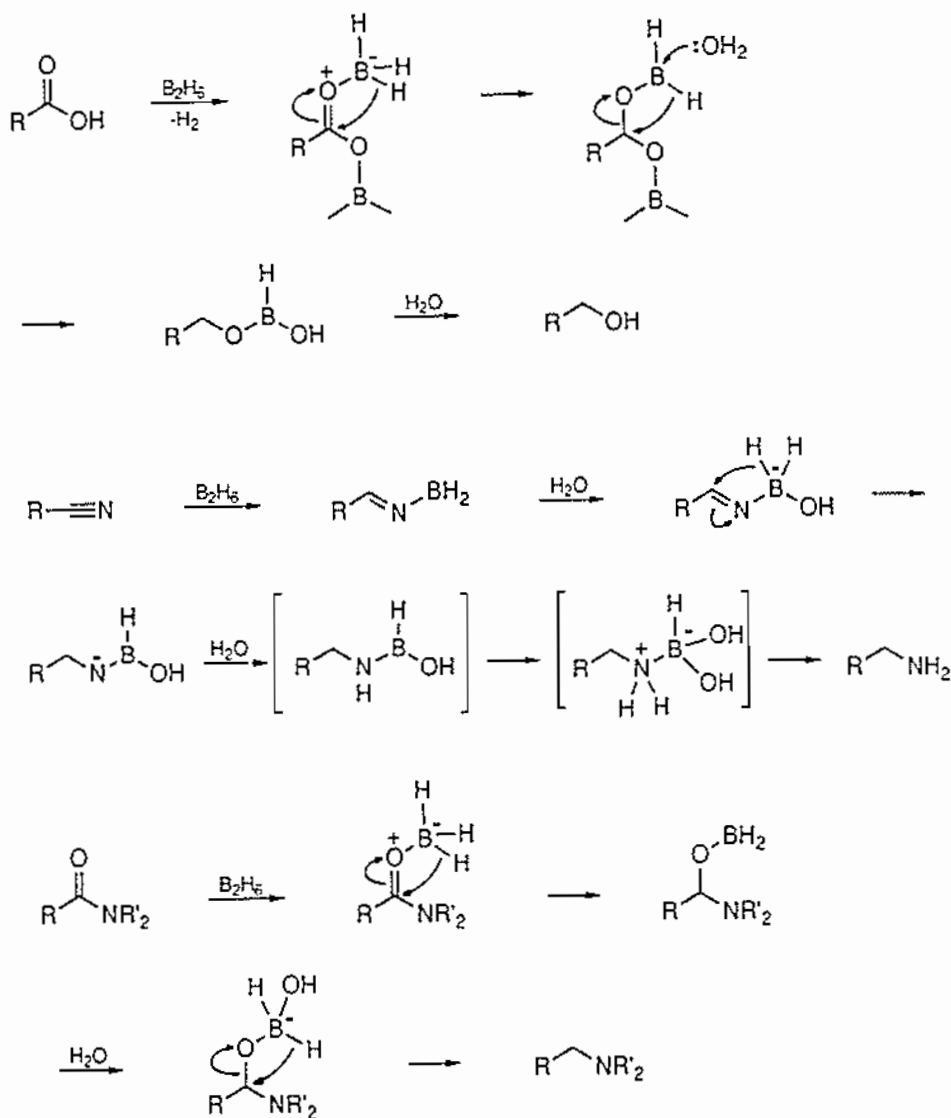


A suitable aluminium compound for this purpose is prepared by the reduction of isobutylene in the presence of aluminium:



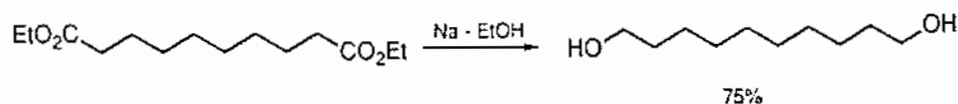
DIBAL is a convenient reagent for reducing carboxylic acids to alcohols. It can also be used for reducing esters to alcohols *via* aldehydes as above, but it is possible to stop reaction at the aldehyde stage (p. 662). Amides and nitriles also yield aldehydes (pp. 665, 666).

Diborane reduces carboxylic acids to alcohols and nitriles and amides to amines, again through complexes:



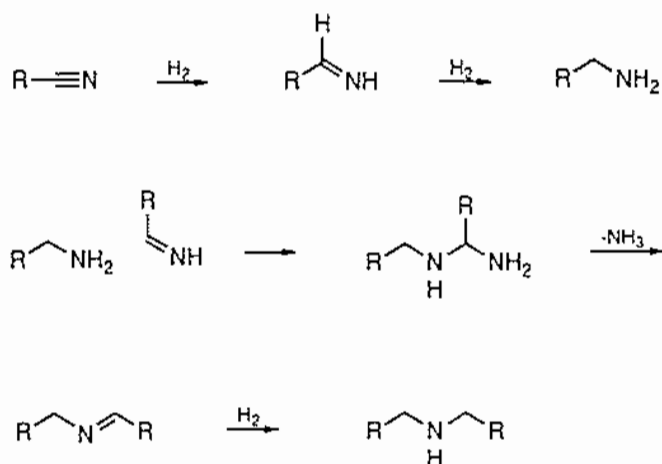
This method has the advantage, compared with lithium aluminium hydride, that ester and nitro groups are not reduced. For example, *p*-nitrobenzoic acid gives *p*-nitrobenzyl alcohol in about 80% yield.

Esters can be reduced by the acyloin method (p. 549) and by the *Bouveault-Blanc method* (sodium in refluxing ethanol), e.g.

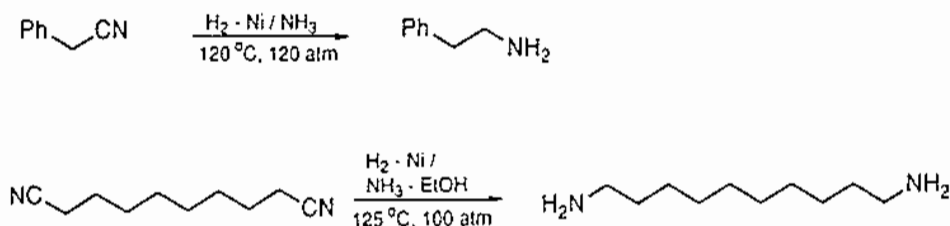


Acid chlorides can be reduced with sodium borohydride in diglyme solution. Acid, ester, amide, $C=C$, and $C\equiv C$ groups in the molecule are not reduced but aldehydic and ketonic groups are. Sodium trimethoxyborohydride, $NaBH(OCH_3)_3$, also reduces $-COCl$ to $-CH_2OH$ without affecting ester groups.

Nitriles can be reduced catalytically. In the absence of added ammonia, considerable quantities of secondary amines are formed since the primary amine can add to the imino intermediate (i.e. the half-reduced stage):



This is circumvented by the addition of an excess of ammonia which assumes the role of nucleophile in place of the primary amine and yields are then high, e.g. over Raney nickel, benzyl cyanide gives phenylethylamine in about 85% yield in liquid ammonia under pressure [3] and 1,8-octanedinitrile gives decamethylenediamine (79–80%) in ammoniacal ethanol [3]:



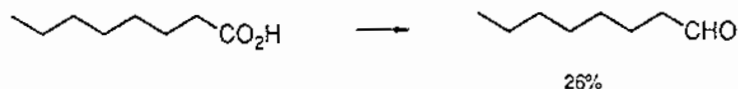
The reduction of nitriles by sodium in ethanol suffers from the disadvantage that secondary amines are formed, as in the ammonia-free catalytic method.

It should be noted that none of the reagents described above is uniquely selective, but in some cases where two methods are available one complements the other. For instance, diborane is the reagent of choice for acids when ester groups are present but not when $C=C$ or $C\equiv C$ bonds are, whereas lithium aluminium hydride is suitable in the latter case but not in the former. Benzylic systems are liable to hydrogenolysis, so that reductions of $ArCO_2H$, $ArCO_2R$, $ArCOCl$, $ArCONR_2$, $ArCH_2OCOR$, and $ArCH_2NHCOR$ are likely to give $ArCH_3$.

(b) *Reduction to aldehydes*

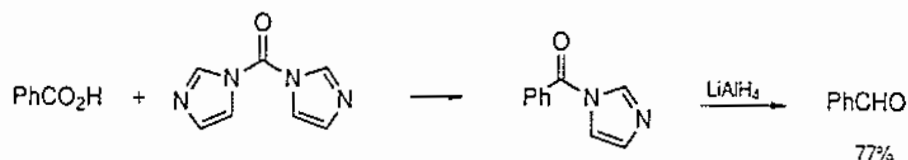
It is often necessary to convert acids and their derivatives into aldehydes, for aldehydes are not readily made by other methods whereas acids are readily available. Selective procedures are required since aldehydes are easily reduced to alcohols, and a number of older, lengthy procedures have been superseded by techniques using modified complex metal hydrides.

(i) *Acids.* Lithium in ethylamine can be used but yields are in general low, e.g.

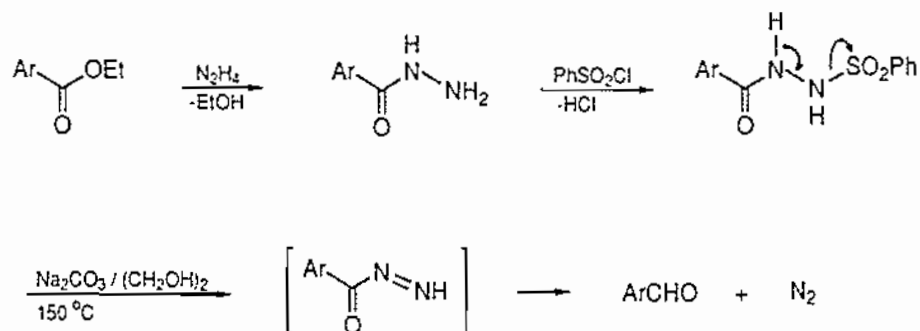


and sometimes zero (e.g. for $\text{Me}_3\text{CCO}_2\text{H}$), but in isolated instances yields of up to 80% have been recorded.

Satisfactory yields can be obtained by reducing the imidazolid of the acid with lithium aluminium hydride, e.g.

(ii) *Esters*

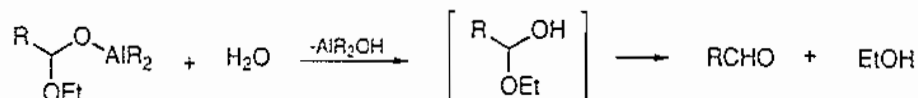
(1) *McFadyen and Stevens' method* consists in converting the ester into its hydrazide, treating this with benzenesulfonyl chloride, and hydrolyzing the product with base. The principle of the method is analogous to that of the Wolff-Kishner reduction (p. 649) and to the method for hydrogenolyzing primary aliphatic amines (p. 648):



Only aromatic aldehydes can be prepared in this way and yields are often low.

(2) *DIBAL* reduces esters to alcohols at room temperature (p. 660). Reaction occurs by way of an aluminium alkoxide intermediate which spontaneously forms aldehyde which, being more readily reduced than the original ester, yields the alcohol. However, at -78°C , the alkoxide is stable enough to accumulate and can

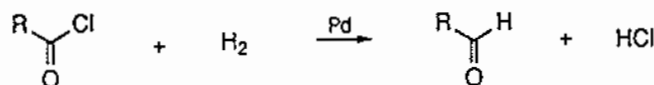
be trapped by added water which also destroys the residual reducing agent. The resulting adduct forms aldehyde:



Amide, nitrile, and $\text{C}\equiv\text{C}$ groups are also reduced.

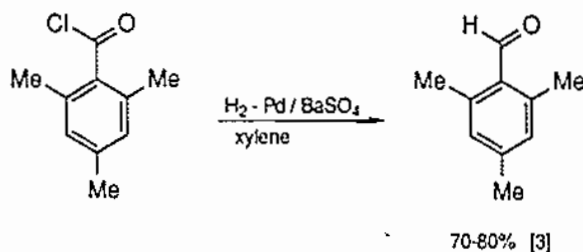
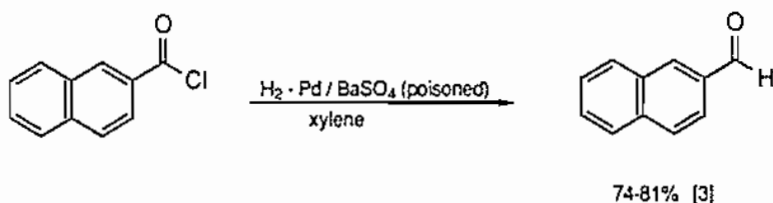
(iii) Acid halides

(1) *Rosenmund's method* consists of reducing the acid chloride with hydrogen on a palladium catalyst supported on barium sulfate or, less commonly, on calcium carbonate or charcoal.

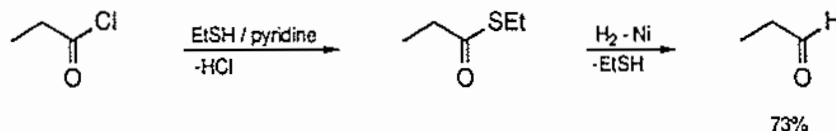


Temperature control is important: the reaction should be conducted at the lowest temperature possible in order to prevent further reduction. Refluxing toluene or xylene (i.e. 110–140°C) is usually suitable and the reaction can be followed by passing the hydrogen chloride into standard base. It is sometimes helpful partially to poison the catalyst: a commonly used poison is obtained by refluxing sulfur in quinoline.

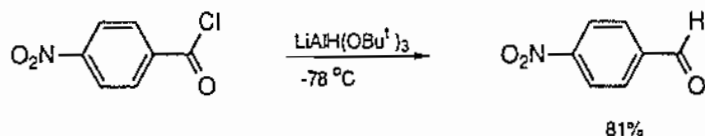
Yields are normally high, even with sterically hindered compounds, e.g.



(2) *Desulfurization* of thiol esters on Raney nickel which has been partially deactivated by boiling with acetone is effective, e.g.

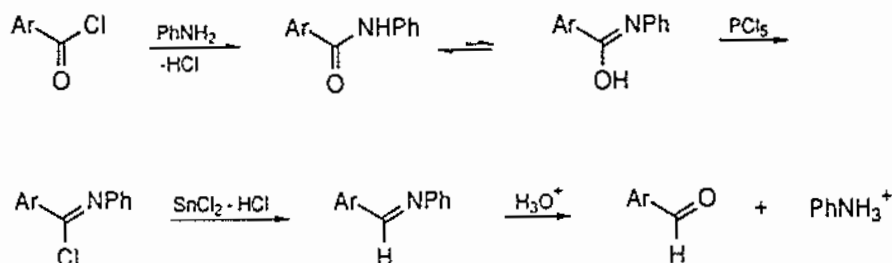


(3) *Lithium tri-*t*-butoxyaluminium hydride*, prepared from lithium aluminium hydride and *t*-butanol in ether, is considerably less reactive than LiAlH_4 since the electron-attracting alkoxy-group stabilizes the negatively charged aluminium ion. It reduces acid chlorides only to aldehydes at -78°C , and does not affect nitro, cyano, and ester groups, e.g.

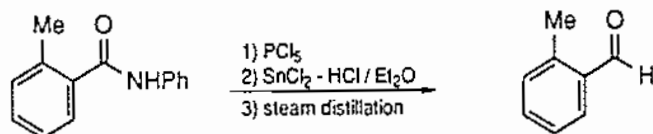


(iv) *Amides*

(1) The *Sonn-Müller* reaction involves conversion of an anilide or toluidide into the imino-chloride which is reduced to the imine with tin(II) chloride; hydrolysis gives the aldehyde.

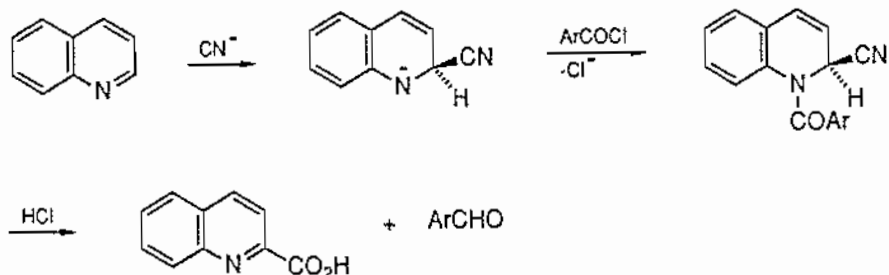


For example, *o*-tolualdehyde can be obtained in up to 70% yield [3]:



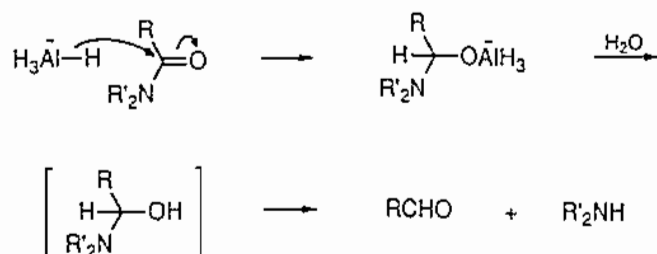
This method is not applicable to aliphatic aldehydes because the intermediate imino-chlorides are unstable.

(2) *The hydrolysis of Reissert compounds* has proved suitable for the preparation of some aromatic aldehydes. The acid chloride is added to quinoline in aqueous potassium cyanide and the resulting Reissert adduct is hydrolyzed with mineral acid:

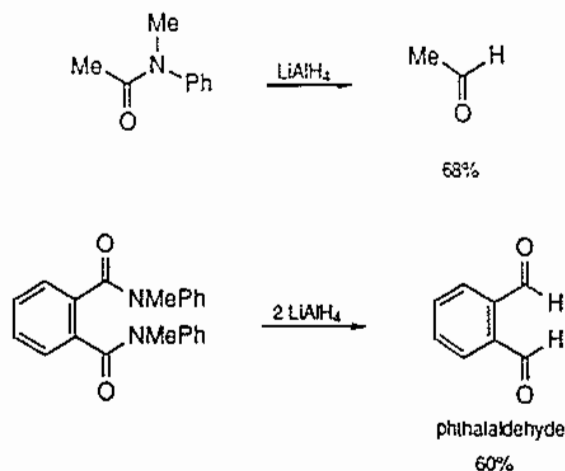


For example, *o*-nitrobenzoyl chloride gives *o*-nitrobenzaldehyde in about 65% yield. The method has also been used for introducing the carboxyl group into quinolines.

(3) *Lithium aluminium hydride* reduces disubstituted amides to aldehydes provided that the temperature is kept low:



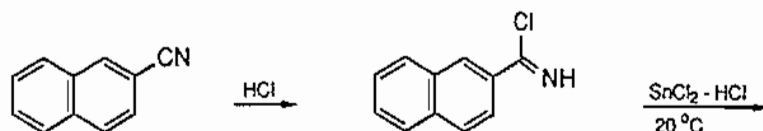
At higher temperatures the first intermediate undergoes elimination, and further reduction to the amine ensues (p. 659). The *N*-methylanilide is usually employed, e.g.



(4) *DIBAL* and *lithium diethoxyaluminium hydride*, $\text{LiAlH}_2(\text{OEt})_2$, from lithium aluminium hydride with two equivalents of ethanol, can be used for the conversion $\text{R}-\text{CONR}'_2 \rightarrow \text{R}-\text{CHO}$.

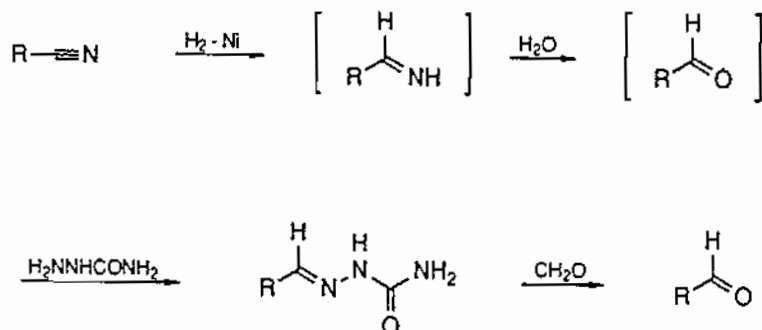
(v) Nitriles

(1) The *Stephen reaction* involves the addition of the nitrile to a suspension of anhydrous tin(II) chloride in diethyl ether saturated with hydrogen chloride, followed by hydrolysis. The reduction occurs at room temperature and is equivalent to that in the *Sonn-Müller reaction* (p. 664). For example, 2-naphthaldehyde can be prepared in about 75% yield [3]:

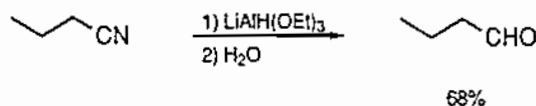




(2) *Hydrogen and Raney nickel* in the presence of semicarbazide and water gives the semicarbazone of the aldehyde from which the aldehyde is liberated by an exchange reaction with formaldehyde. Over-reduction is here prevented by the trapping of the aldehyde.



(3) *DIBAL* and *lithium triethoxyaluminium hydride*, $\text{LiAlH}(\text{OEt})_3$ (from lithium aluminium hydride and three equivalents of ethanol) can be used, e.g.

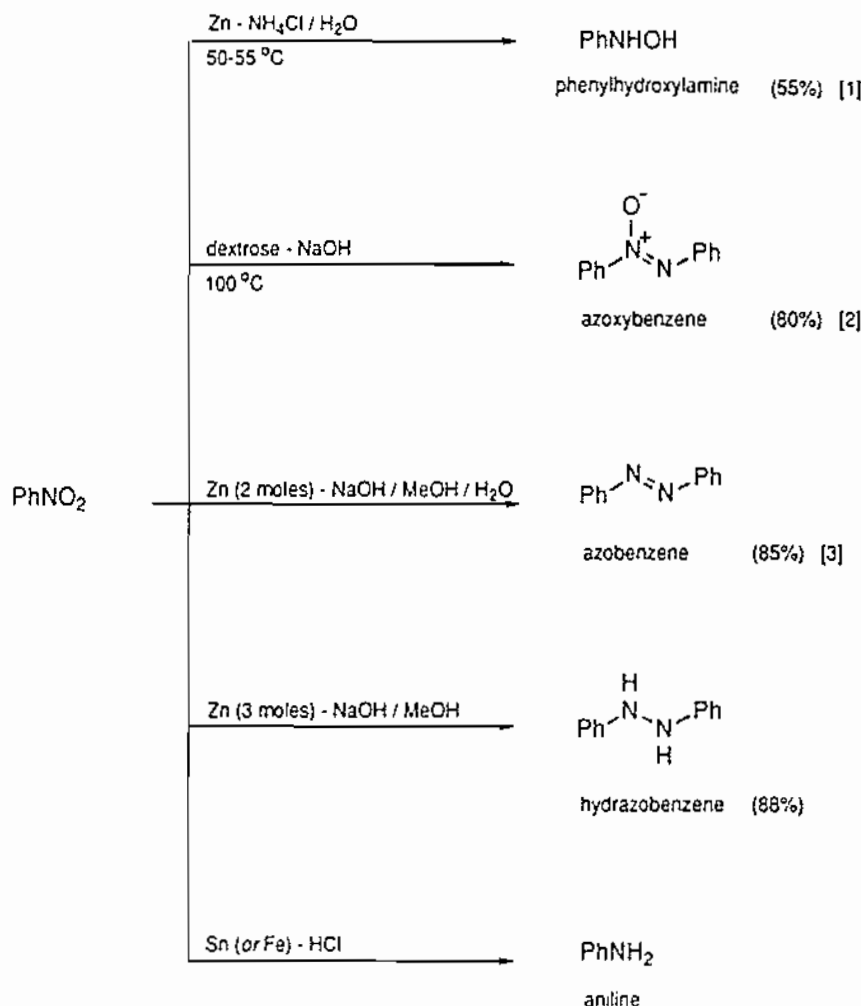


20.7 Systems containing nitrogen

(a) Nitro-compounds

The reduction of aliphatic nitro-compounds to amines is of little importance because the amino group is easily introduced in a number of other ways (e.g. from the halide and an amine, p. 302). Lithium aluminium hydride and catalytic reduction on Raney nickel are both suitable methods, e.g. 2-nitrobutane is reduced to 2-aminobutane in 85% yield by the former reagent (compare the mode of reduction of aromatic nitro-compounds by LiAlH_4 ; below).

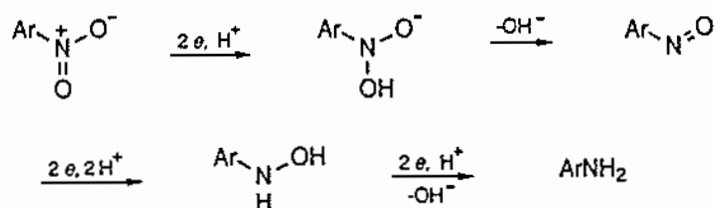
The reduction of aromatic nitro-compounds is much more important because the nitro group can be introduced into a wide variety of aromatic systems by nitration whereas the amino group can only be introduced directly into those aromatic compounds which are strongly activated to nucleophiles (p. 397). Several types of reduction product are obtainable, as illustrated for nitrobenzene:



In neutral solution buffered with ammonium chloride, phenylhydroxylamine is the main product. In basic solution, using a weak reducing agent such as dextrose or glucose, the intermediate nitrosobenzene reacts with phenylhydroxylamine to give azoxybenzene, which the more powerful reducing agents convert into azobenzene and hydrazobenzene. Lithium aluminium hydride gives azobenzene quantitatively and, in the presence of Lewis acids, brings about further reduction to hydrazobenzene. Metals in acid solution bring about reduction to aniline; industrially, iron and dilute hydrochloric acid are used, the product is neutralized with lime, and aniline is isolated by distillation in steam.

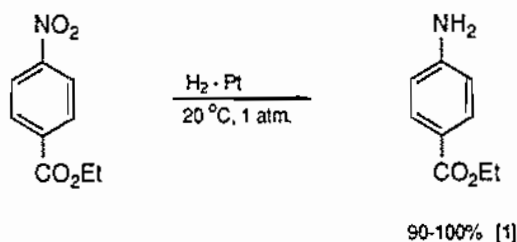
The only member of the series which cannot be isolated is nitrosobenzene, but this is obtainable by the oxidation of phenylhydroxylamine (p. 622).

Controlled electrolytic reduction occurs in exact stages:

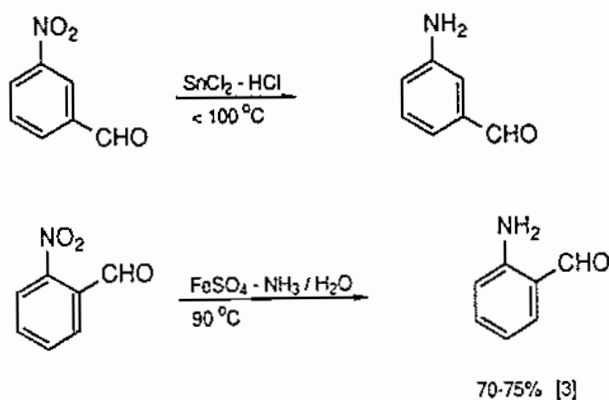


It is usually the amino group which is required from the reduction and the problem is to find a selective reagent which leaves other reducible groups intact, so that the standard reagent for reducing nitrobenzene to aniline, tin and hydrochloric acid, is unsuitable for many substituted nitro-compounds.

Catalytic methods are suitable provided that $C=C$ and $C\equiv C$ bonds are not present: e.g.

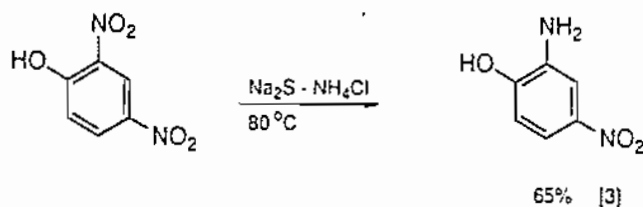


and transfer-hydrogenation by hydrazine on palladium is also very effective. Electron-transfer reagents of mild type can be used, e.g.

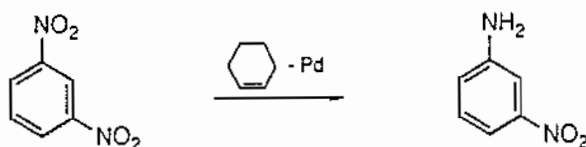


The complement to the last two reactions, reduction of the aldehydic group without reduction of the nitro group, can be effected with sodium borohydride.

It is also possible to reduce one nitro group in the presence of another. The older method employed ammonium sulfide, e.g.

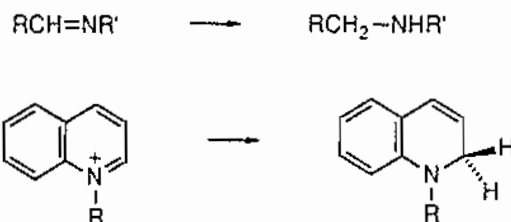


but catalytic methods are usually more efficient. For example, *m*-dinitrobenzene gives *m*-nitroaniline almost quantitatively when reduced on palladium with cyclohexene as the hydrogen donor (p. 632):

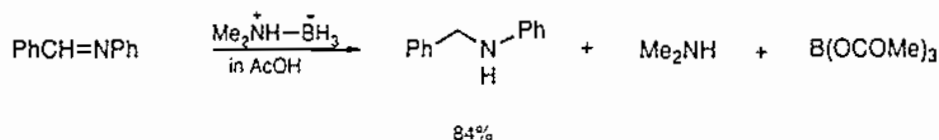


(b) *Imines*

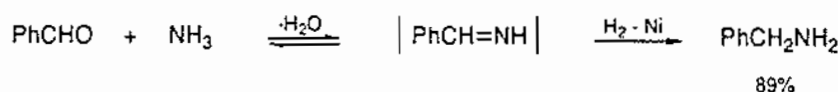
Imines of the type $\text{RCH}=\text{NR}'$ and $\text{RCH}=\text{N}^+\text{R}'_2$ can be selectively reduced with sodium borohydride:



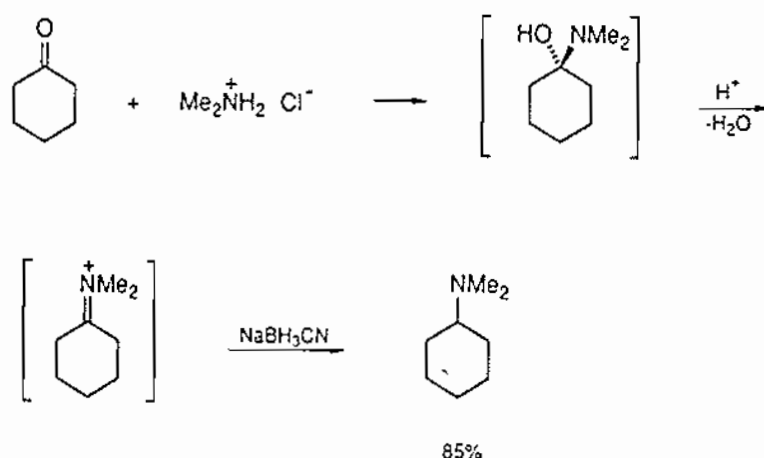
An alternative reagent is dimethylamine borane, which reduces $\text{C}=\text{N}$ without reducing acids, esters, or nitro groups. Yields are high, e.g.



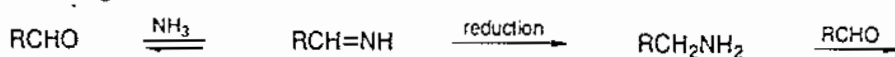
(i) *Reductive amination.* When the reaction between an aldehyde or ketone and ammonia or a primary or secondary amine is carried out in the presence of a suitable reducing agent, the imine or immonium ion which is formed is reduced *in situ* to an amine. The usual reducing agents are hydrogen on Raney nickel, e.g.



or sodium cyanoborohydride, e.g.



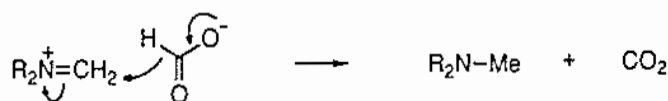
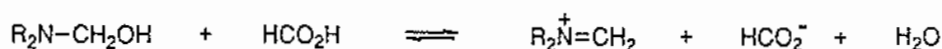
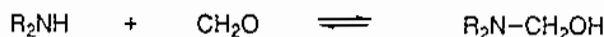
A side reaction can be disadvantageous: as the amine is formed, it can react with more of the carbonyl compound to give a new imine and, by reduction, a new amine, e.g.





This can be minimized by using a large excess of ammonia or primary amine.

A variation of the method suitable for making tertiary amines in which there is at least one *N*-methyl group is to heat an amine with formaldehyde and formic acid under reflux (c. 100°C) (*Eschweiler-Clarke reaction*). With a secondary amine, the intermediate immonium ion is reduced by formate ion:

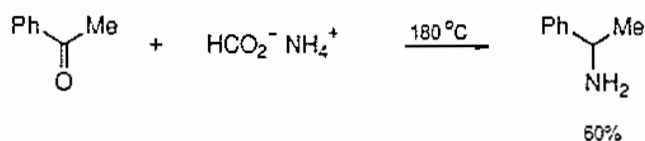


When the starting amine is primary, successive condensations and reductions occur:



For example, the conversion $\text{PhCH}_2\text{CH}_2\text{NH}_2 \rightarrow \text{PhCH}_2\text{CH}_2\text{NMe}_2$ occurs in about 80% yield [3].

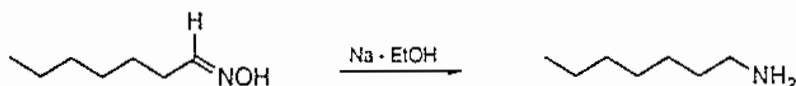
The *Leuckart reaction* is a variant of the Eschweiler-Clarke reaction in which aldehydes or ketones other than formaldehyde are heated with ammonium formate or a formamide at 180–200°C, e.g.



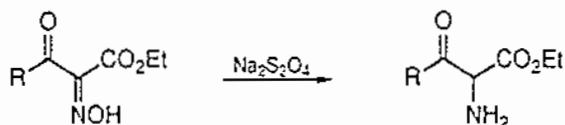
However, while it is a more general procedure than the Eschweiler-Clarke process, yields are usually lower.

(c) Oximes

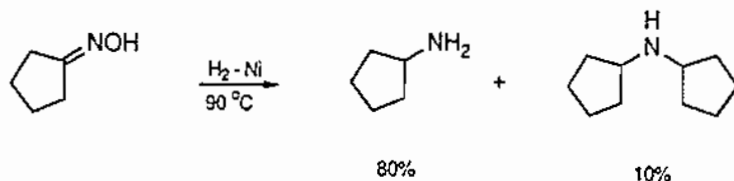
Electron-transfer reagents are normally employed, as in the formation of 1-heptylamine in 60% yield from heptanal oxime with sodium in ethanol [2],



and in the Knorr synthesis of pyrroles (p. 686), where sodium dithionite is preferably used:

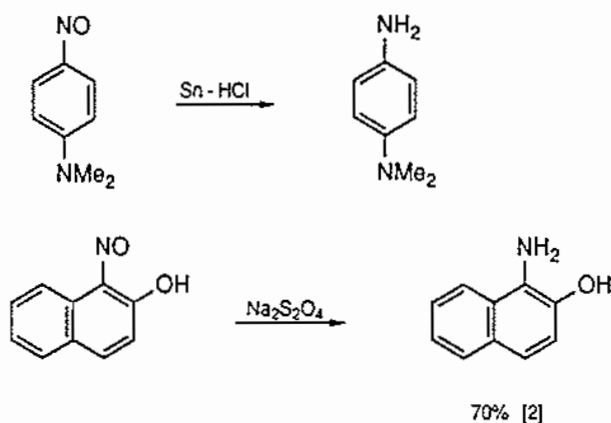


Catalytic methods are reasonably efficient but tend to produce some of the secondary amine, e.g.



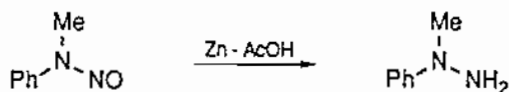
(d) Nitroso-compounds

C-Nitroso-compounds can be converted into hydroxylamines by controlled electrolytic reduction (p. 667). Reduction to amines is normally brought about by electron-transfer reagents, e.g.

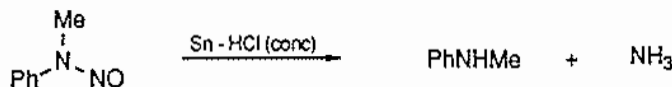


These reactions are commonly used for obtaining amino groups in aromatic nuclei which are strongly activated towards electrophiles, since it is often easier to nitrosate these compounds than to nitrate them (see the synthesis of pteridine, p. 719).

N-Nitroso-compounds are reduced by mild electron-transfer reagents to substituted hydrazines, e.g. *N*-nitroso-*N*-methylaniline with zinc in acetic acid gives 55% of *N*-methyl-*N*-phenylhydrazine [2]:

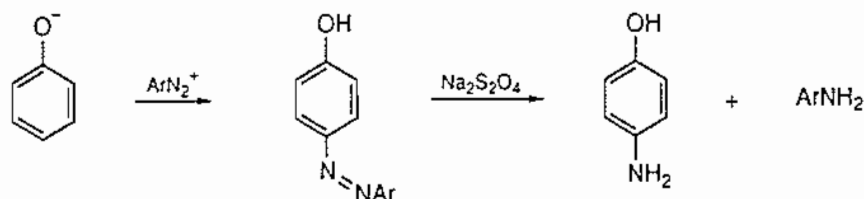


Stronger reducing agents rupture the N—N bond, e.g.



(e) *Azo-compounds*

Azo-compounds are reduced to hydrazo-compounds by lithium aluminium hydride in the presence of a Lewis acid, as already described (p. 667). A more useful type of reduction is brought about by sodium dithionite which cleaves the $\text{N}=\text{N}$ bond to give amines.* Since azo-compounds can be obtained by diazonium-coupling both to strongly activated aromatic compounds and to activated methylene compounds (section 13.4), this provides a useful route for the introduction of the amino group, e.g.



An alternative method, nitrosation followed by reduction, is also available.

The reduction of diazonium salts both to arylhydrazines and to hydrocarbons has been described earlier (pp. 420, 413).

20.8

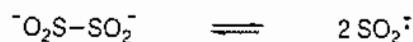
Systems containing sulfur

Catalytic reduction of sulfur-containing compounds removes the sulfur as hydrogen sulfide. Reduction with retention of the sulfur atom requires selective methods; the less active electron-transfer reagents are usually employed.

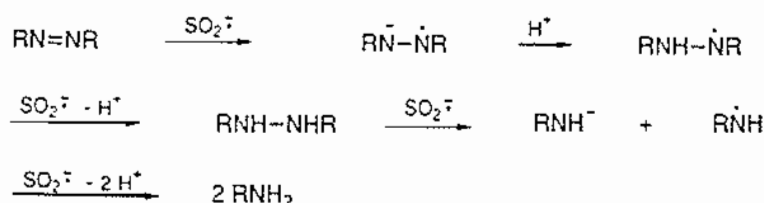
(a) *Disulfides*

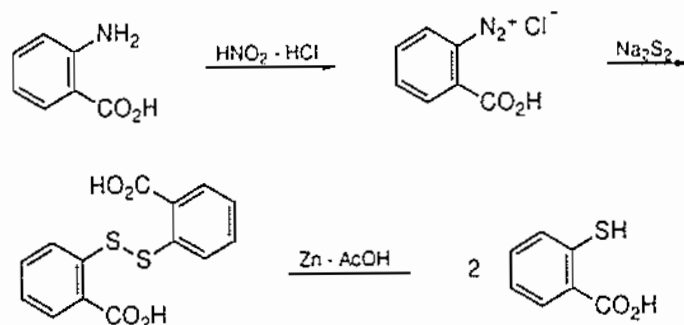
Zinc in refluxing acetic acid is normally used. For example, thiosalicylic acid can be obtained from anthranilic acid in 75–80% overall yield in this way [2]:

* The dithionite ion is in equilibrium in solution with the sulfur dioxide anion-radical,



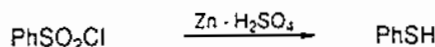
and it is the latter which is the reducing agent, acting as a one-electron donor, probably as follows:



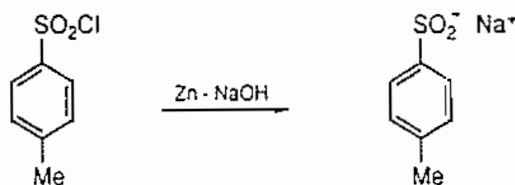


(b) Sulfonyl chlorides

Sulfonic acids are not easily reduced, but the readily derived chlorides can be reduced to sulfinic acids and to thiols. There is a resemblance to reduction of the nitro group (p. 667): zinc or tin in mineral acid gives the thiol, whereas in neutral or alkaline solution intermediate products are obtainable. For example, benzene-sulfonyl chloride with zinc in sulfuric acid gives thiophenol in 96% yield [1],



and toluene-*p*-sulfonyl chloride with zinc in aqueous sodium hydroxide gives sodium toluene-*p*-sulfinate in 64% yield [1],

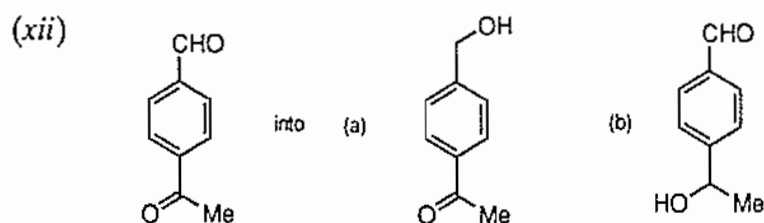
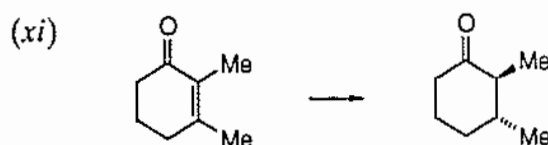
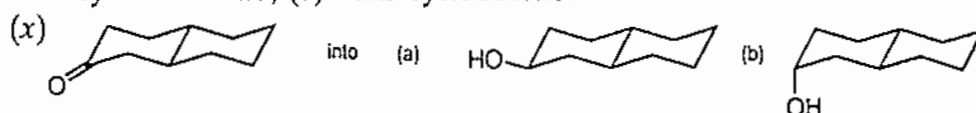


1. How would you carry out the following transformations?

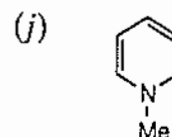
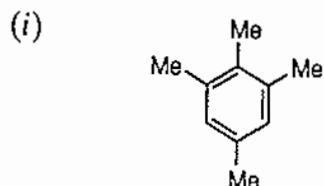
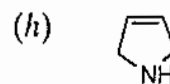
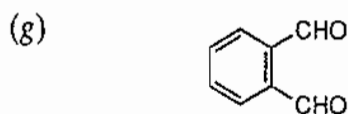
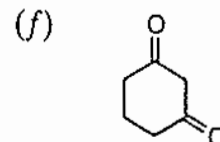
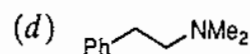
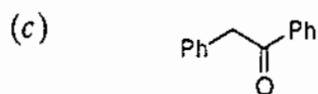
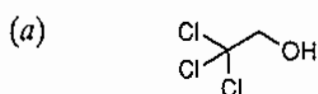
- (i) $\text{R}-\text{CH}=\text{CH}_2$ into (a) $\text{R}-\text{CH}_2-\text{CH}_2\text{OH}$, (b) $\text{R}-\text{CH}(\text{OH})-\text{CH}_3$, (c) $\text{R}-\text{CH}_2-\text{CH}_2-\text{NH}_2$.
- (ii) $\text{R}-\text{C}\equiv\text{CH}$ into (a) $\text{R}-\text{CO}-\text{CH}_3$, (b) $\text{R}-\text{CH}_2-\text{CHO}$.
- (iii) $\text{R}-\text{C}\equiv\text{C}-\text{R}$ into the corresponding (a) *cis*- and (b) *trans*-alkenes.
- (iv) $\text{EtO}_2\text{C}-\text{CH}_2-\text{CH}_2-\text{CO}_2\text{H}$ into (a) $\text{EtO}_2\text{C}-\text{CH}_2-\text{CH}_2-\text{CH}_2\text{OH}$ and (b) $\text{HOCH}_2-\text{CH}_2-\text{CH}_2-\text{CO}_2\text{H}$.
- (v) $\text{Ph}-\text{CH}=\text{CH}-\text{CO}_2\text{H}$ into (a) $\text{Ph}-\text{CH}_2-\text{CH}_2-\text{CO}_2\text{H}$, (b) $\text{Ph}-\text{CH}=\text{CH}-\text{CH}_2\text{OH}$, (c) $\text{Ph}-\text{CH}=\text{CH}-\text{CHO}$, (d) $\text{Ph}-\text{CH}_2-\text{CH}_2-\text{CH}_2\text{OH}$, (e) γ -cyclohexylpropanol.
- (vi) *m*-Nitrobenzoyl chloride into *m*-nitrobenzaldehyde.
- (vii) *m*-Nitrobenzaldehyde into (a) *m*-nitrobenzyl alcohol, (b) *m*-amino-benzaldehyde, (c) *m*-nitrotoluene.
- (viii) 1-Methylcyclohexene into *trans*-2-methylcyclohexanol.

Problems

(ix) $\text{EtO}_2\text{C}-(\text{CH}_2)_8-\text{CO}_2\text{Et}$ into (a) decamethylenediol, (b) cyclodecanone, (c) *trans*-cyclodecene.

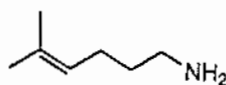


2. Outline a synthesis, which includes a reductive method, of each of the following, from readily available compounds:

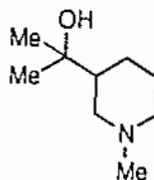


3. How would you attempt to convert acrylonitrile into adiponitrile ($\text{NC}-(\text{CH}_2)_4-\text{CN}$) (which is required for the manufacture of nylon)?

4. An attempt to make the *N*-dimethyl derivative of the amine (I) by treatment with aqueous formaldehyde in the presence of formic acid gave the compound (II). Discuss.



(I)



(II)

5. Summarize the uses of hydrazine and its derivatives in effecting the reduction of various types of organic groupings.